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(54) **SYSTEMS AND METHODS OF ADJUSTABLE SUSPENSIONS FOR OFF-ROAD RECREATIONAL VEHICLES**

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See application file for complete search history.

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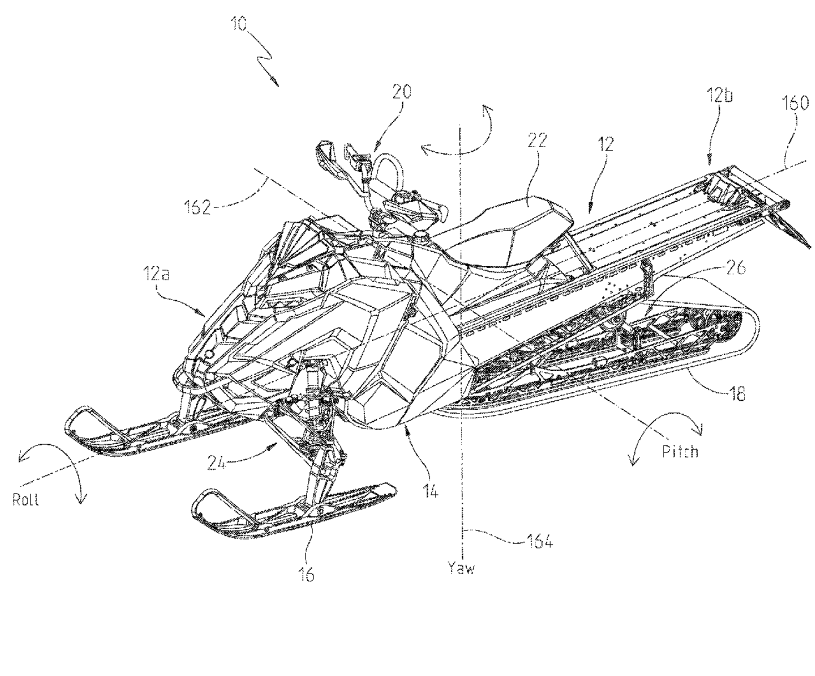
(57) **ABSTRACT**

A damping control system is provided for an off-road recreational vehicles having a suspension located between a ground engaging member and a vehicle frame and including at least one adjustable shock absorber having an adjustable damping characteristic based on an input from a sensor.

33 Claims, 46 Drawing Sheets



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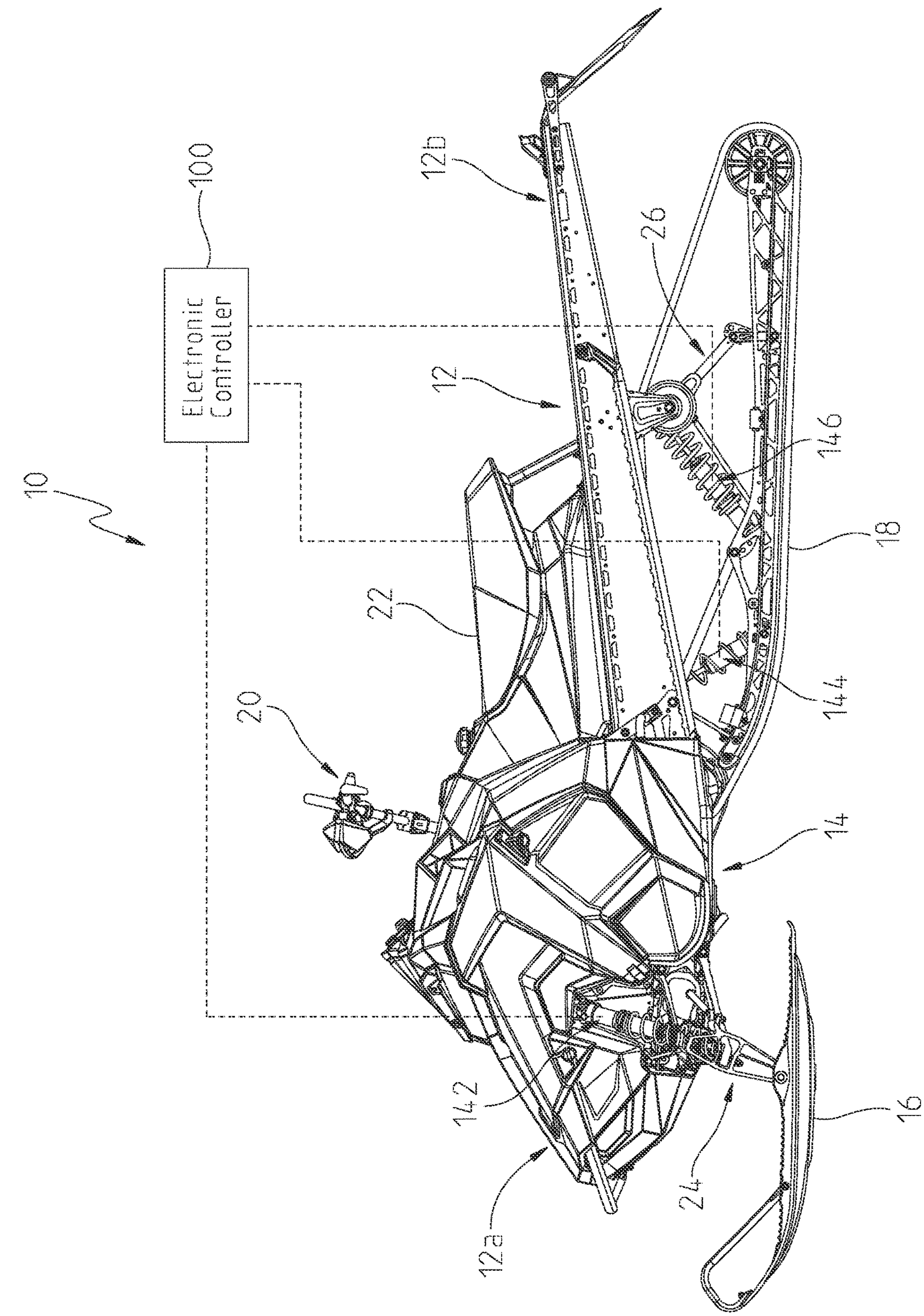
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12b
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12
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18
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16
24
12a

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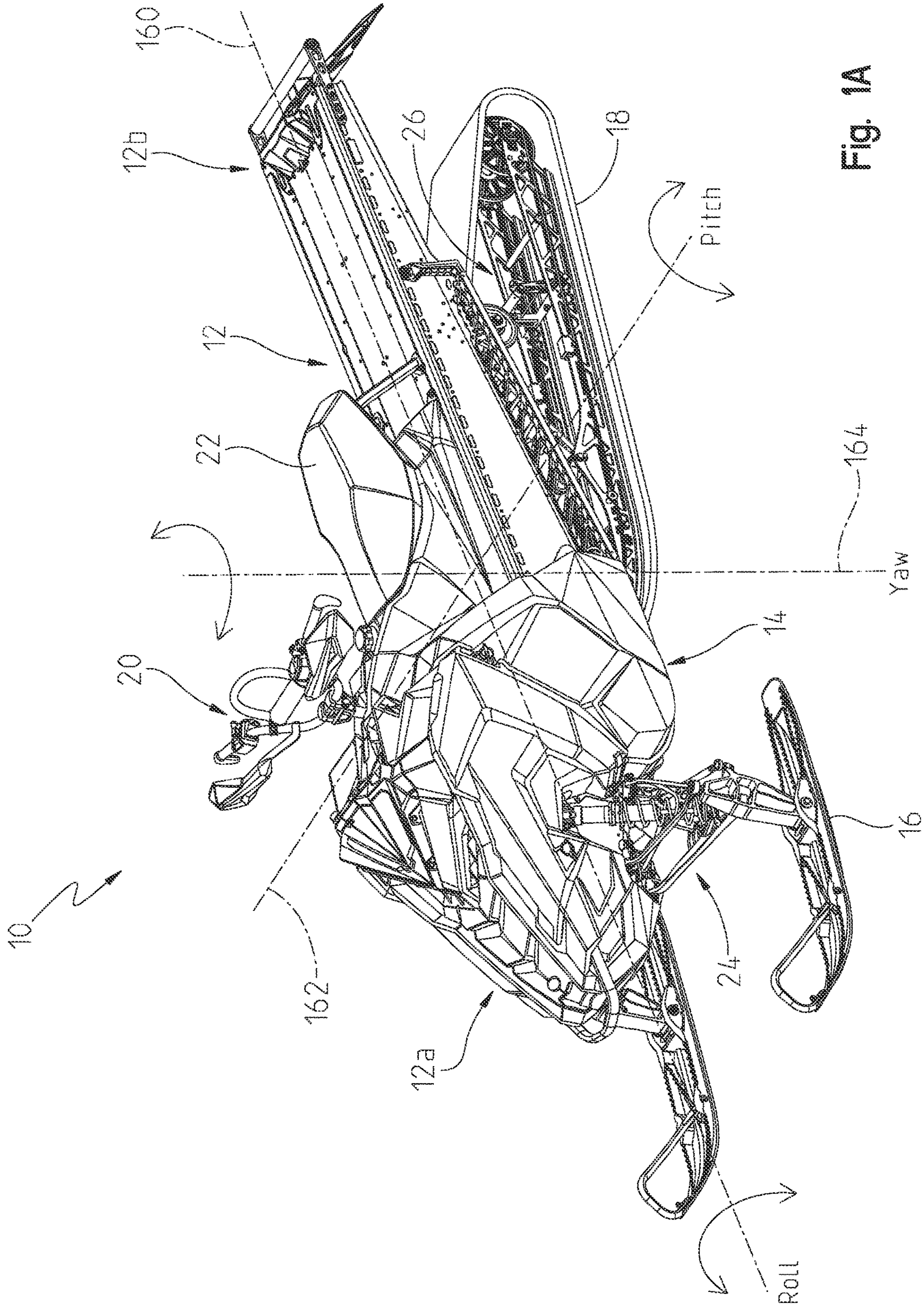

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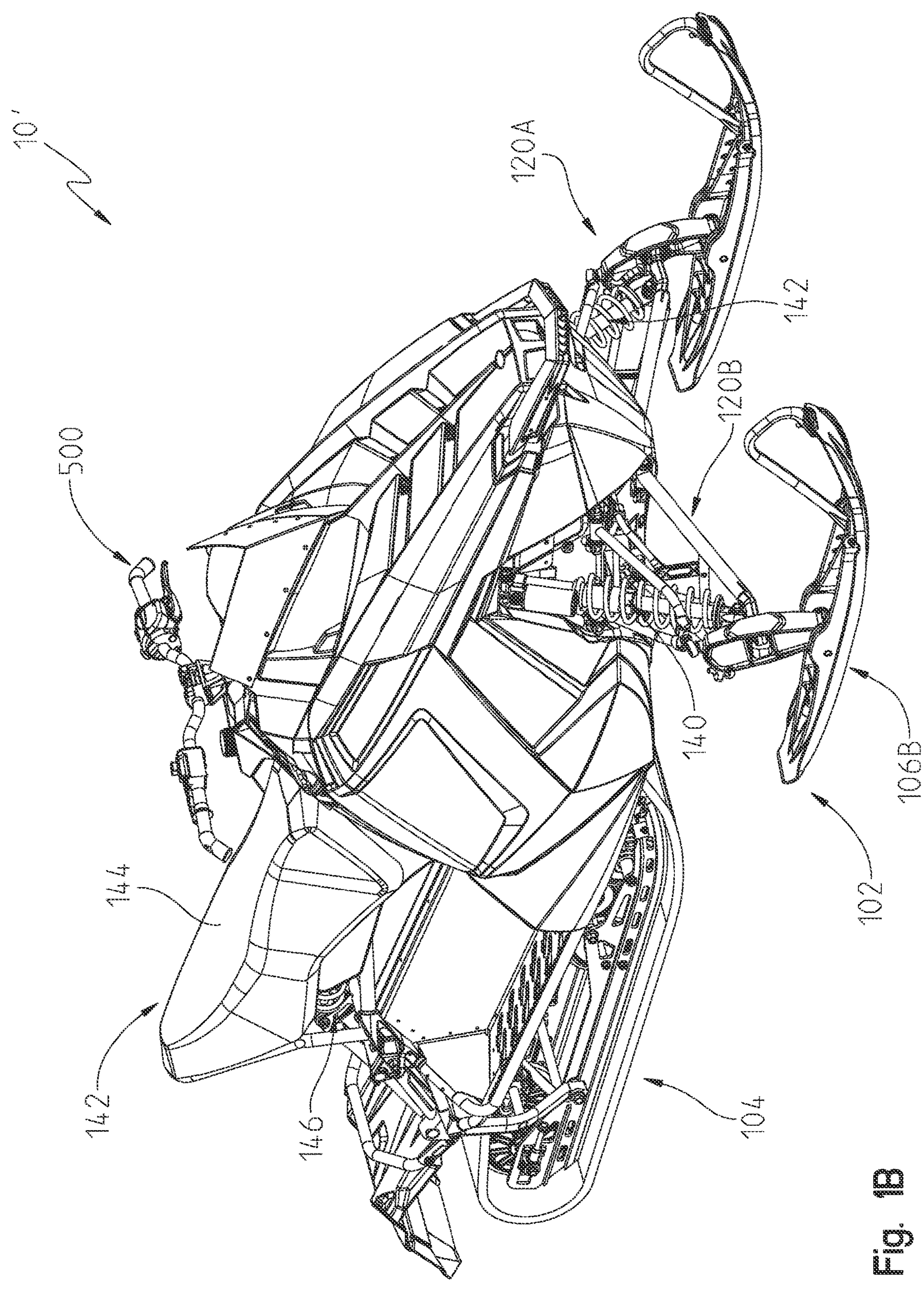

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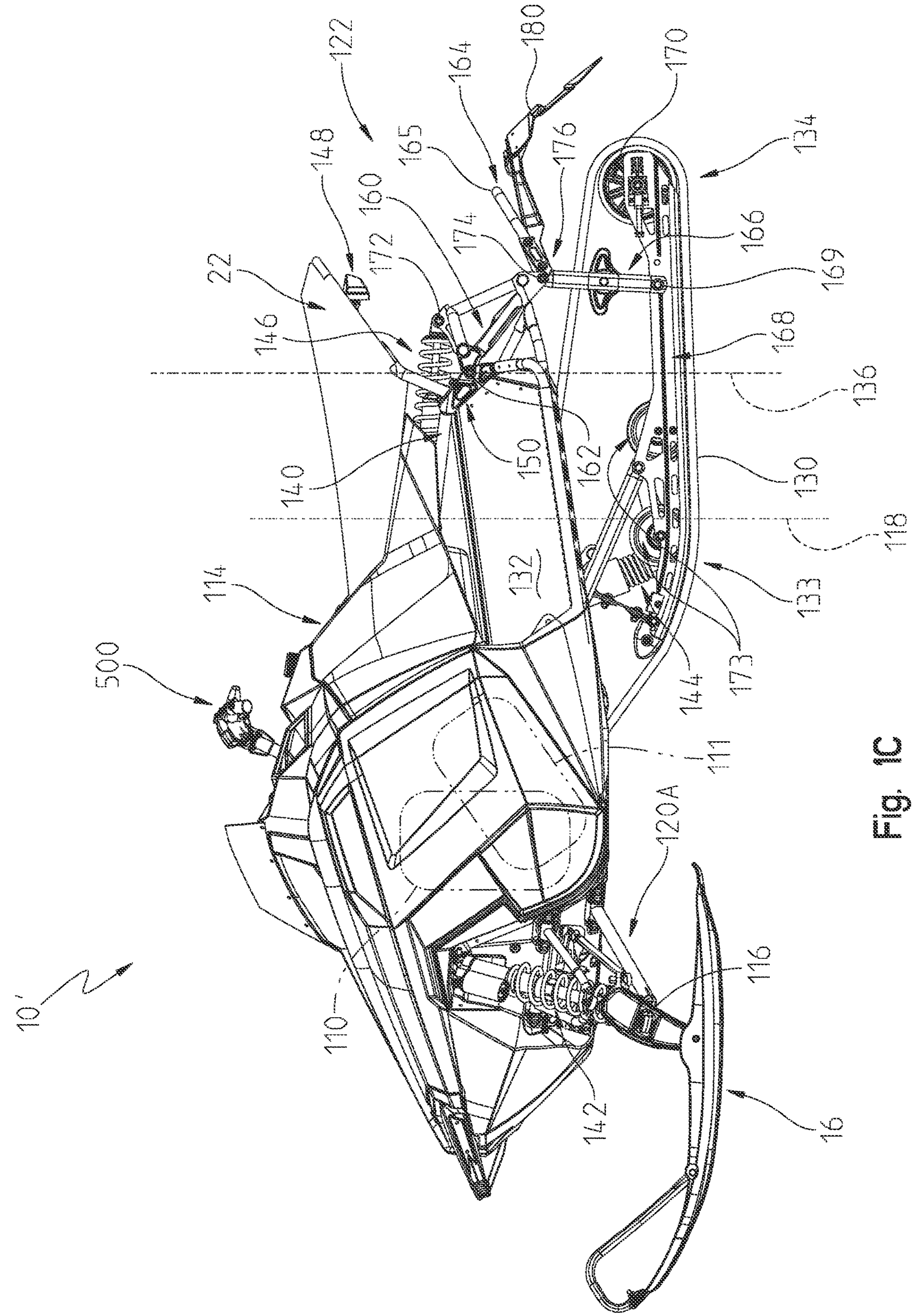
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142
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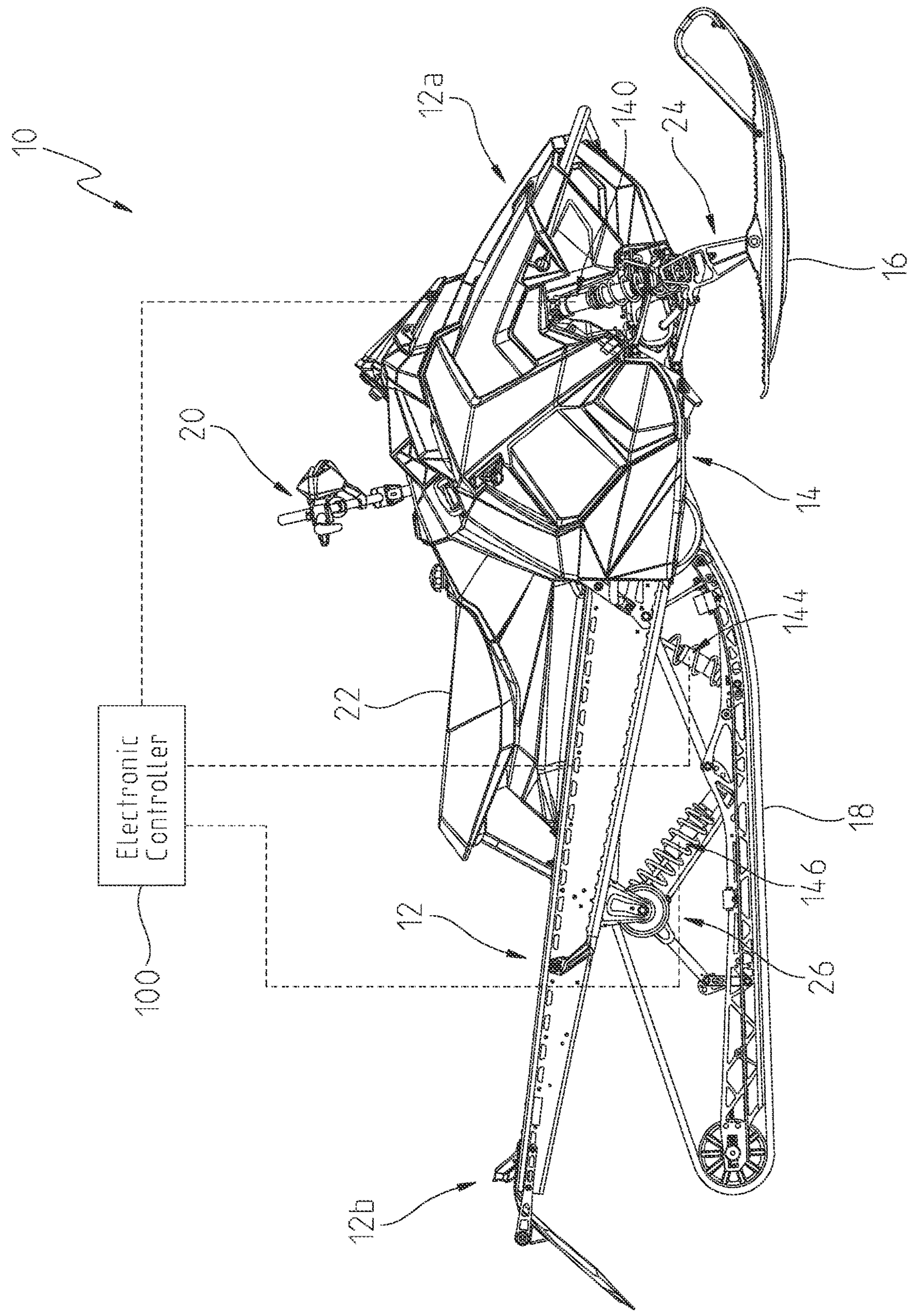
Fig. 1C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12b

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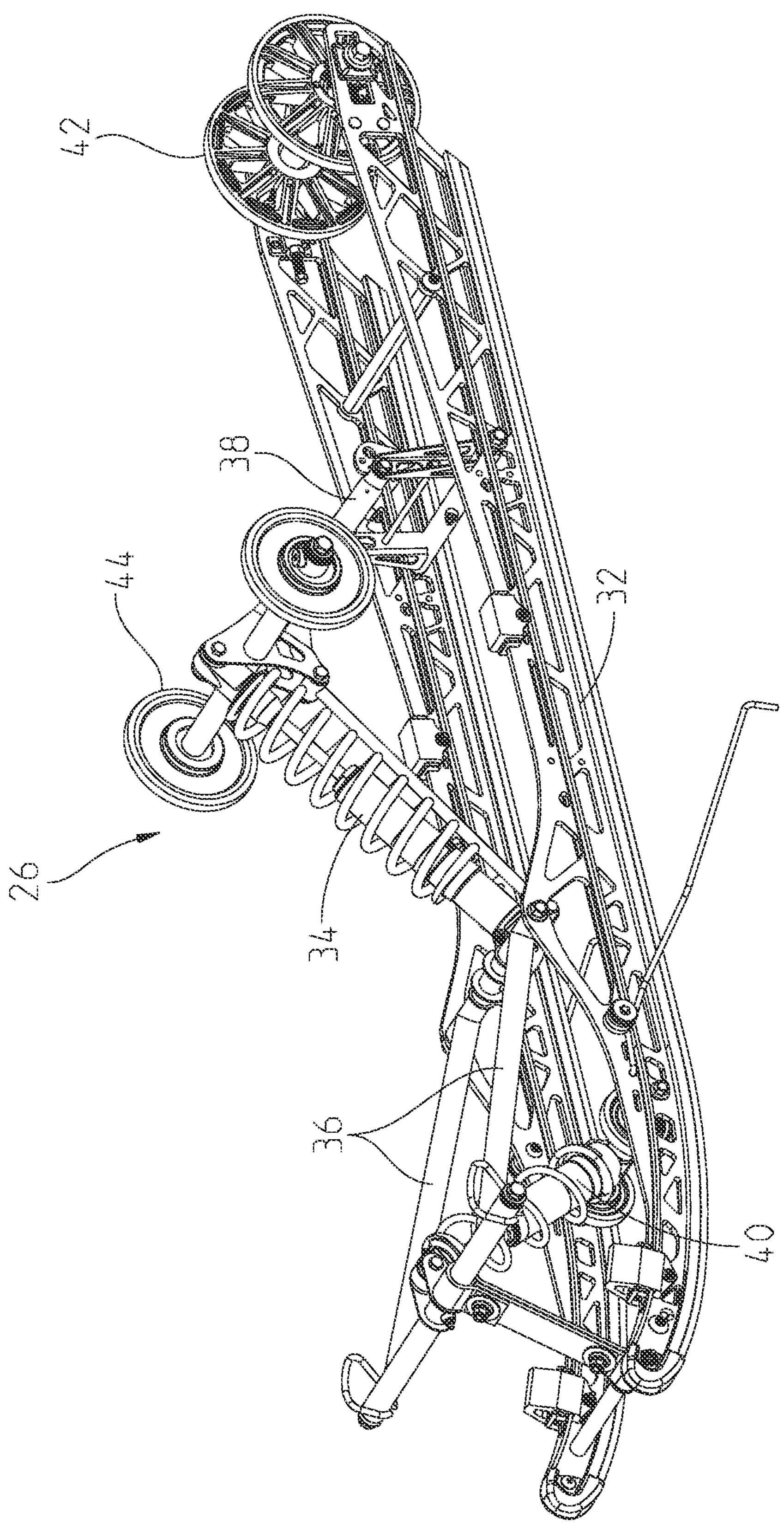

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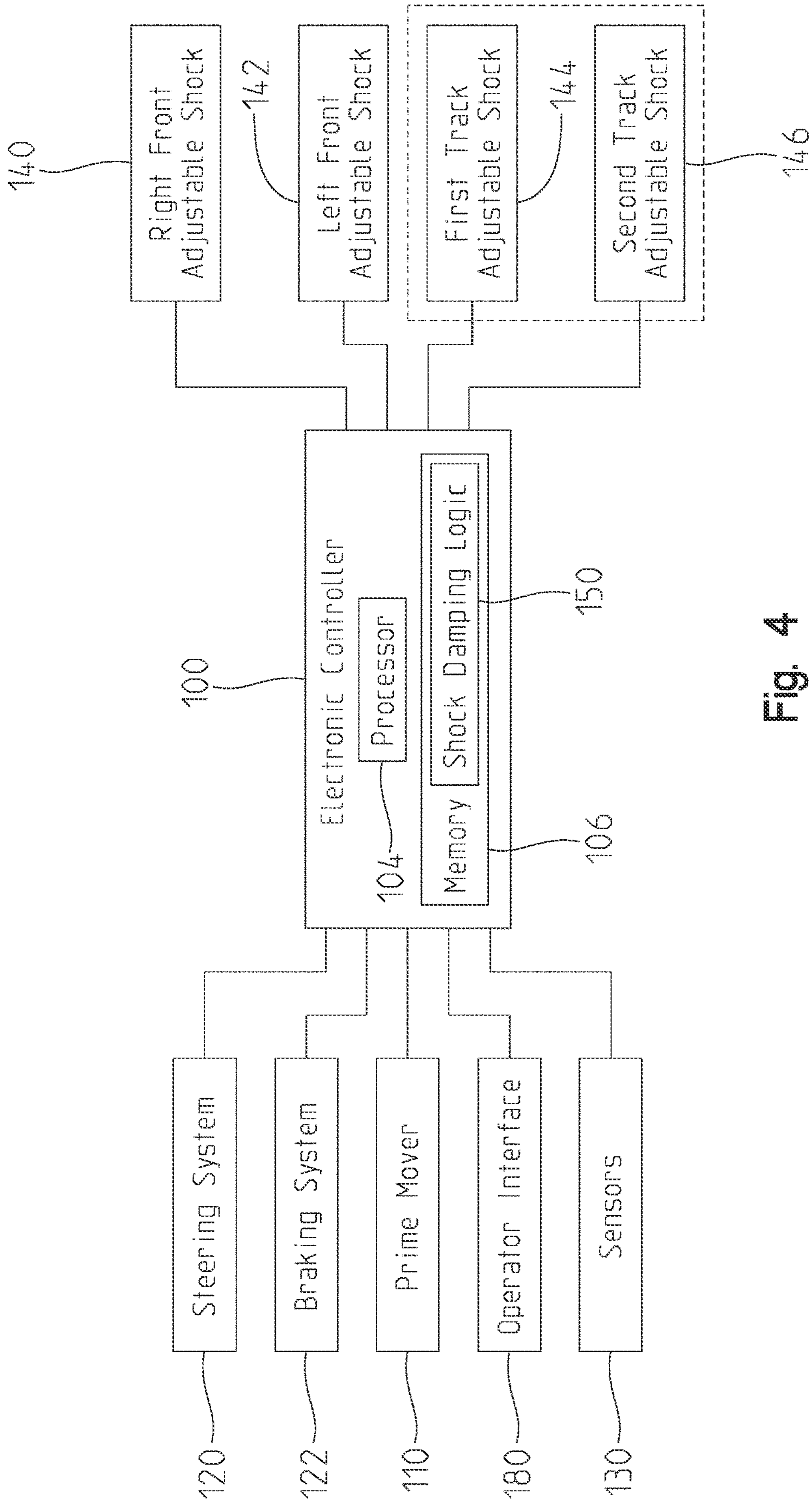
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Left Front Adjustable Shock
142
First Track Adjustable Shock
144
Second Track Adjustable Shock
146
Electronic Controller
100
Processor
104
Shock Damping Logic
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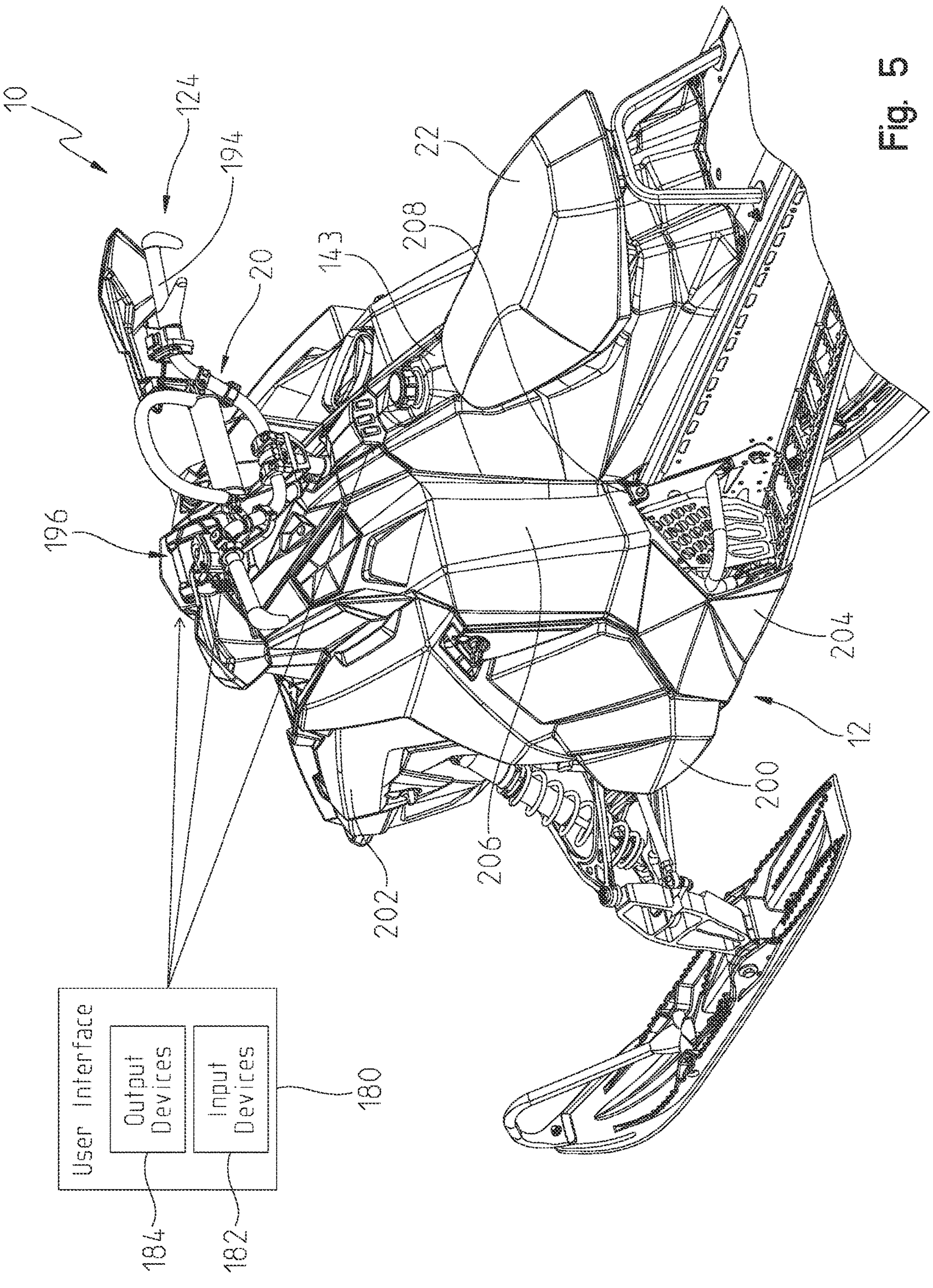
Memory
106
Steering System
120
Braking System
122
Prime Mover
110
Operator Interface
180
Sensors
130

Fig. 4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User Interface
Output Devices
Input Devices
180
184
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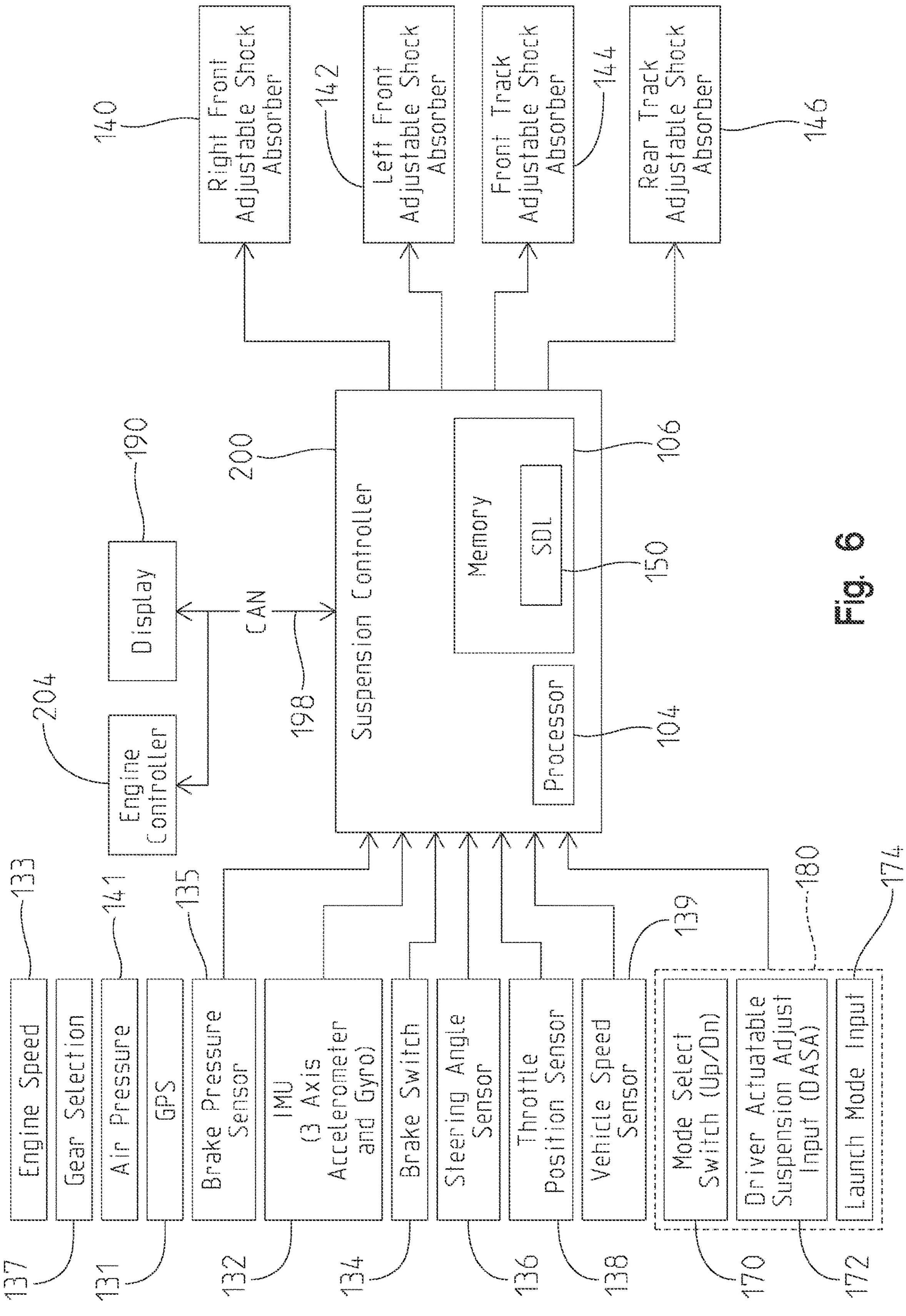
Fig. 5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Gear Selection
Air Pressure
GPS
Brake Pressure Sensor
IMU (3 Axis Accelerometer and Gyro)
Brake Switch
Steering Angle Sensor
Throttle Position Sensor
Vehicle Speed Sensor
Mode Select Switch (Up/Dn)
Driver Actuatable Suspension Adjust Input (DASA)
Launch Mode Input
137
131
132
134
136
138
170
172
133
141
135
139
180
174
Engine Controller
204
Display
190
CAN
198
Suspension Controller
200
Memory
SDL
Processor
106
150
104
Right Front Adjustable Shock Absorber
Left Front Adjustable Shock Absorber
Front Track Adjustable Shock Absorber
Rear Track Adjustable Shock Absorber
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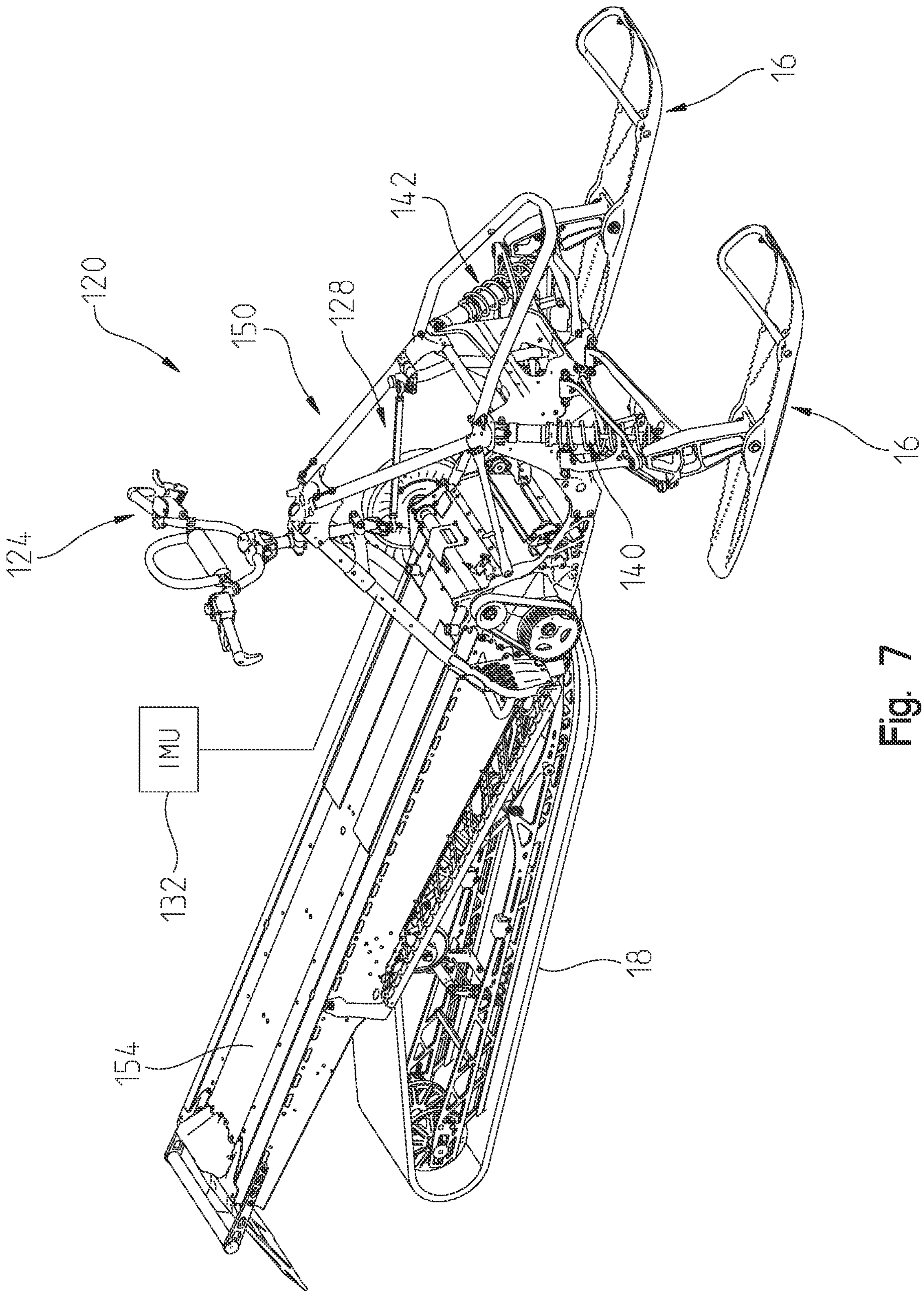
142
144
146

Fig. 6



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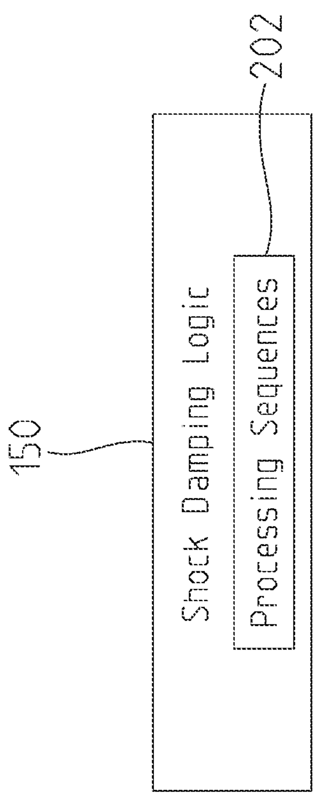
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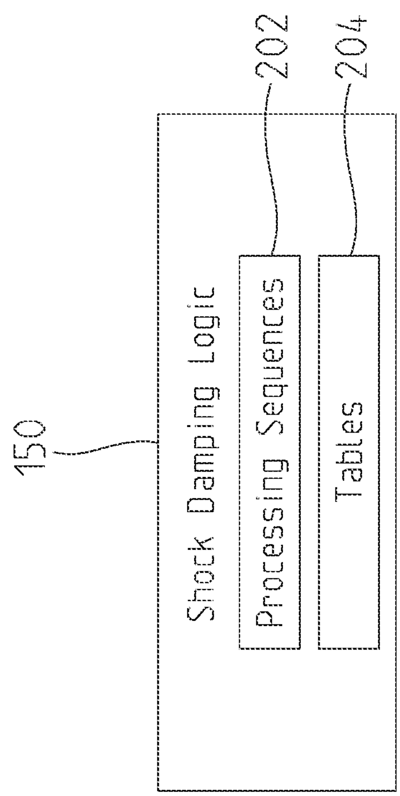
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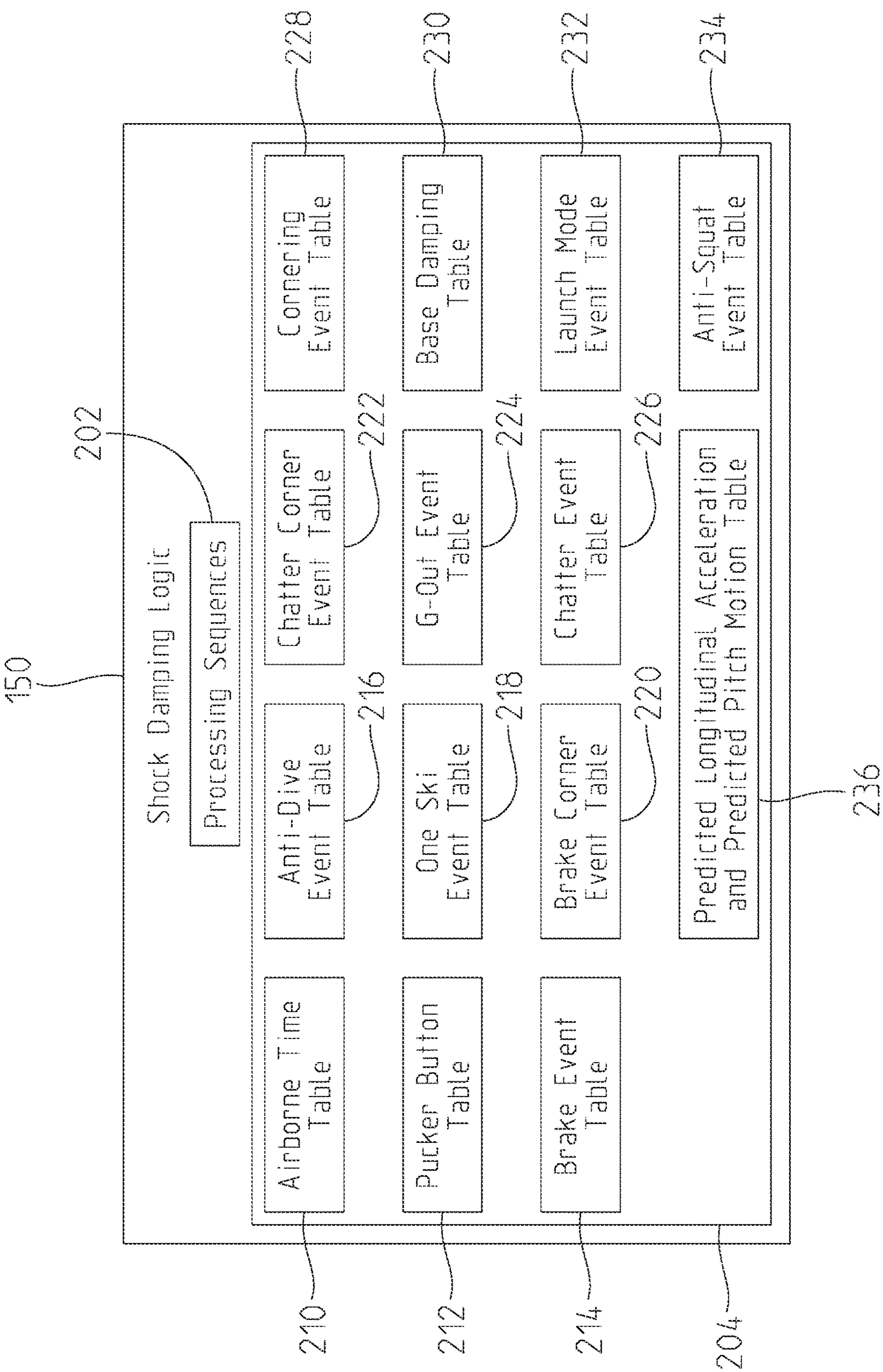
Fig. 8B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Pucker Button Table
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Brake Event Table
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Anti-Dive Event Table
218
One Ski Event Table
220
Brake Corner Event Table
222
Chatter Corner Event Table
224
G-Out Event Table
226
Chatter Event Table
228
Cornering Event Table
230
Base Damping Table
232
Launch Mode Event Table
234
Anti-Squat Event Table
236
Predicted Longitudinal Acceleration and Predicted Pitch Motion Table

Fig. 8C

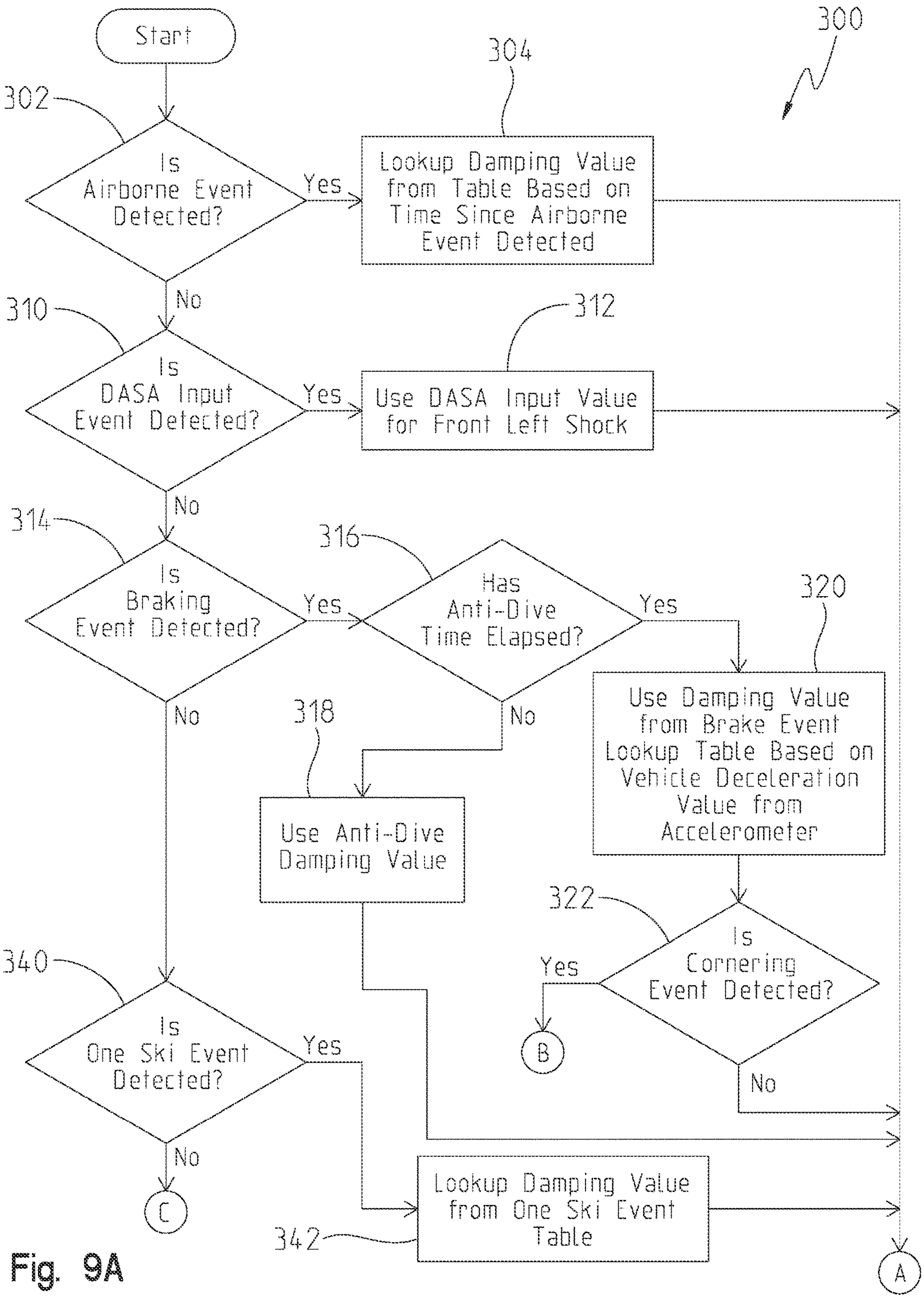
300
Start
302
Is Airborne Event Detected?
Yes
No
304
Lookup Damping Value from Table Based on Time Since Airborne Event Detected
310
Is DASA Input Event Detected?
312
Use DASA Input Value for Front Left Shock
314
Is Braking Event Detected?
316
Has Anti-Dive Time Elapsed?
320
Use Damping Value from Brake Event Lookup Table Based on Vehicle Deceleration Value from Accelerometer
318
Use Anti-Dive Damping Value
322
Is Cornering Event Detected?
B
340
Is One Ski Event Detected?
C
342
Lookup Damping Value from One Ski Event Table
A

Fig. 9A

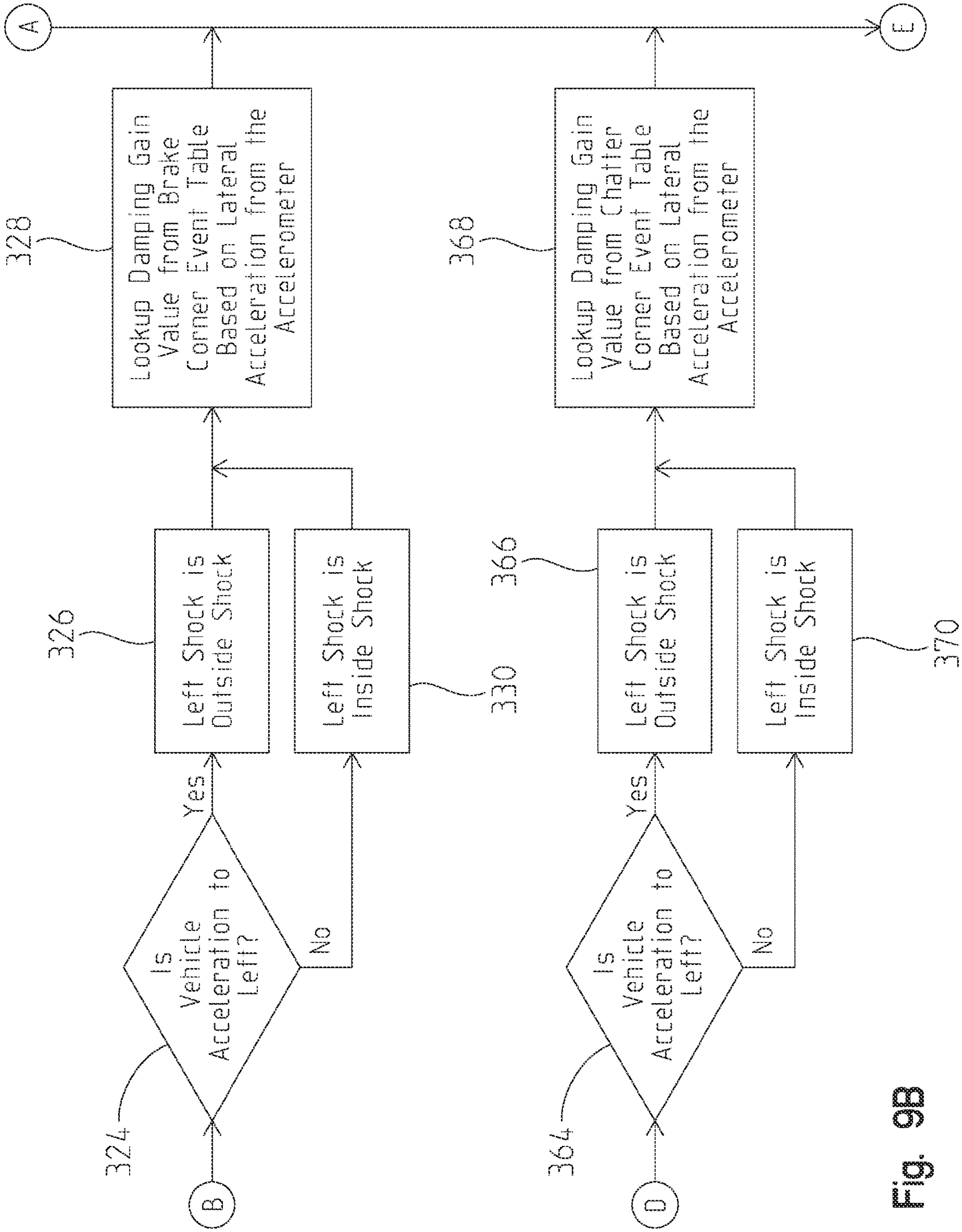
A
E
328
Lookup Damping Gain Value from Brake Corner Event Table Based on Lateral Acceleration from the Accelerometer
368
Lookup Damping Gain Value from Chatter Corner Event Table Based on Lateral Acceleration from the Accelerometer
326
Left Shock is Outside Shock
330
Left Shock is Inside Shock
366
Left Shock is Outside Shock
370
Left Shock is Inside Shock
Yes
No
Is Vehicle Acceleration to Left?
324
Yes
No
Is Vehicle Acceleration to Left?
364
B
D

Fig. 9B

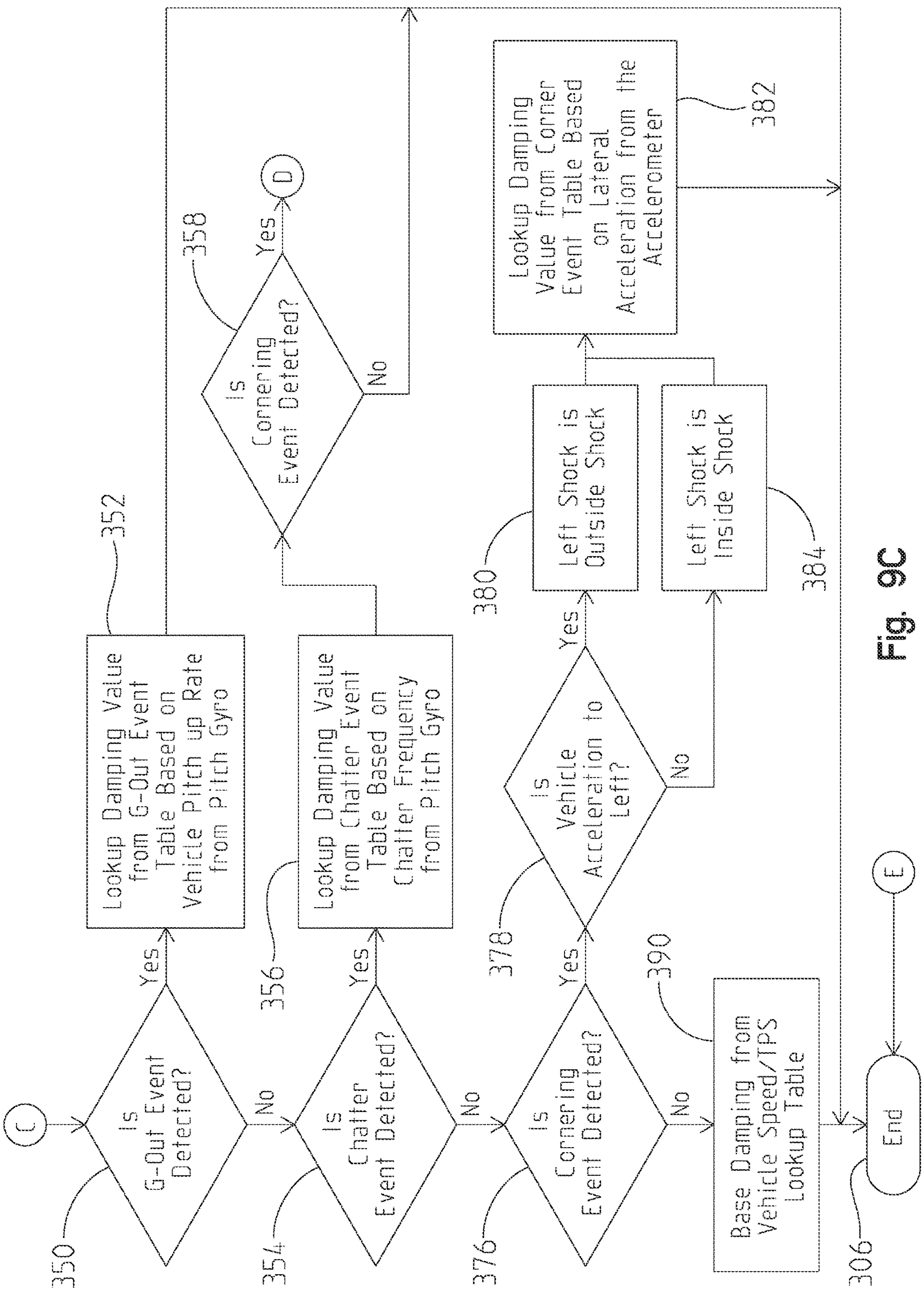
C
Is G-Out Event Detected?
350
Yes
No
Lookup Damping Value from G-Out Event Table Based on Vehicle Pitch up Rate from Pitch Gyro
352
Is Chatter Event Detected?
354
Yes
No
Lookup Damping Value from Chatter Event Table Based on Chatter Frequency from Pitch Gyro
356
Is Cornering Event Detected?
358
Yes
No
D
Is Cornering Event Detected?
376
Yes
No
Is Vehicle Acceleration to Left?
378
Yes
No
Left Shock is Outside Shock
380
Left Shock is Inside Shock
384
Lookup Damping Value from Corner Event Table Based on Lateral Acceleration from the Accelerometer
382
Base Damping from Vehicle Speed/TPS Lookup Table
390
E
End
306

Fig. 9C

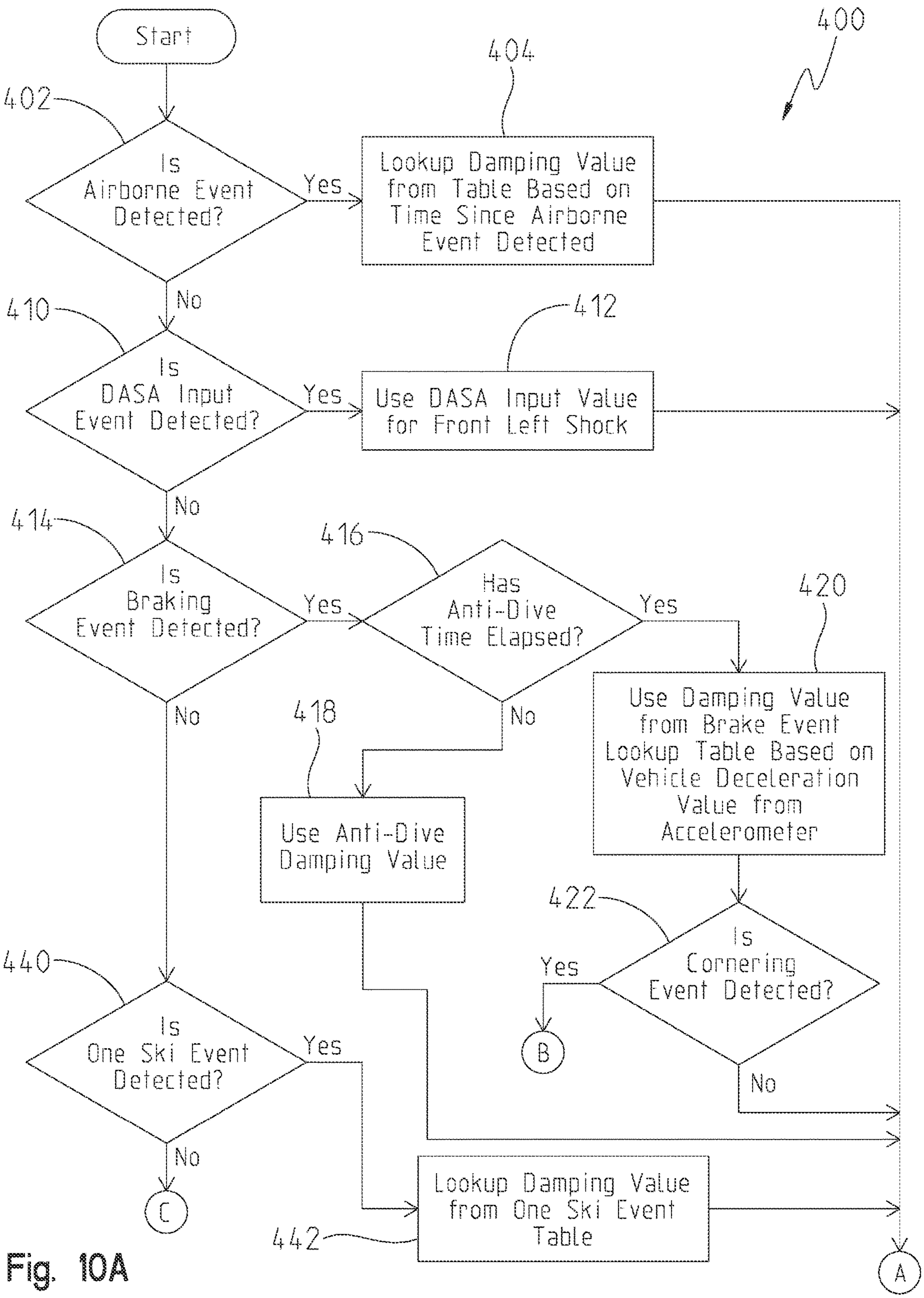
Start
400
402
Is Airborne Event Detected?
Yes
404
Lookup Damping Value from Table Based on Time Since Airborne Event Detected
No
410
Is DASA Input Event Detected?
Yes
412
Use DASA Input Value for Front Left Shock
No
414
Is Braking Event Detected?
Yes
416
Has Anti-Dive Time Elapsed?
Yes
420
Use Damping Value from Brake Event Lookup Table Based on Vehicle Deceleration Value from Accelerometer
No
No
418
Use Anti-Dive Damping Value
422
Is Cornering Event Detected?
Yes
B
No
440
Is One Ski Event Detected?
Yes
No
C
442
Lookup Damping Value from One Ski Event Table
A

Fig. 10A

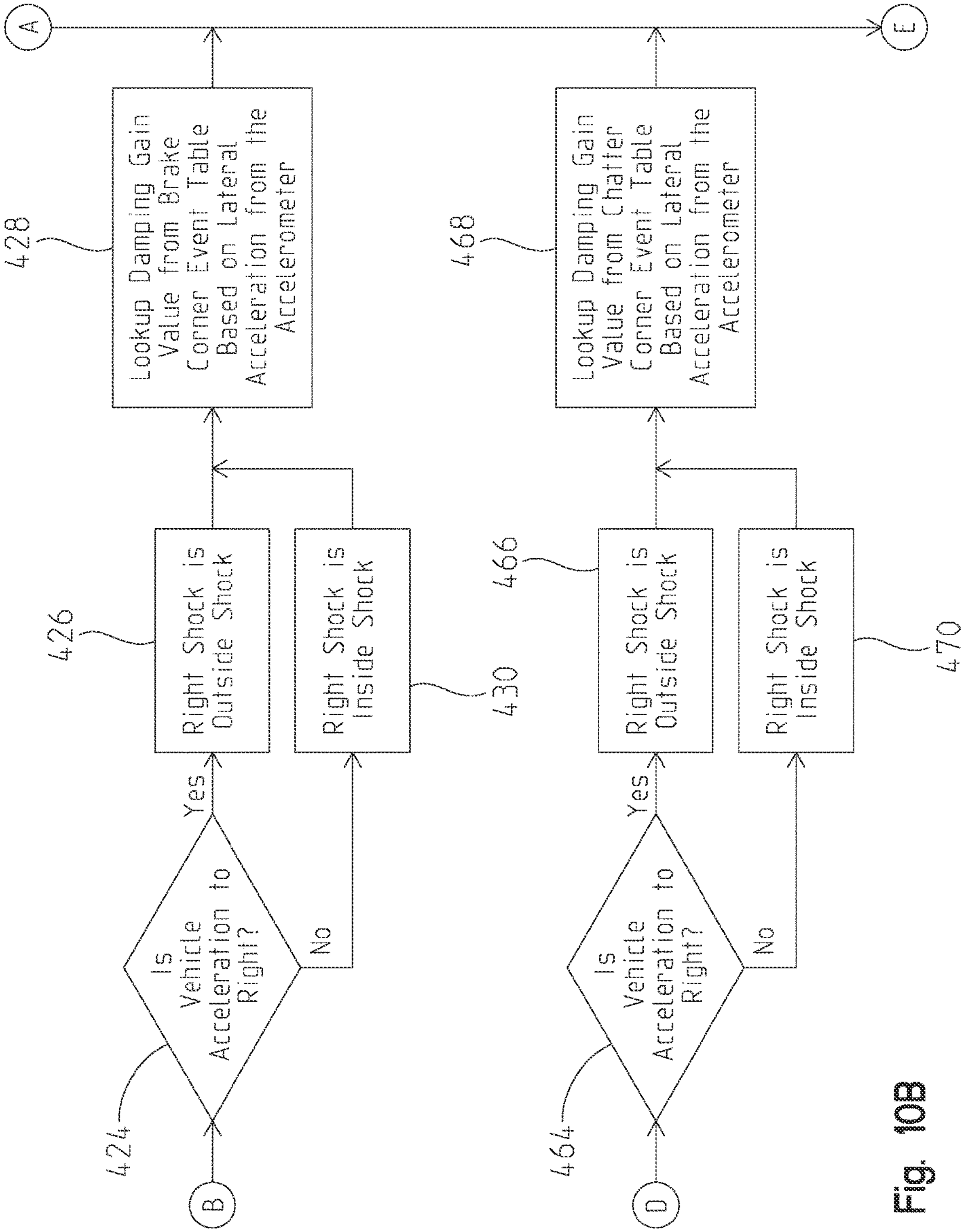
B
424
Is Vehicle Acceleration to Right?
Yes
No
426
Right Shock is Outside Shock
430
Right Shock is Inside Shock
428
Lookup Damping Gain Value from Brake Corner Event Table Based on Lateral Acceleration from the Accelerometer
A
D
464
Is Vehicle Acceleration to Right?
Yes
No
466
Right Shock is Outside Shock
470
Right Shock is Inside Shock
468
Lookup Damping Gain Value from Chatter Corner Event Table Based on Lateral Acceleration from the Accelerometer
E

Fig. 10B

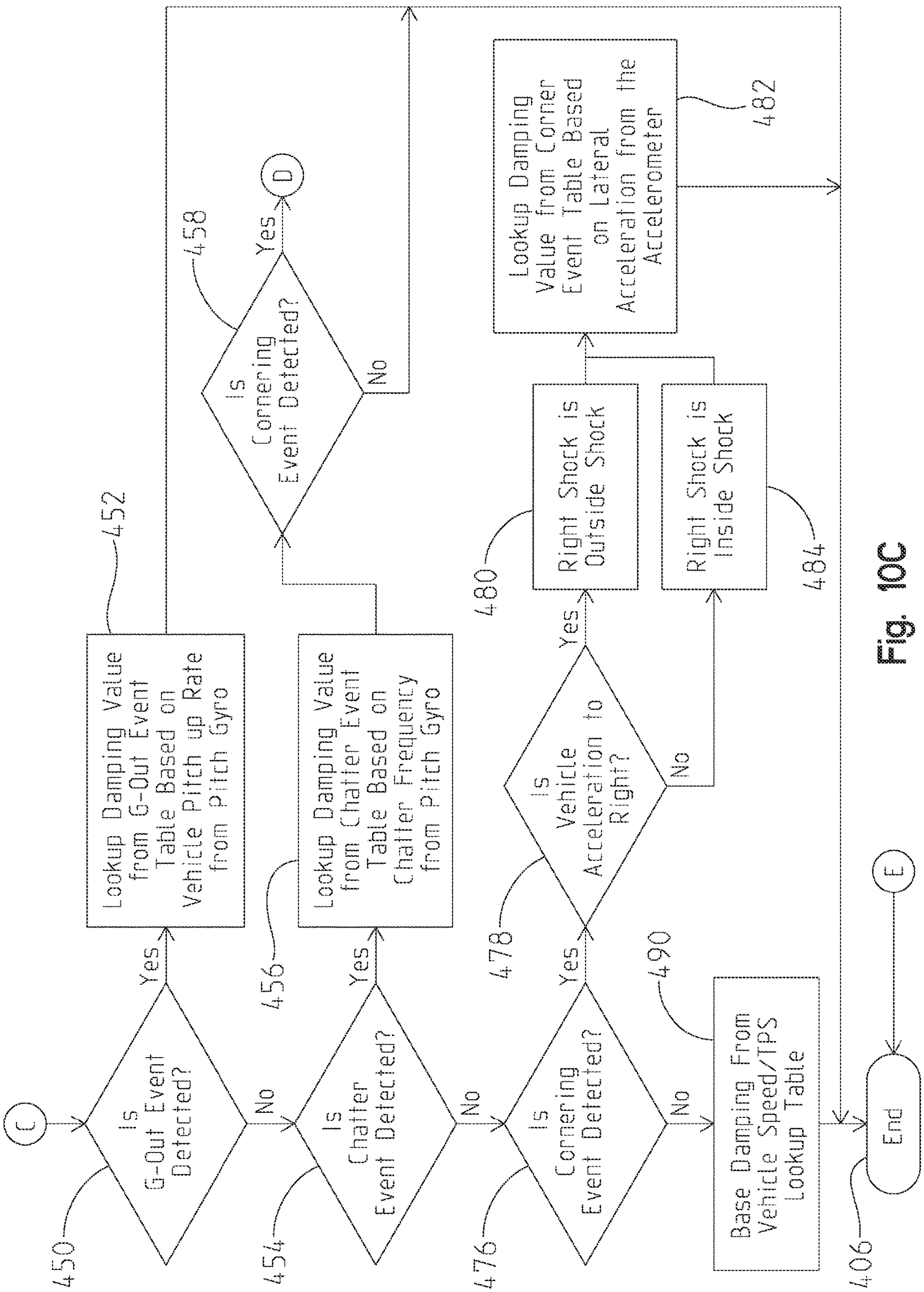
C
Is G-Out Event Detected?
450
Yes
No
Lookup Damping Value from G-Out Event Table Based on Vehicle Pitch up Rate from Pitch Gyro
452
Is Chatter Event Detected?
454
Yes
No
Lookup Damping Value from Chatter Event Table Based on Chatter Frequency from Pitch Gyro
456
Is Cornering Event Detected?
458
Yes
No
D
Is Cornering Event Detected?
476
Yes
No
Is Vehicle Acceleration to Right?
478
Yes
No
Right Shock is Outside Shock
480
Right Shock is Inside Shock
484
Lookup Damping Value from Corner Event Table Based on Lateral Acceleration from the Accelerometer
482
Base Damping From Vehicle Speed/TPS Lookup Table
490
E
End
406

Fig. 10C

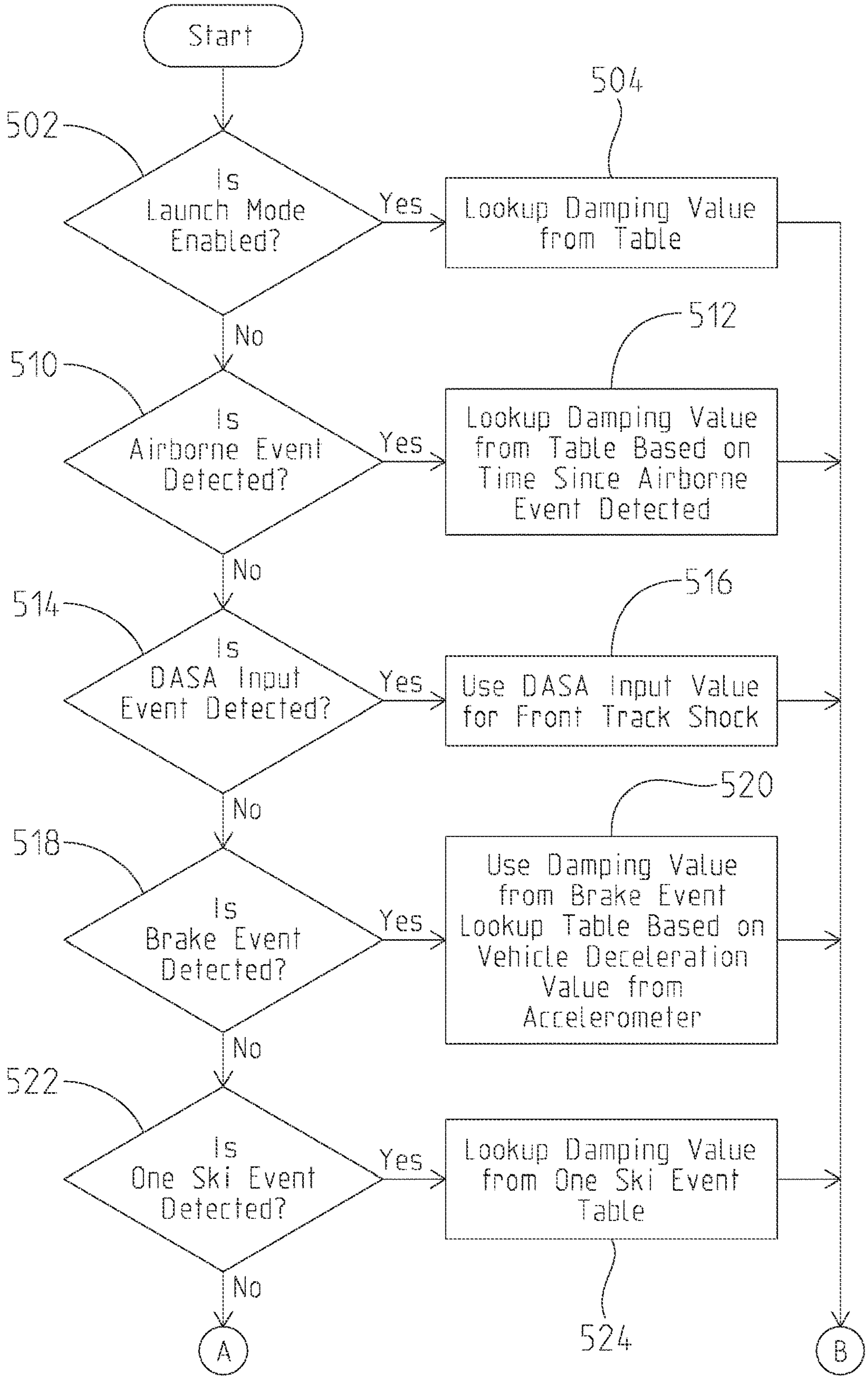
Start
502
Is Launch Mode Enabled?
Yes
504
Lookup Damping Value from Table
No
510
Is Airborne Event Detected?
Yes
512
Lookup Damping Value from Table Based on Time Since Airborne Event Detected
No
514
Is DASA Input Event Detected?
Yes
516
Use DASA Input Value for Front Track Shock
No
518
Is Brake Event Detected?
Yes
520
Use Damping Value from Brake Event Lookup Table Based on Vehicle Deceleration Value from Accelerometer
No
522
Is One Ski Event Detected?
Yes
524
Lookup Damping Value from One Ski Event Table
No
A
B

Fig. 11A

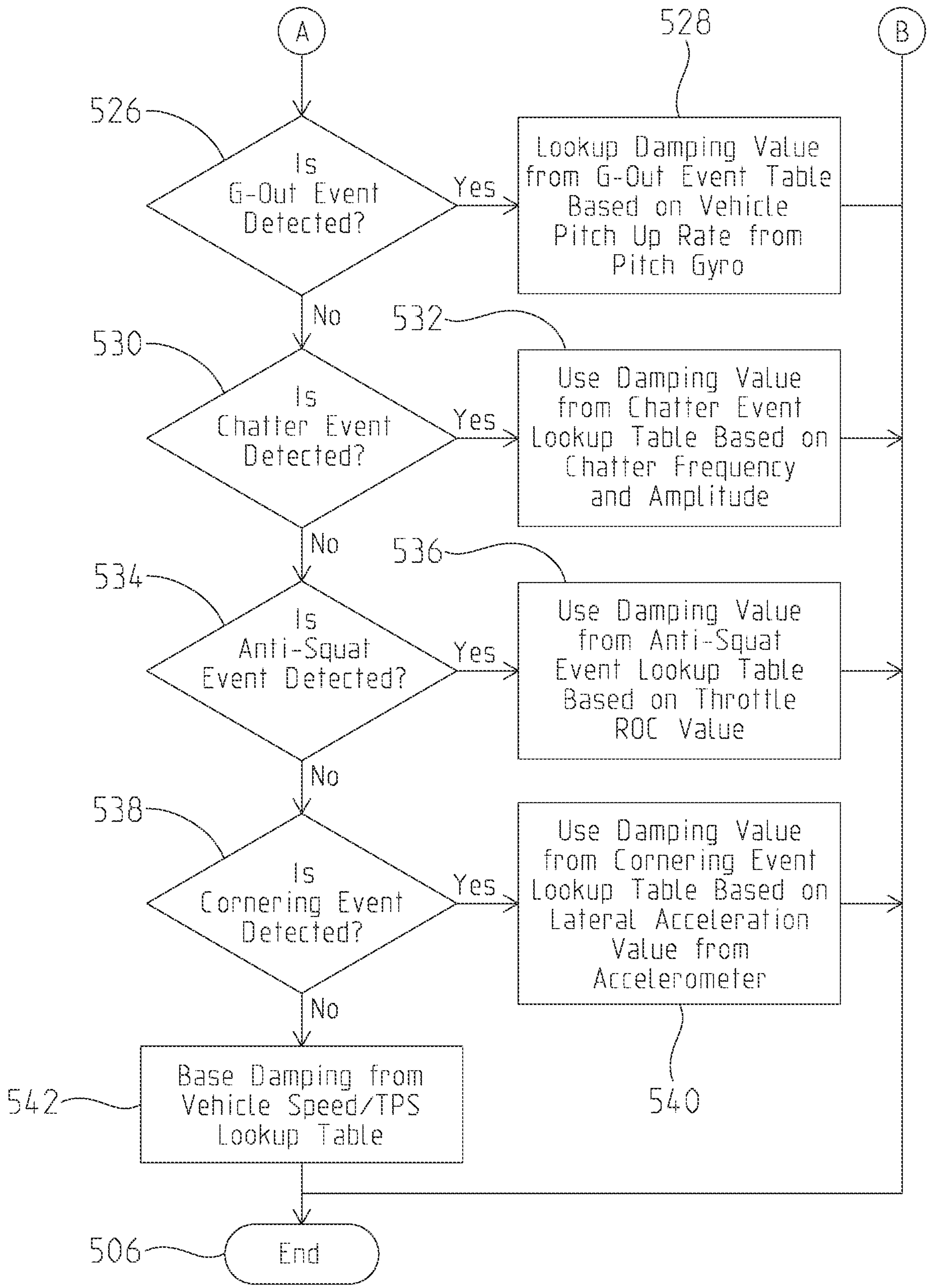
A
B
528
526
Is
G-Out Event
Detected?
Yes
Lookup Damping Value from G-Out Event Table Based on Vehicle Pitch Up Rate from Pitch Gyro
No
532
530
Is
Chatter Event
Detected?
Yes
Use Damping Value from Chatter Event Lookup Table Based on Chatter Frequency and Amplitude
No
536
534
Is
Anti-Squat
Event Detected?
Yes
Use Damping Value from Anti-Squat Event Lookup Table Based on Throttle ROC Value
No
538
Is
Cornering Event
Detected?
Yes
Use Damping Value from Cornering Event Lookup Table Based on Lateral Acceleration Value from Accelerometer
No
542
Base Damping from Vehicle Speed/TPS Lookup Table
540
506
End

Fig. 11B

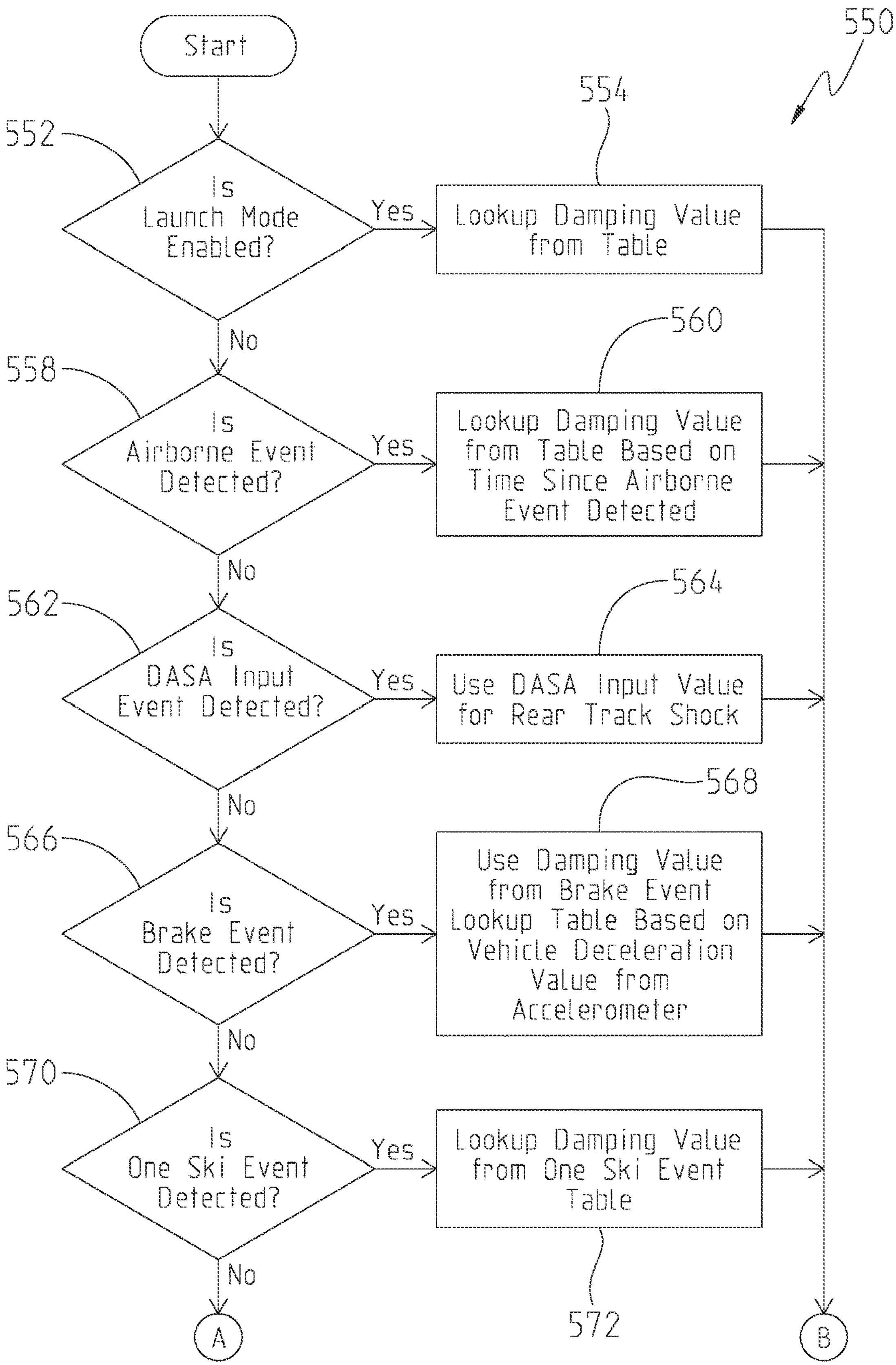
550
Start
552
Is Launch Mode Enabled?
Yes
554
Lookup Damping Value from Table
No
558
Is Airborne Event Detected?
Yes
560
Lookup Damping Value from Table Based on Time Since Airborne Event Detected
No
562
Is DASA Input Event Detected?
Yes
564
Use DASA Input Value for Rear Track Shock
No
566
Is Brake Event Detected?
Yes
568
Use Damping Value from Brake Event Lookup Table Based on Vehicle Deceleration Value from Accelerometer
No
570
Is One Ski Event Detected?
Yes
Lookup Damping Value from One Ski Event Table
572
No
A
B

Fig. 12A

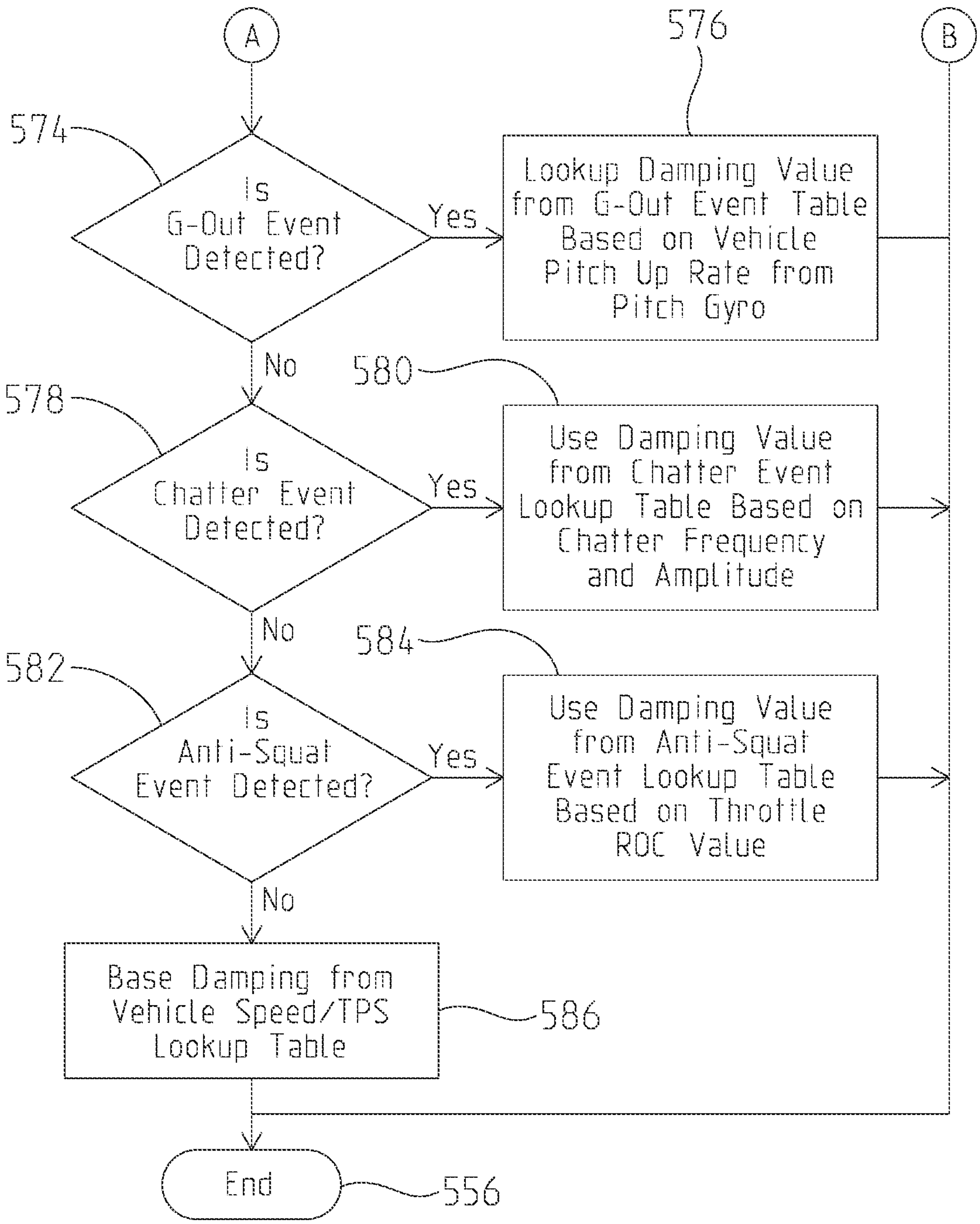
A
576
B
574
Is G-Out Event Detected?
Yes
Lookup Damping Value from G-Out Event Table Based on Vehicle Pitch Up Rate from Pitch Gyro
No
580
578
Is Chatter Event Detected?
Yes
Use Damping Value from Chatter Event Lookup Table Based on Chatter Frequency and Amplitude
No
584
582
Is Anti-Squat Event Detected?
Yes
Use Damping Value from Anti-Squat Event Lookup Table Based on Throttle ROC Value
No
Base Damping from Vehicle Speed/TPS Lookup Table
586
End
556

Fig. 12B

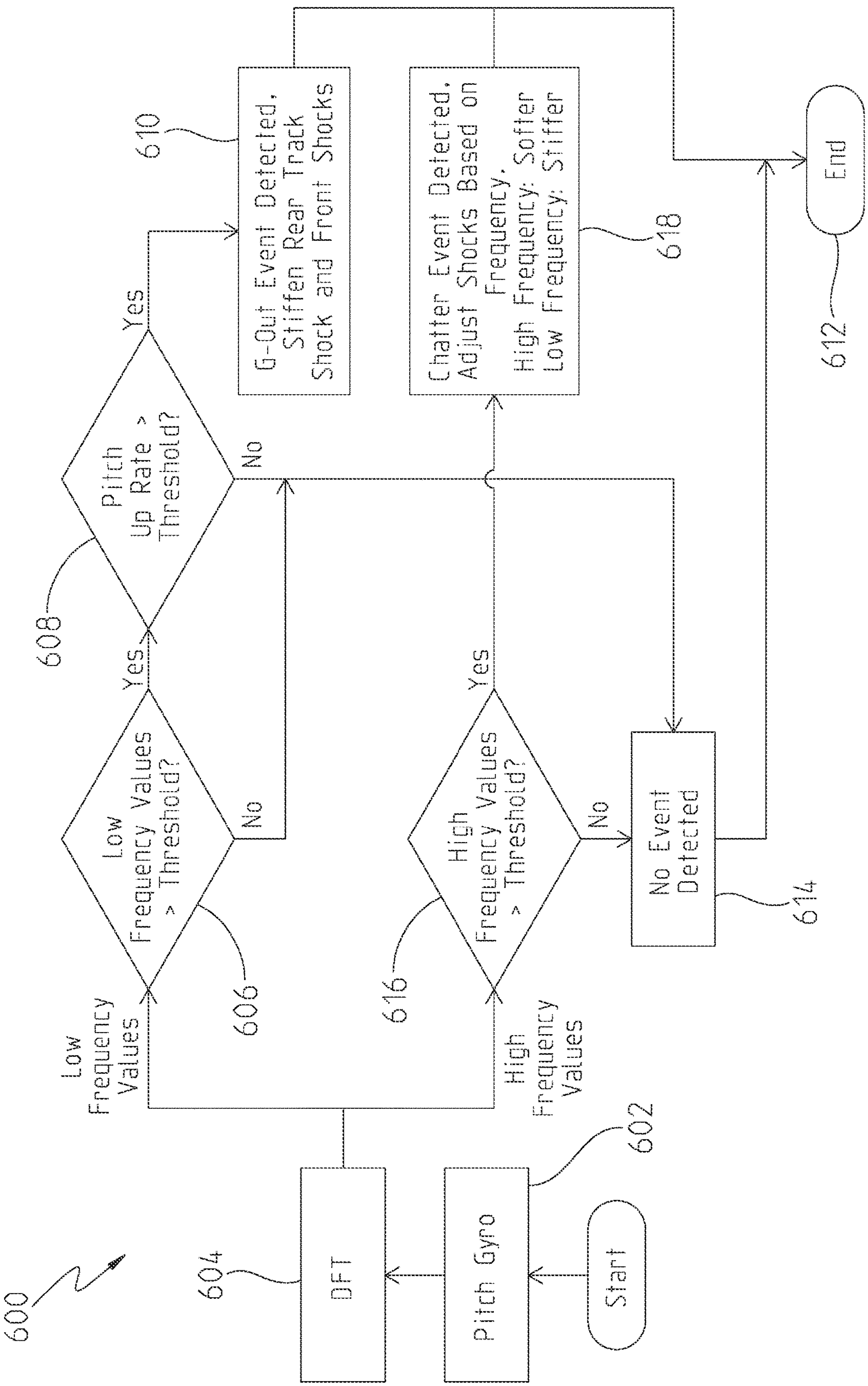
600
Start
Pitch Gyro
602
DFT
604
Low Frequency Values
High Frequency Values
Low Frequency Values > Threshold?
606
High Frequency Values > Threshold?
616
Yes
No
Pitch Up Rate > Threshold?
608
G-Out Event Detected, Stiffen Rear Track Shock and Front Shocks
610
Chatter Event Detected, Adjust Shocks Based on Frequency. High Frequency: Softer Low Frequency: Stiffer
618
No Event Detected
614
End
612

Fig. 13

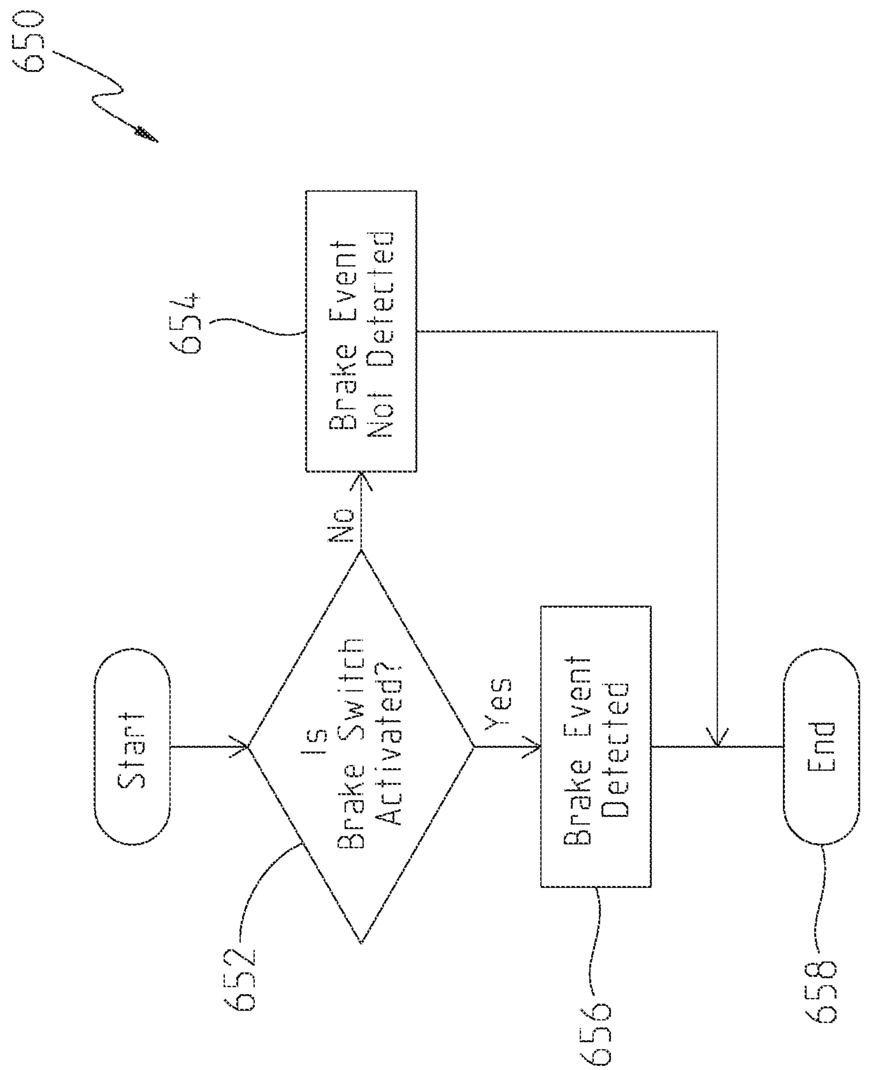
650
Start
Is
Brake Switch
Activated?
652
No
Brake Event
Not Detected
654
Yes
Brake Event
Detected
656
End
658

Fig. 14

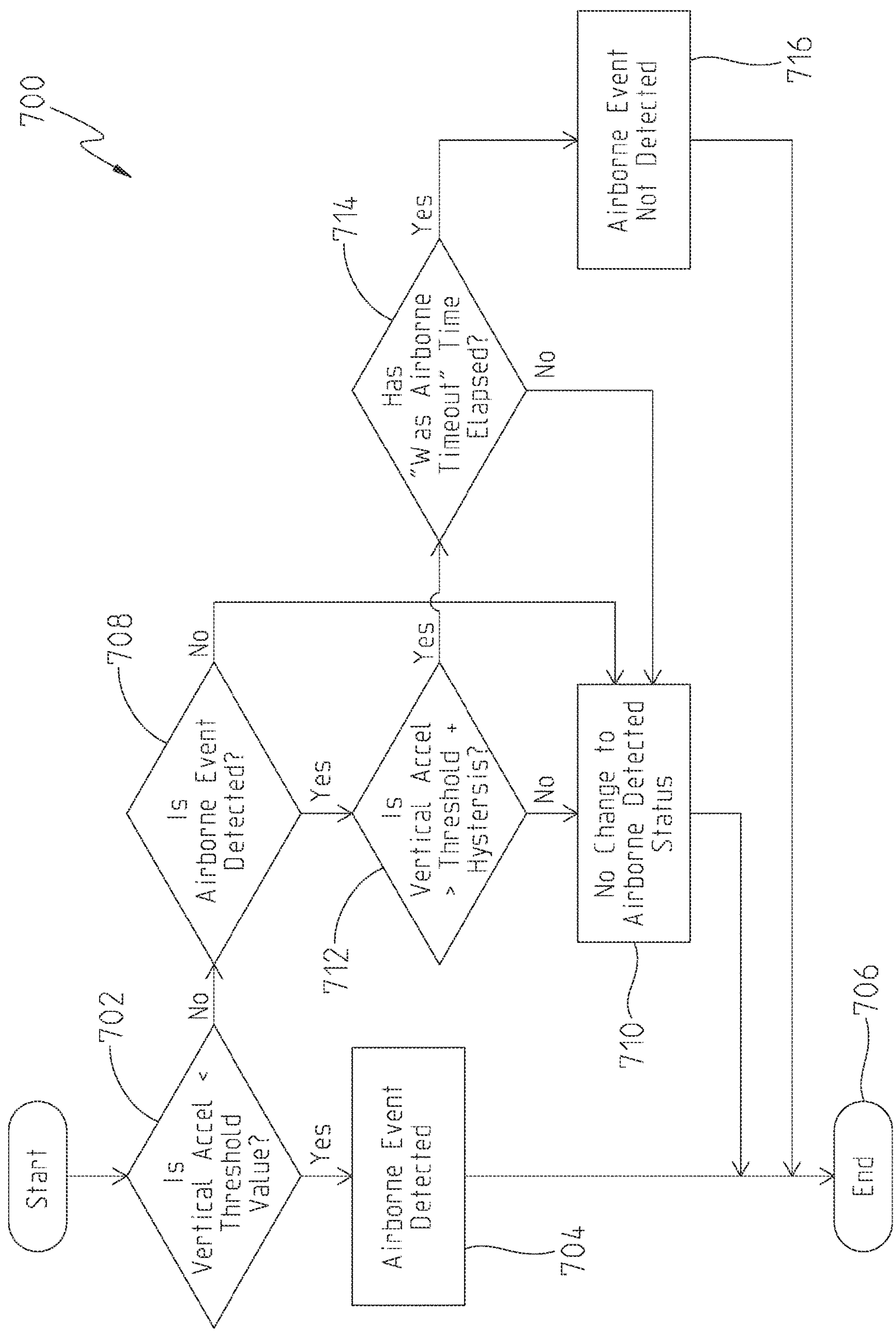
700
Start
Is Vertical Accel < Threshold Value?
702
Yes
No
Airborne Event Detected
704
Is Airborne Event Detected?
708
Yes
No
Is Vertical Accel > Threshold + Hysteresis?
712
Yes
No
No Change to Airborne Detected Status
710
Has "Was Airborne Timeout" Time Elapsed?
714
Yes
No
Airborne Event Not Detected
716
End
706

Fig. 15

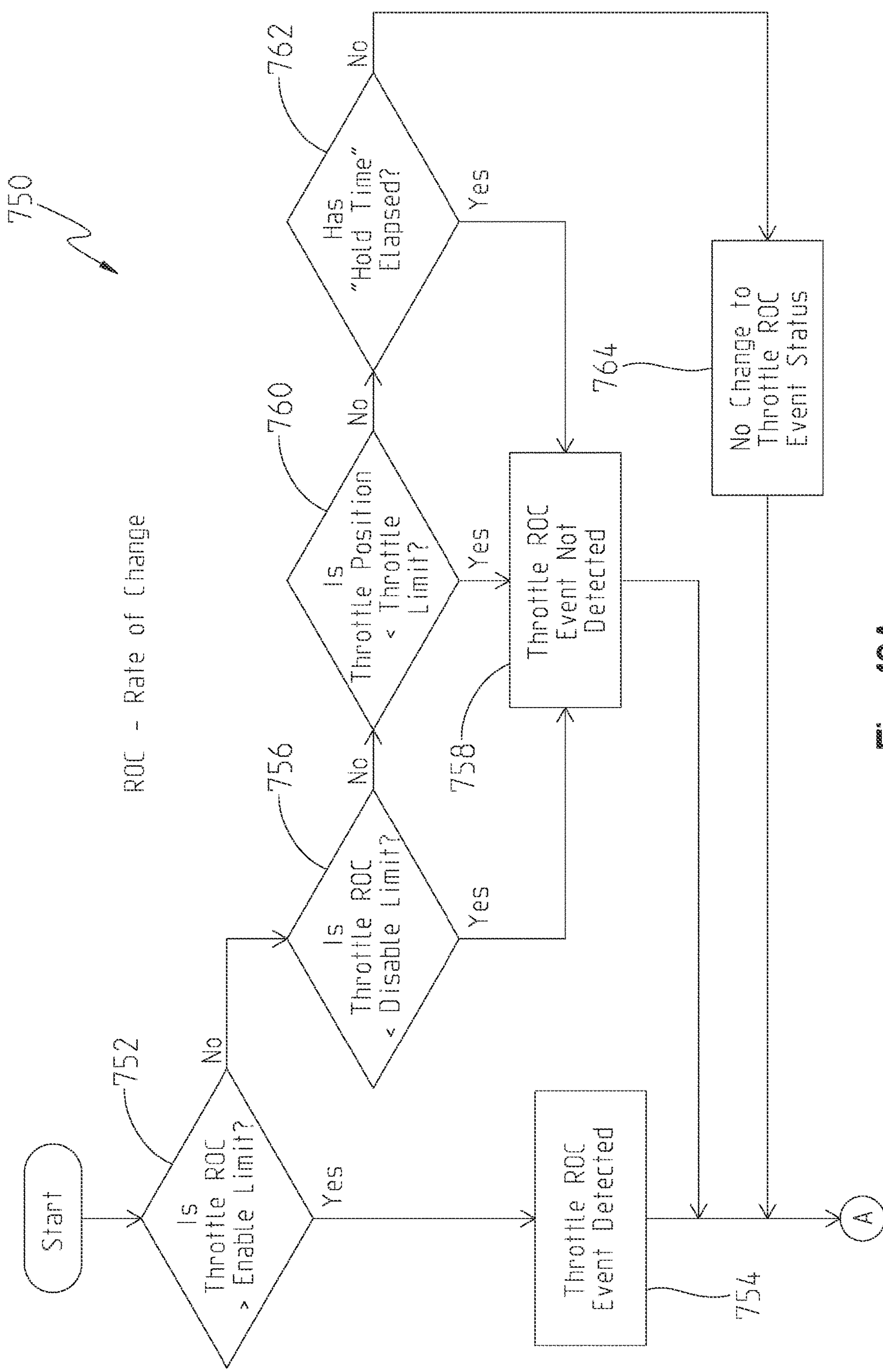
750
ROC - Rate of Change
Start
752
Is Throttle ROC > Enable Limit?
Yes
No
Throttle ROC Event Detected
754
756
Is Throttle ROC < Disable Limit?
Yes
No
758
Throttle ROC Event Not Detected
760
Is Throttle Position < Throttle Limit?
Yes
No
762
Has "Hold Time" Elapsed?
Yes
No
764
No Change to Throttle ROC Event Status
A

Fig. 16A

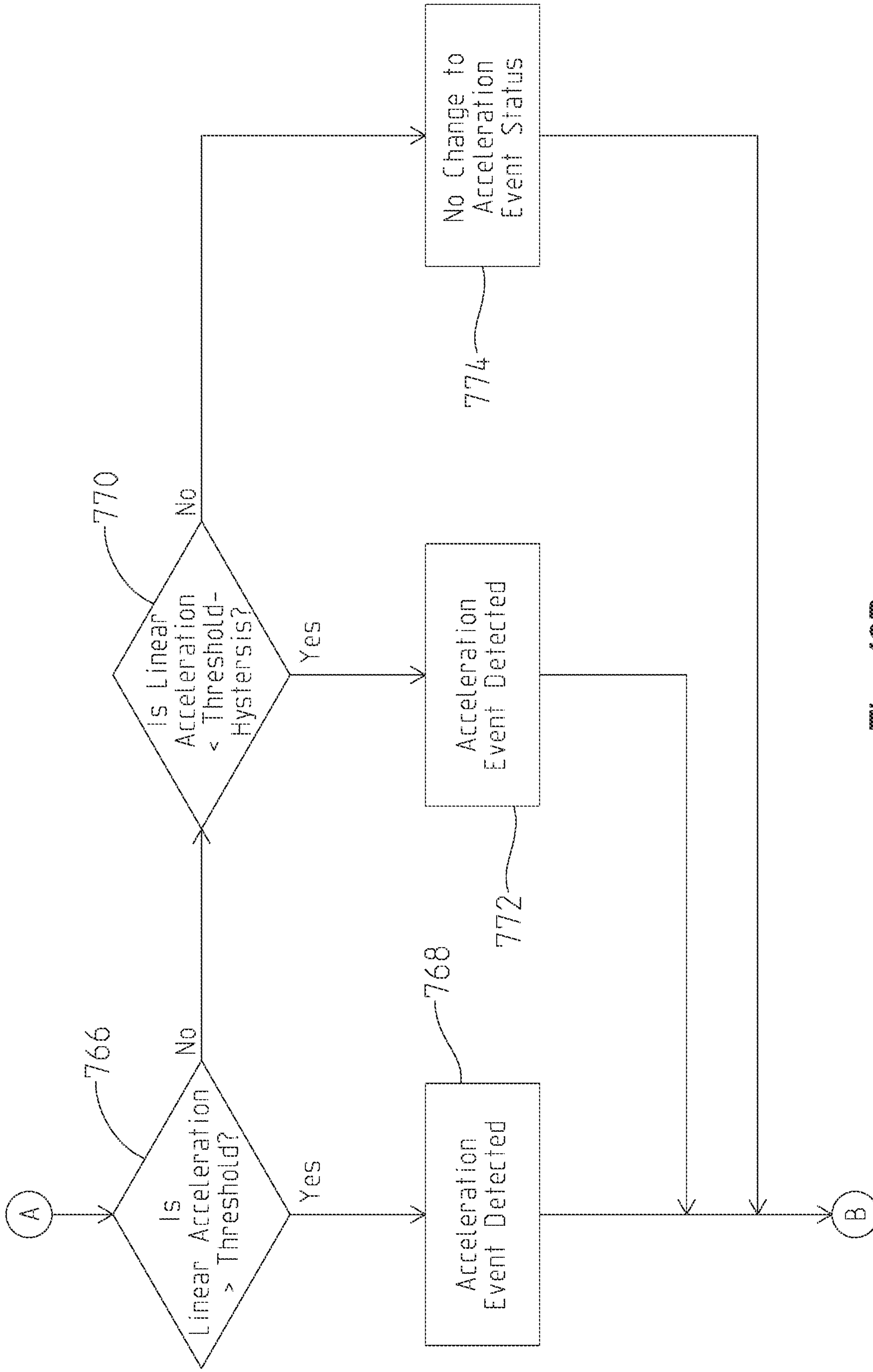
A
Is Linear Acceleration > Threshold?
766
Yes
No
Acceleration Event Detected
768
Is Linear Acceleration < Threshold-Hysteresis?
770
Yes
No
Acceleration Event Detected
772
No Change to Acceleration Event Status
774
B

Fig. 16B

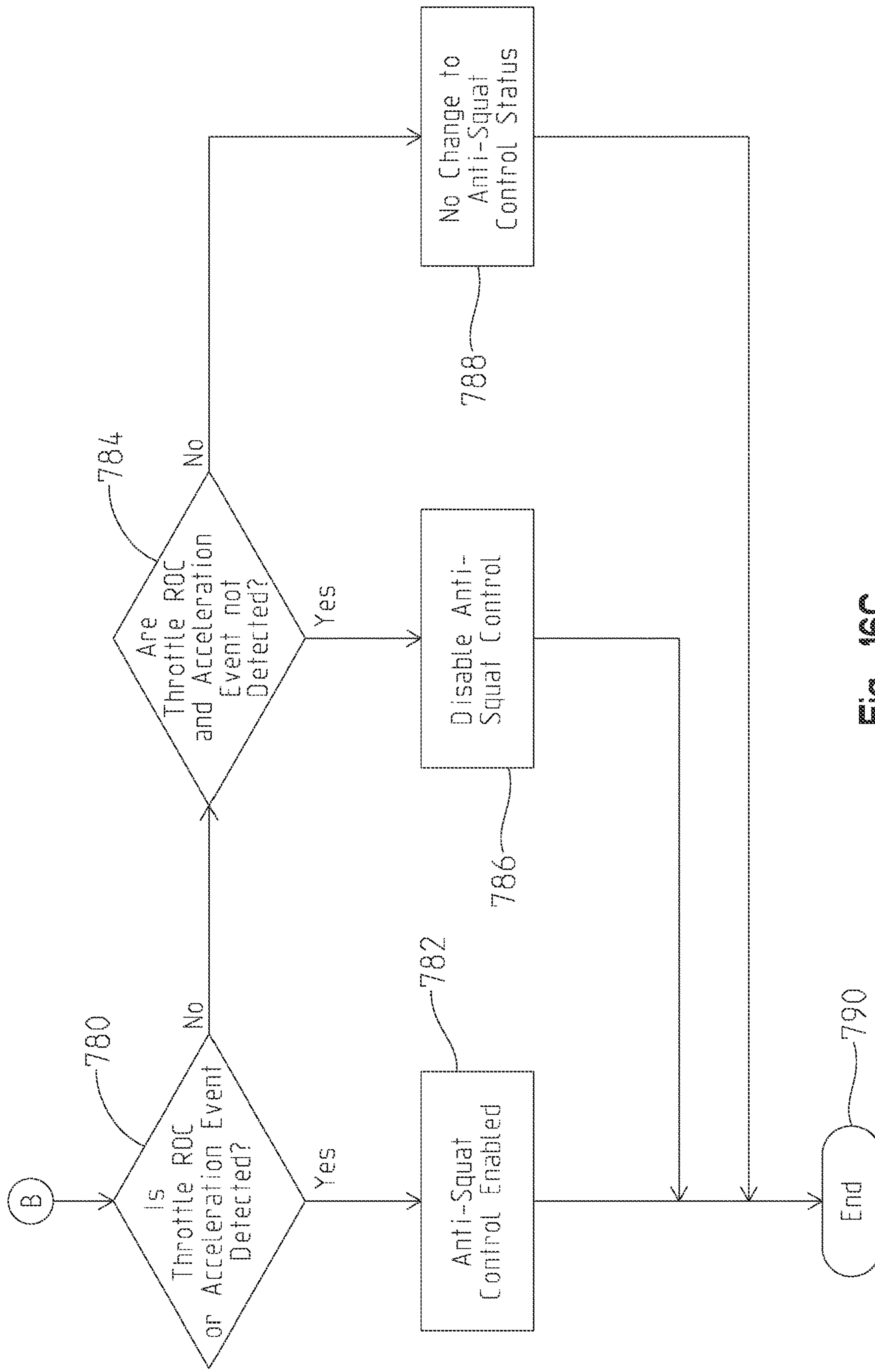
B
Is Throttle ROC or Acceleration Event Detected?
780
Yes
No
Anti-Squat Control Enabled
782
Are Throttle ROC and Acceleration Event not Detected?
784
Yes
No
Disable Anti-Squat Control
786
No Change to Anti-Squat Control Status
788
End
790

Fig. 16C

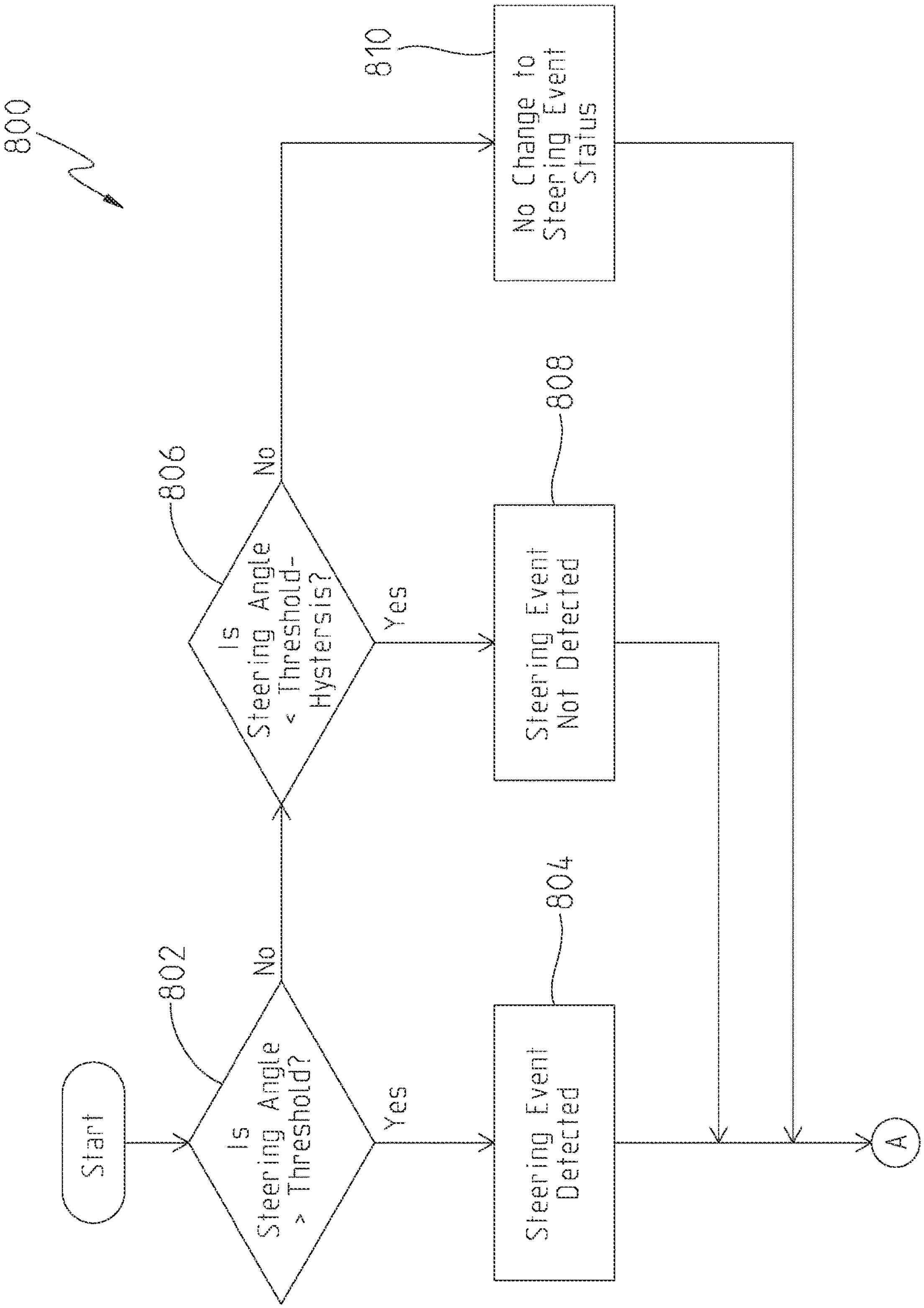
800
Start
802
Is Steering Angle > Threshold?
Yes
No
804
Steering Event Detected
806
Is Steering Angle < Threshold- Hysteresis?
Yes
No
808
Steering Event Not Detected
810
No Change to Steering Event Status
A

Fig. 17A

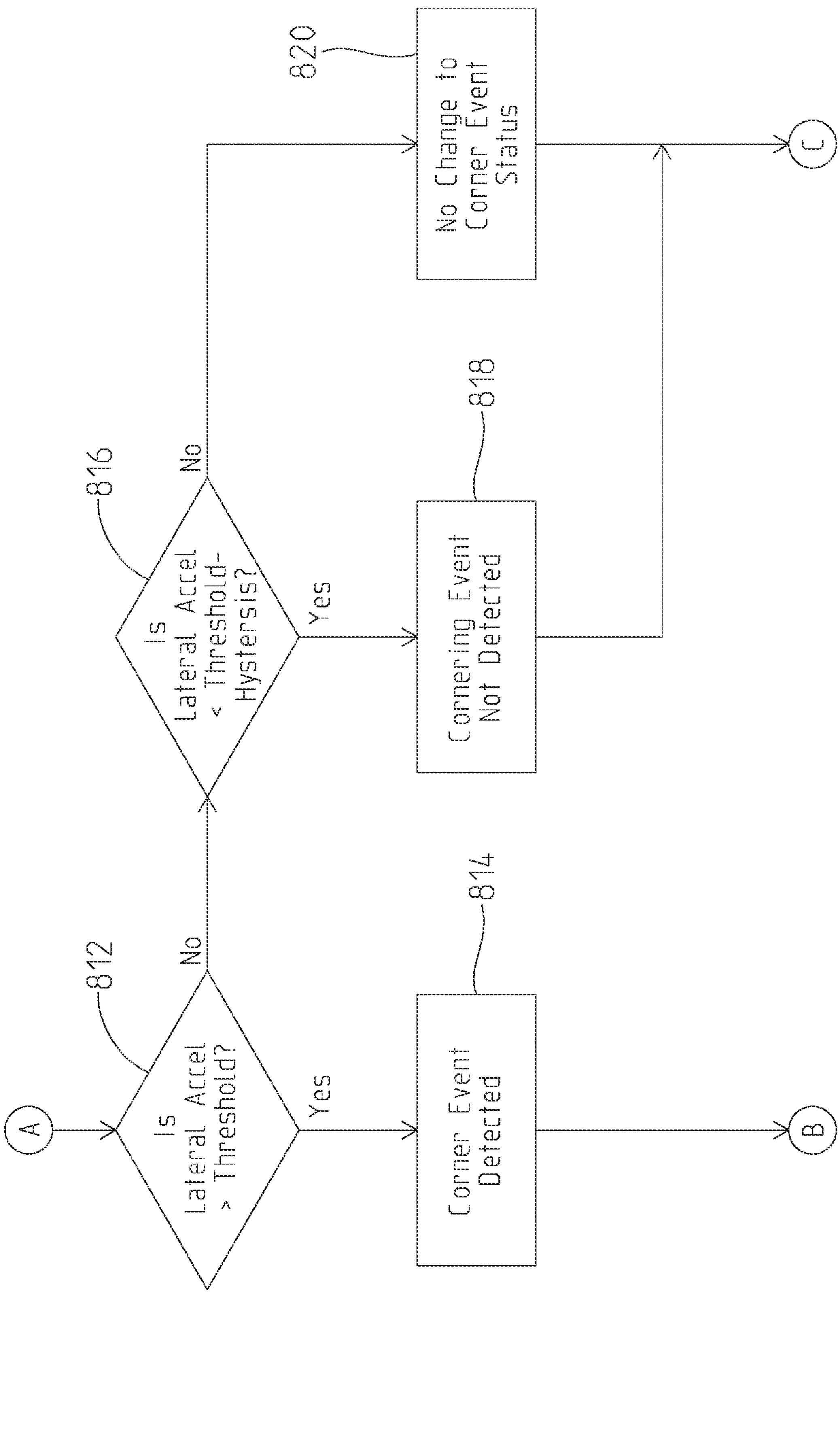
A
Is Lateral Accel > Threshold?
812
Yes
No
Corner Event Detected
814
B
Is Lateral Accel < Threshold-Hysteresis?
816
Yes
No
Cornering Event Not Detected
818
No Change to Corner Event Status
820
C

Fig. 17B

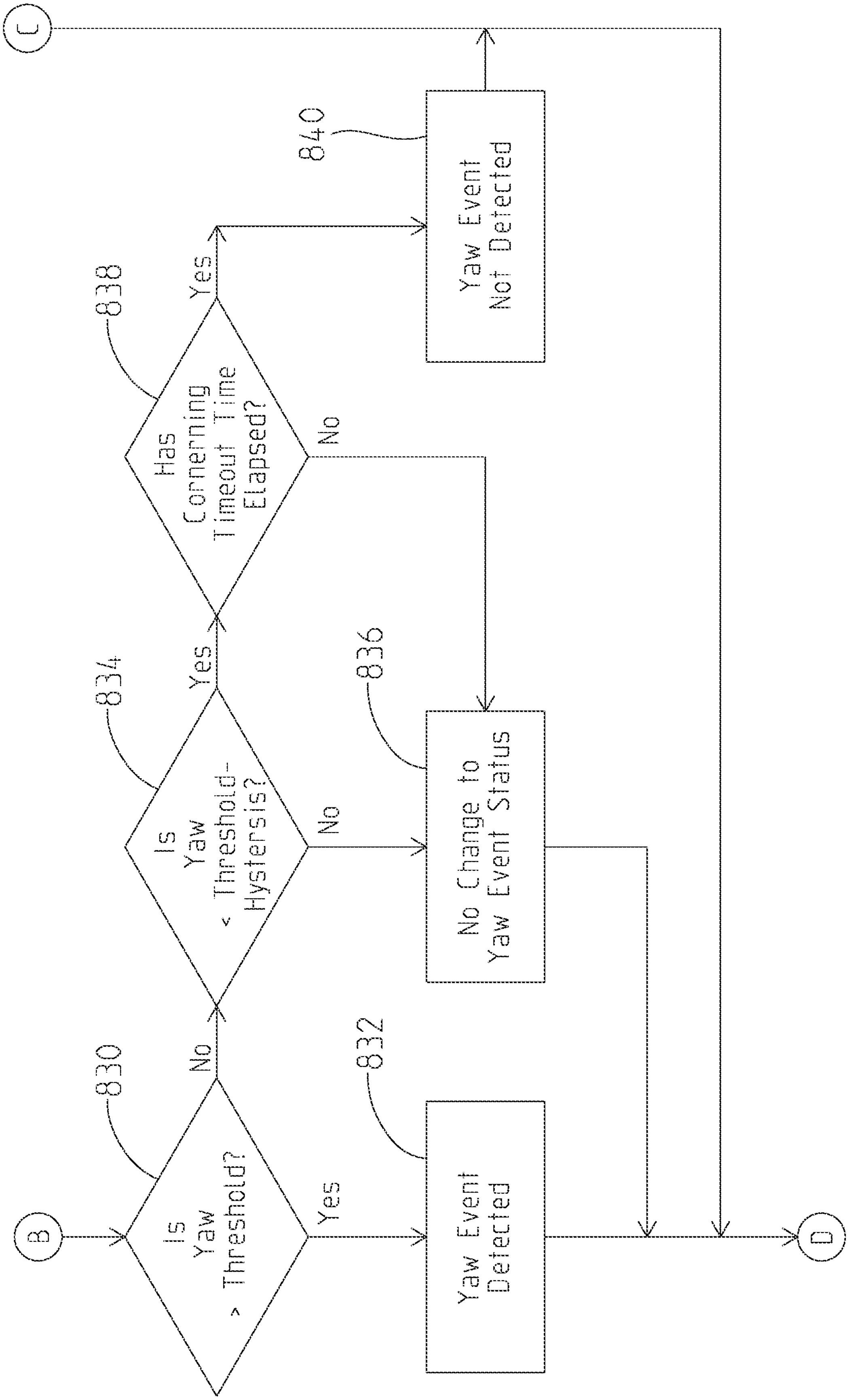
B
Is Yaw > Threshold?
830
Yes
No
Yaw Event Detected
832
Is Yaw < Threshold-Hysteresis?
834
Yes
No
No Change to Yaw Event Status
836
Has Cornerning Timeout Time Elapsed?
838
Yes
No
Yaw Event Not Detected
840
C
D

Fig. 17C

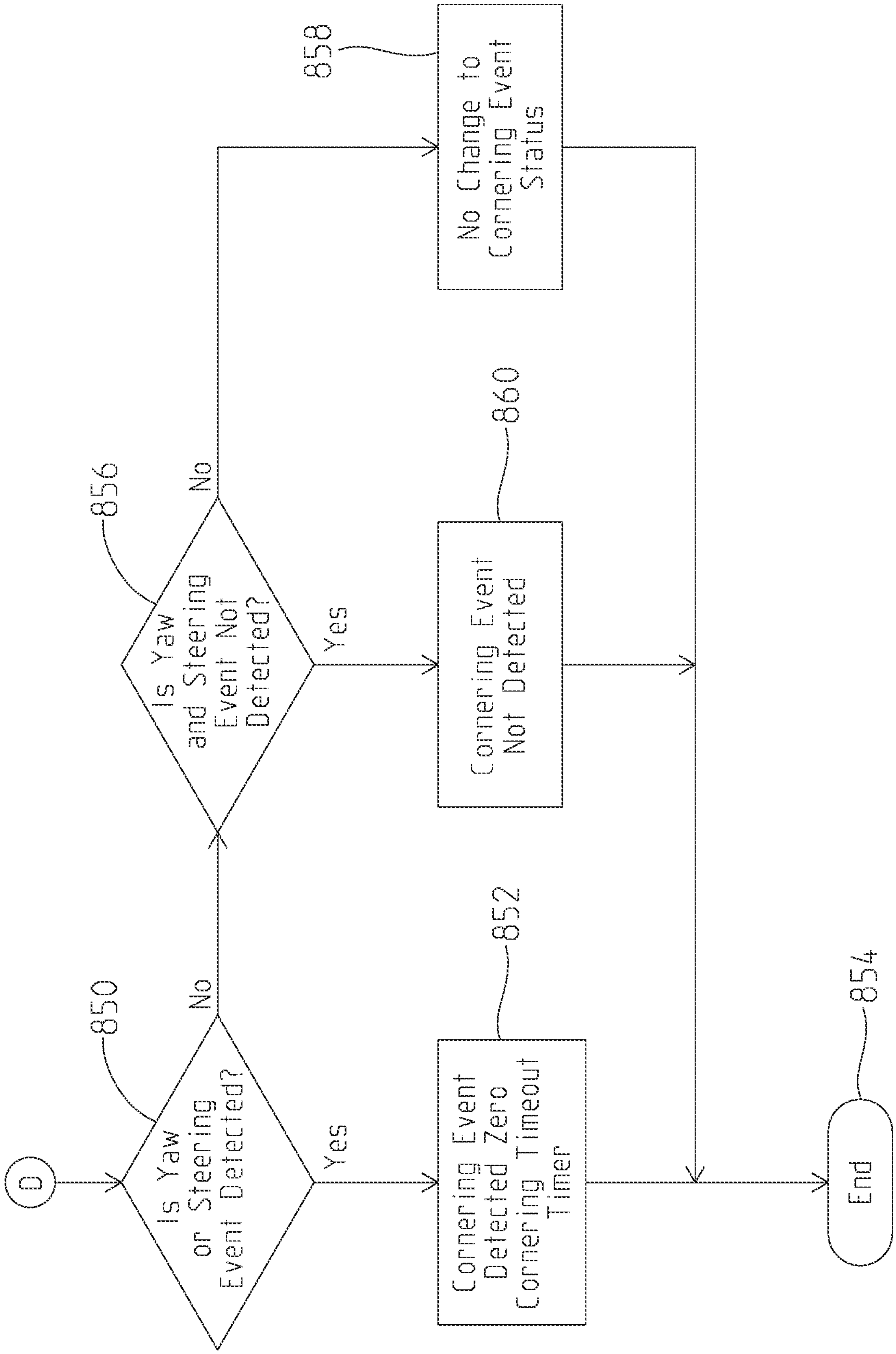
D
Is Yaw or Steering Event Detected?
850
Yes
No
Cornering Event Detected Zero Cornering Timeout Timer
852
Is Yaw and Steering Event Not Detected?
856
Yes
No
Cornering Event Not Detected
860
No Change to Cornering Event Status
858
End
854

Fig. 17D

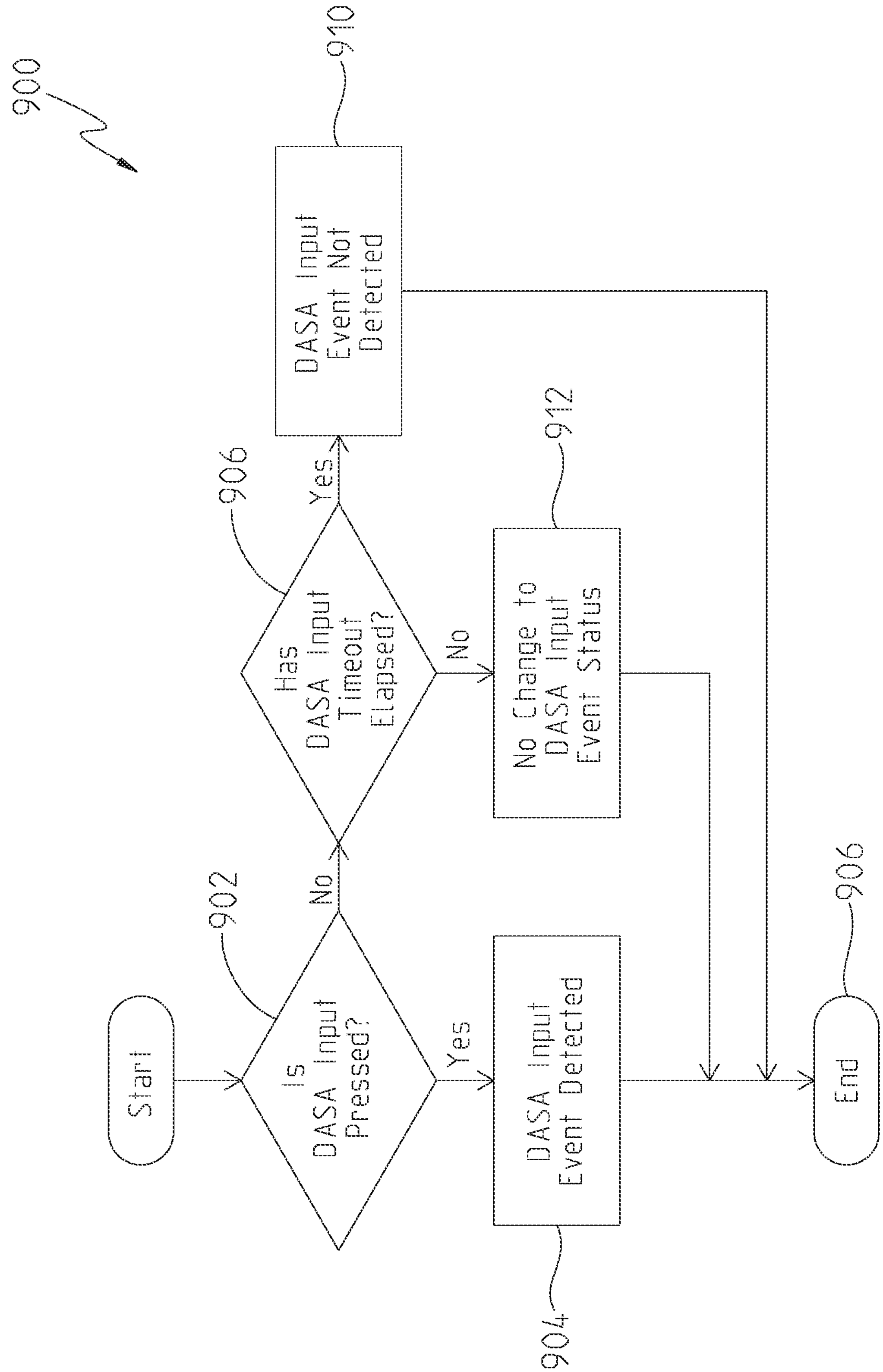
900
Start
902
Is DASA Input Pressed?
Yes
No
DASA Input Event Detected
904
906
Has DASA Input Timeout Elapsed?
Yes
No
DASA Input Event Not Detected
910
No Change to DASA Input Event Status
912
End
906

Fig. 18

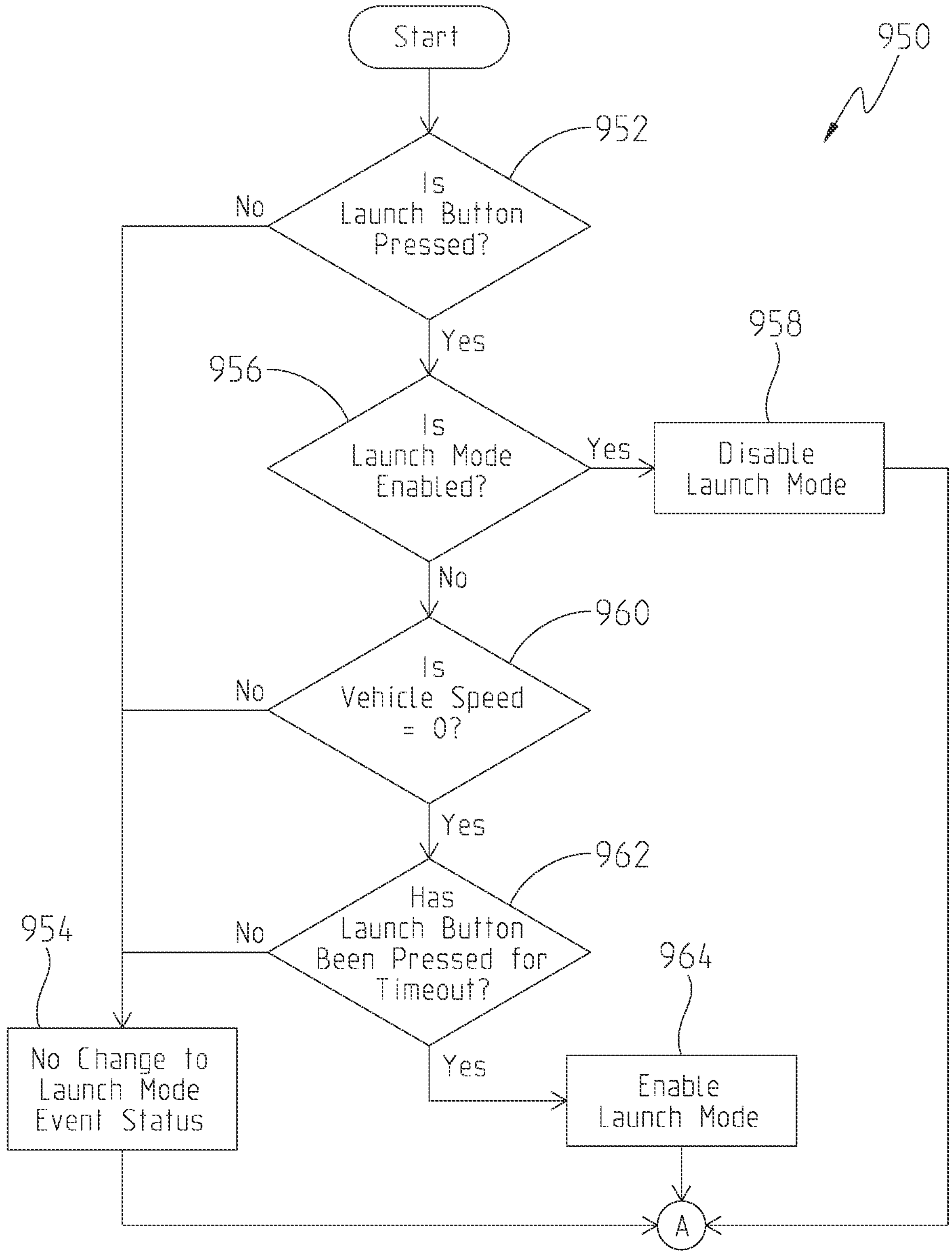
Start
950
952
Is Launch Button Pressed?
No
Yes
956
Is Launch Mode Enabled?
Yes
958
Disable Launch Mode
No
960
Is Vehicle Speed = 0?
No
Yes
962
Has Launch Button Been Pressed for Timeout?
No
954
No Change to Launch Mode Event Status
Yes
964
Enable Launch Mode
A

Fig. 19A

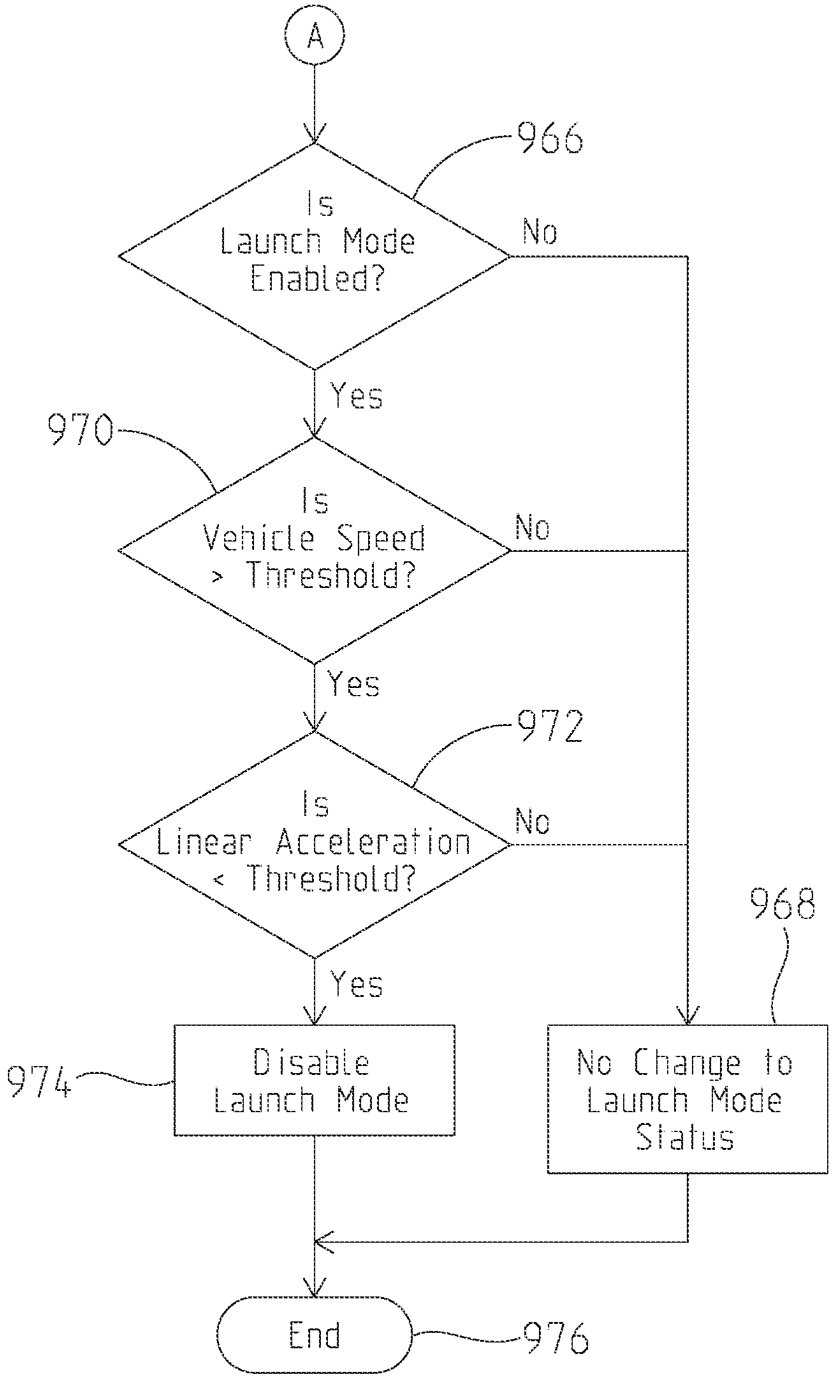
A
966
Is
Launch Mode
Enabled?
No
Yes
970
Is
Vehicle Speed
> Threshold?
No
Yes
972
Is
Linear Acceleration
< Threshold?
No
968
Yes
974
Disable
Launch Mode
No Change to
Launch Mode
Status
End
976

Fig. 19B

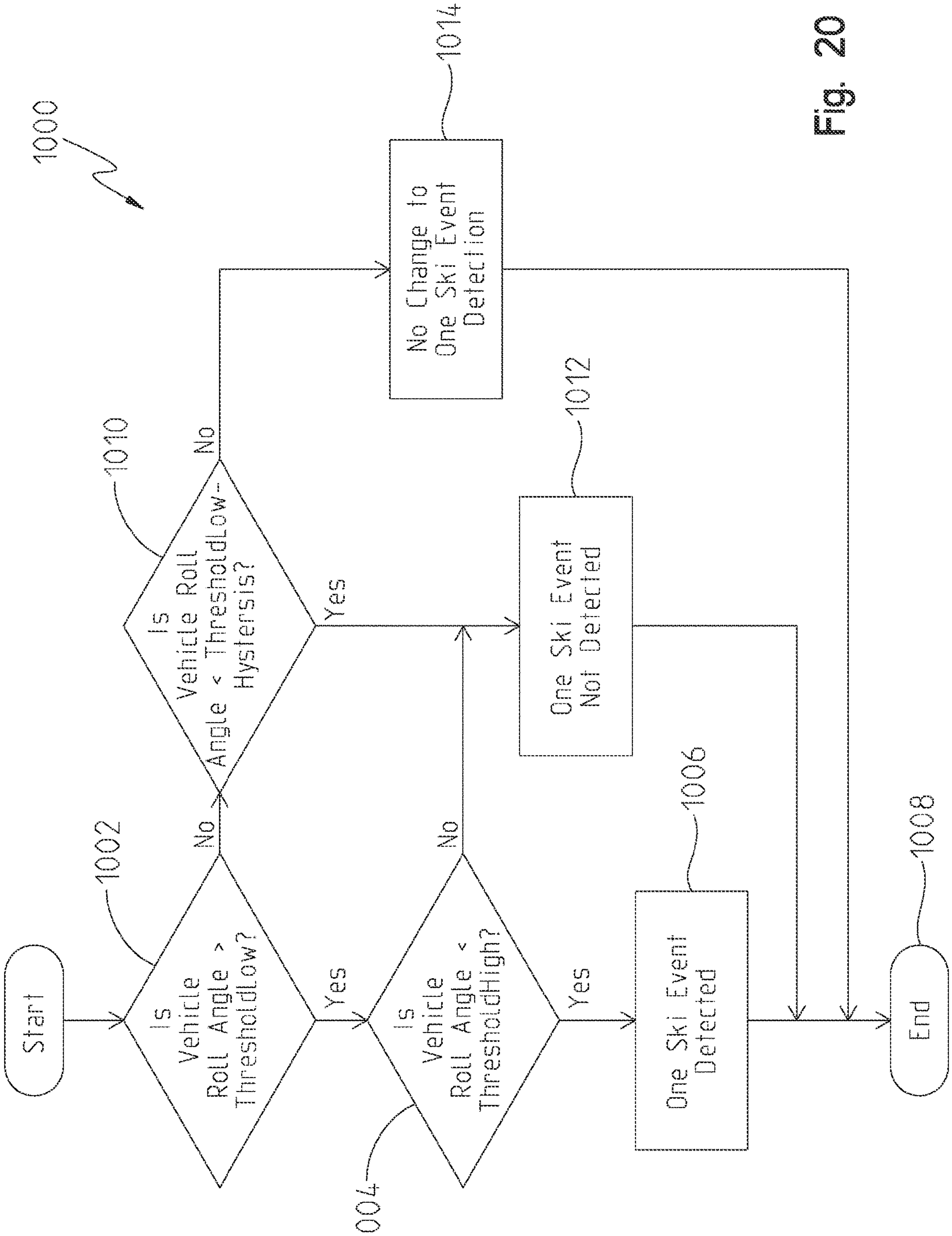
1000
Start
1002
Is Vehicle Roll Angle > ThresholdLow?
No
Yes
1004
Is Vehicle Roll Angle < ThresholdHigh?
No
Yes
1006
One Ski Event Detected
1008
End
1010
Is Vehicle Roll Angle < ThresholdLow-Hystersis?
No
Yes
1012
One Ski Event Not Detected
1014
No Change to One Ski Event Detection

Fig. 20

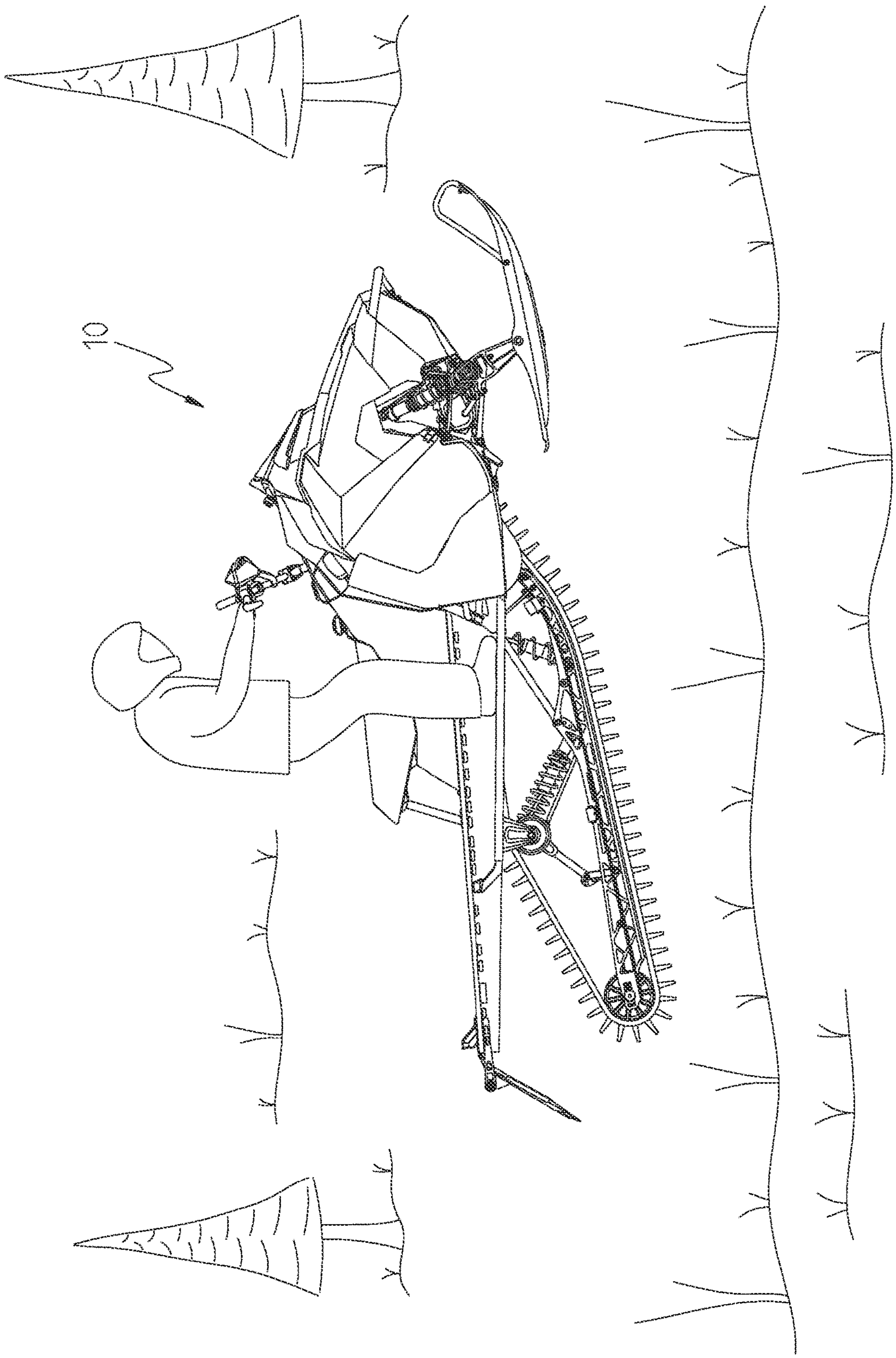

Fig. 21

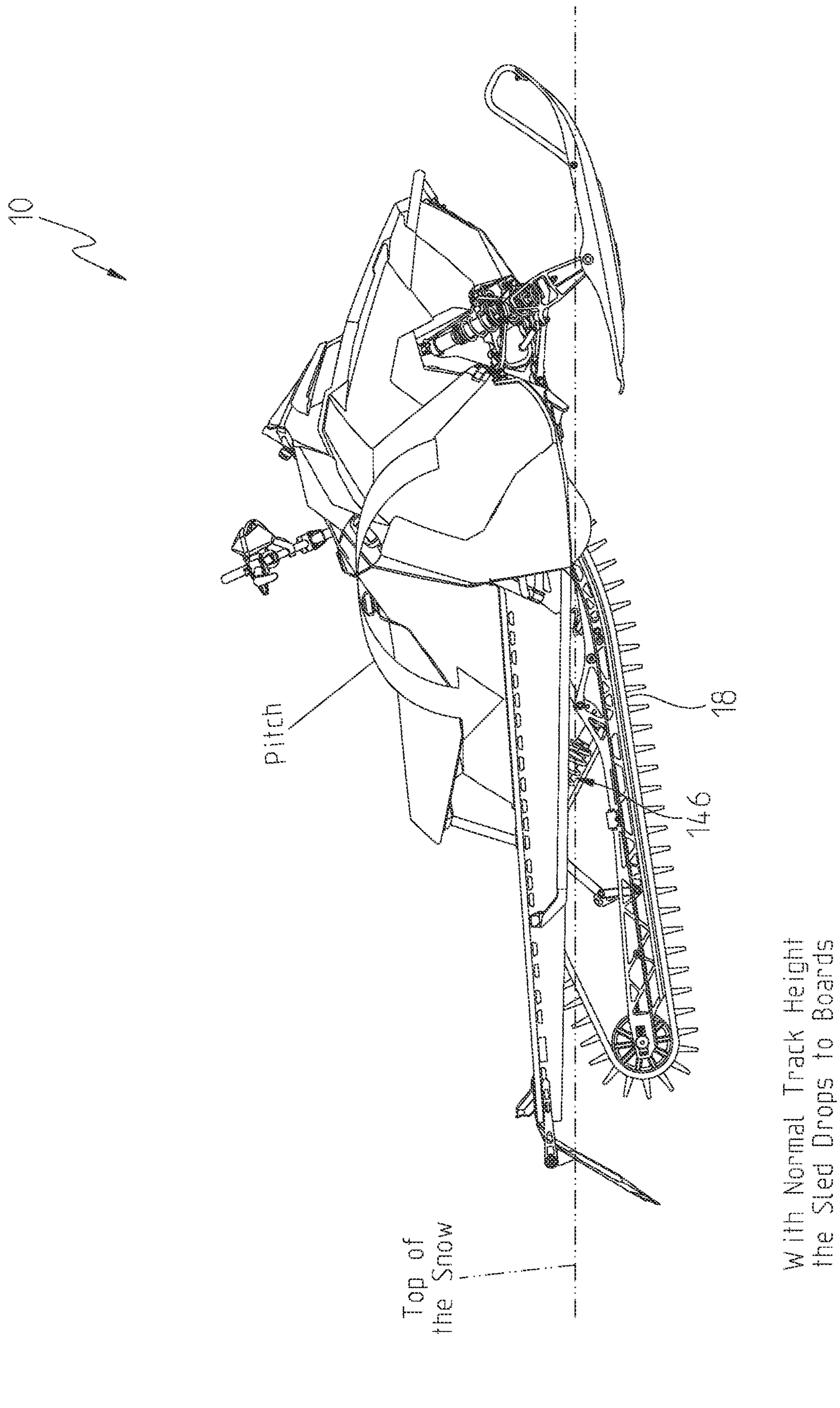
10
Pitch
18
146
Top of the Snow
With Normal Track Height the Sled Drops to Boards

Fig. 22

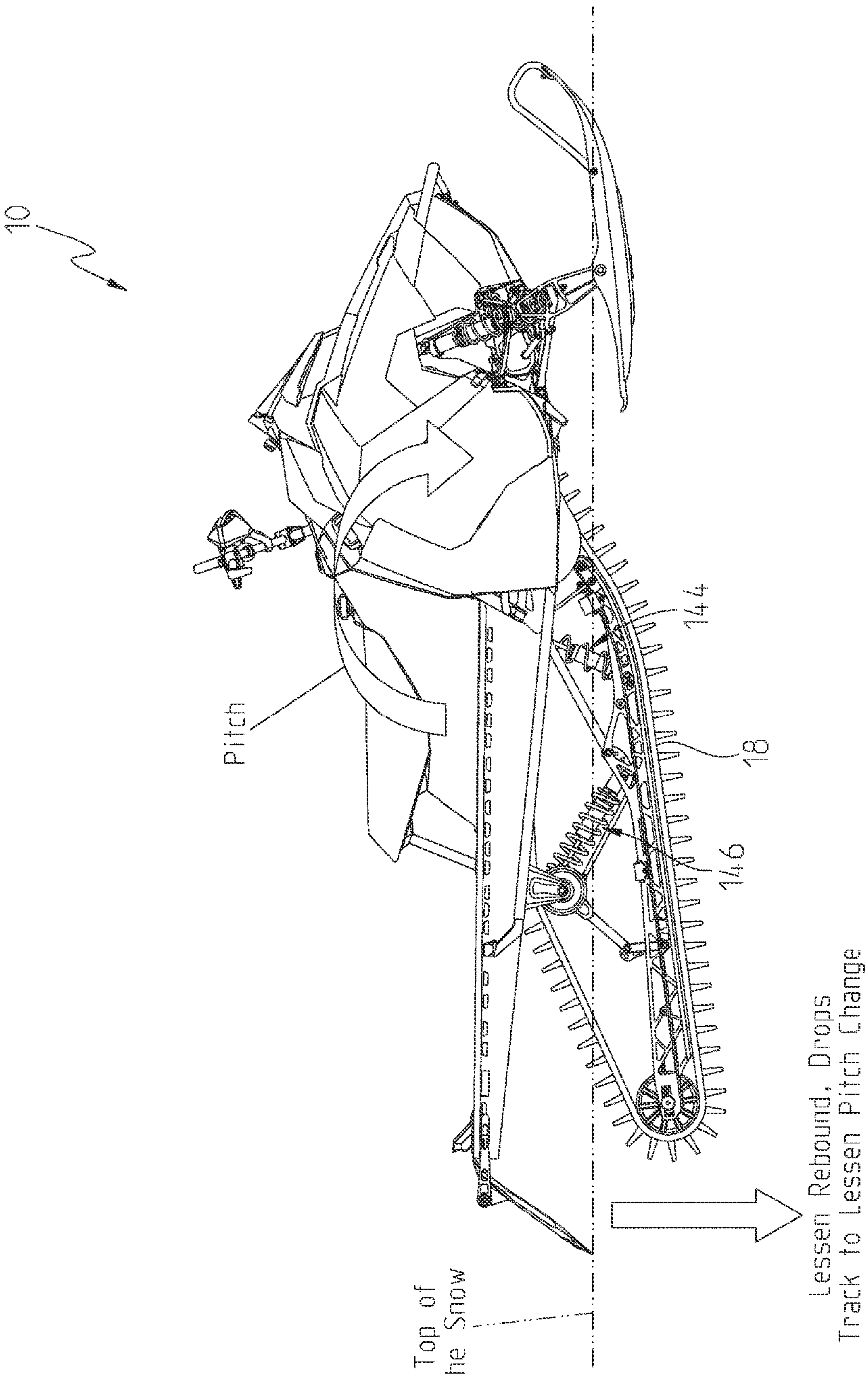
10
Pitch
144
18
146
Top of the Snow
Lessen Rebound, Drops Track to Lessen Pitch Change

Fig. 23

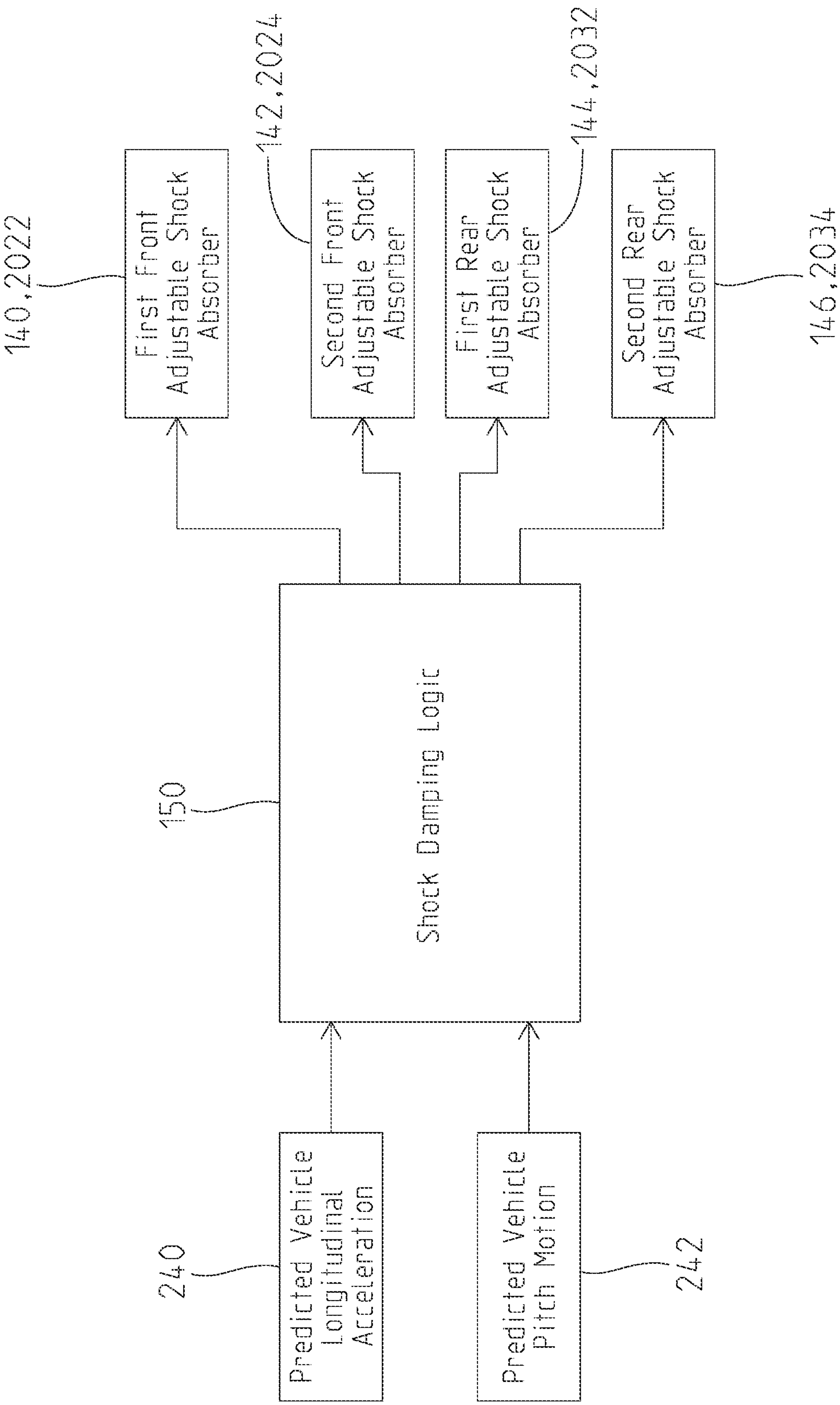
140,2022
First Front Adjustable Shock Absorber
142,2024
Second Front Adjustable Shock Absorber
144,2032
First Rear Adjustable Shock Absorber
146,2034
Second Rear Adjustable Shock Absorber
150
Shock Damping Logic
240
Predicted Vehicle Longitudinal Acceleration
242
Predicted Vehicle Pitch Motion

Fig. 24

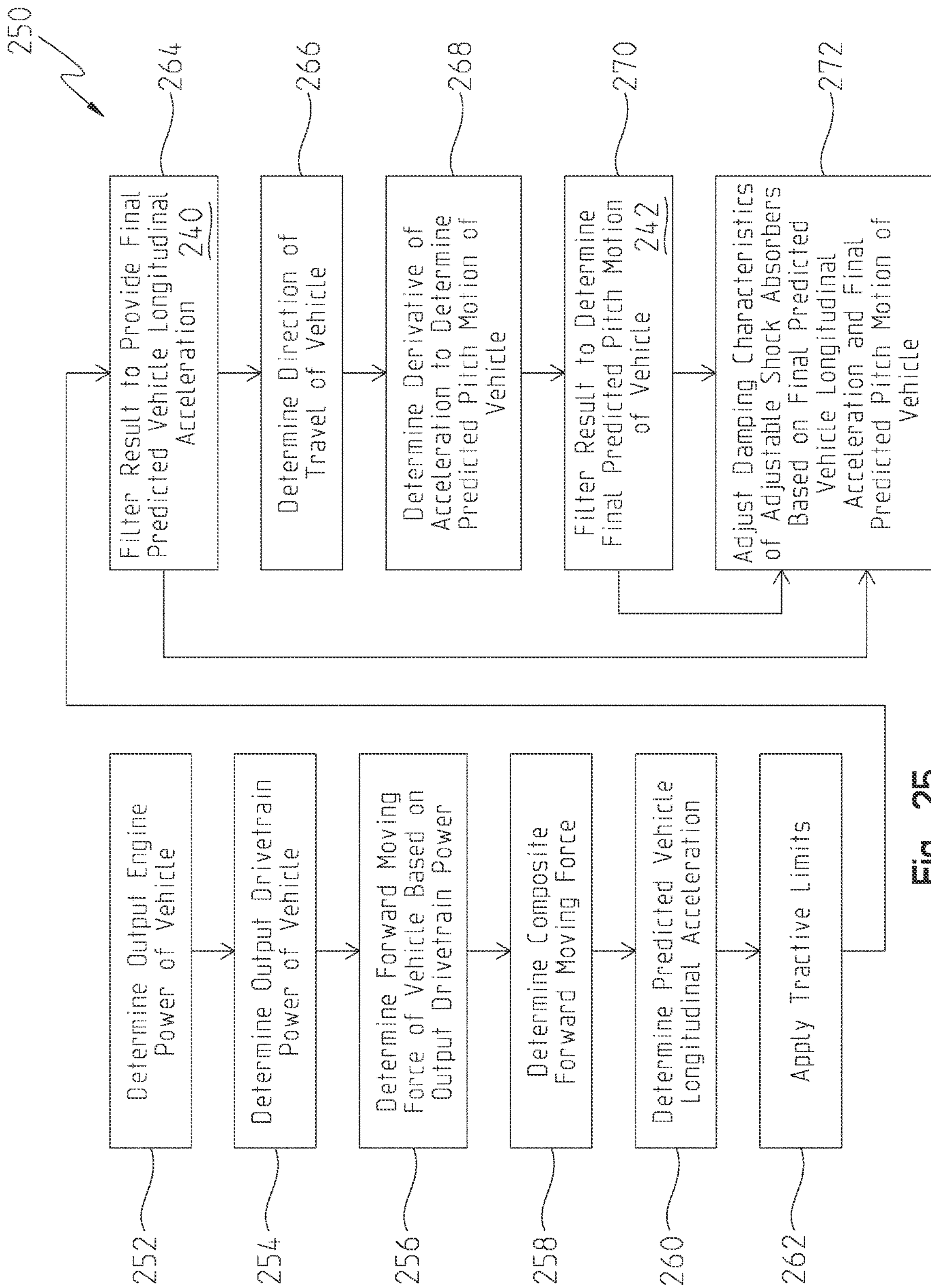
250
Determine Output Engine Power of Vehicle
252
Determine Output Drivetrain Power of Vehicle
254
Determine Forward Moving Force of Vehicle Based on Output Drivetrain Power
256
Determine Composite Forward Moving Force
258
Determine Predicted Vehicle Longitudinal Acceleration
260
Apply Tractive Limits
262
Filter Result to Provide Final Predicted Vehicle Longitudinal Acceleration 240
264
Determine Direction of Travel of Vehicle
266
Determine Derivative of Acceleration to Determine Predicted Pitch Motion of Vehicle
268
Filter Result to Determine Final Predicted Pitch Motion of Vehicle 242
270
Adjust Damping Characteristics of Adjustable Shock Absorbers Based on Final Predicted Vehicle Longitudinal Acceleration and Final Predicted Pitch Motion of Vehicle
272

Fig. 25

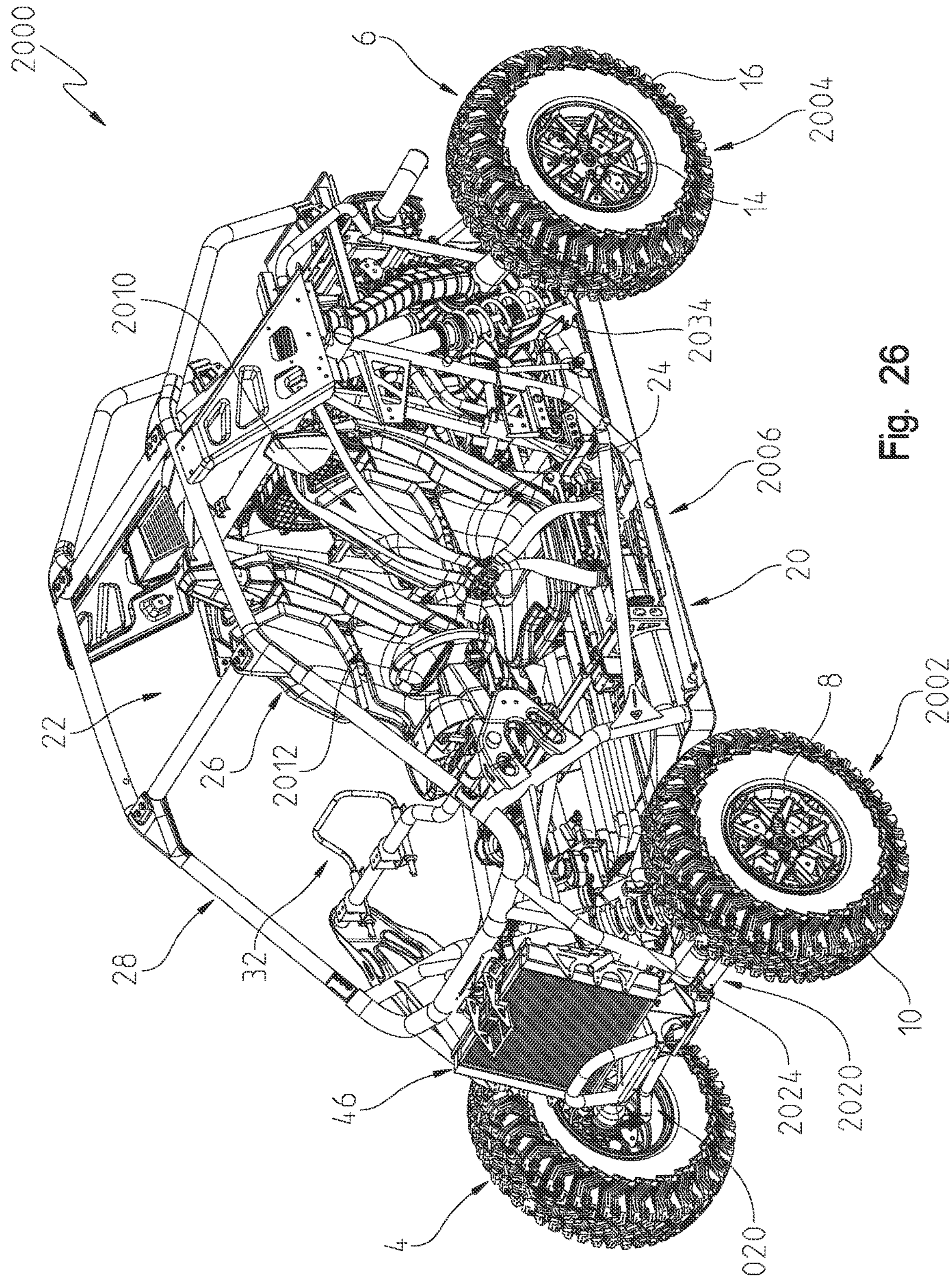
2000
6
16
2004
14
2010
2034
24
2006
20
2002
8
22
26
2012
28
32
10
2020
2024
2020
46
2020
4

Fig. 26

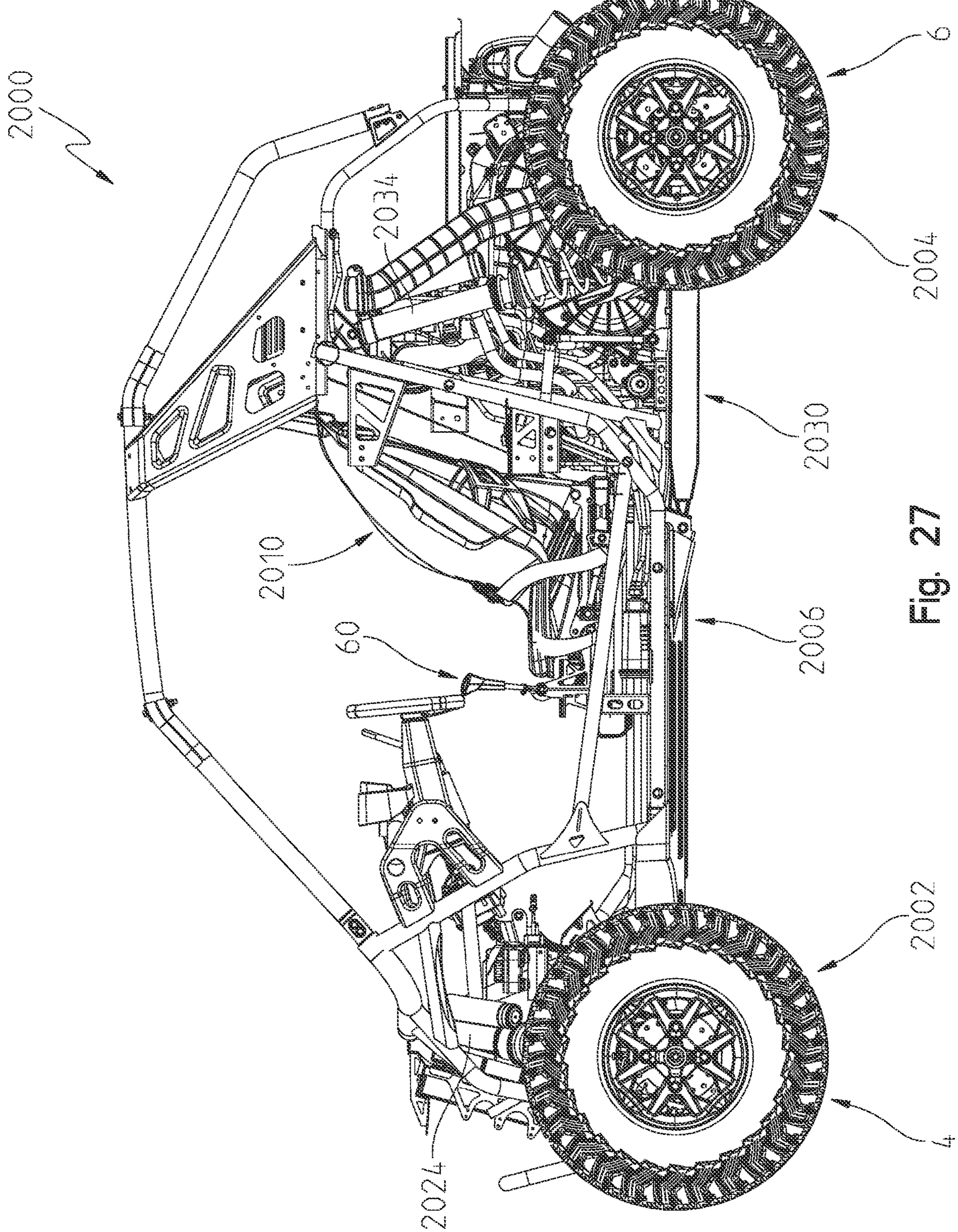

Fig. 27

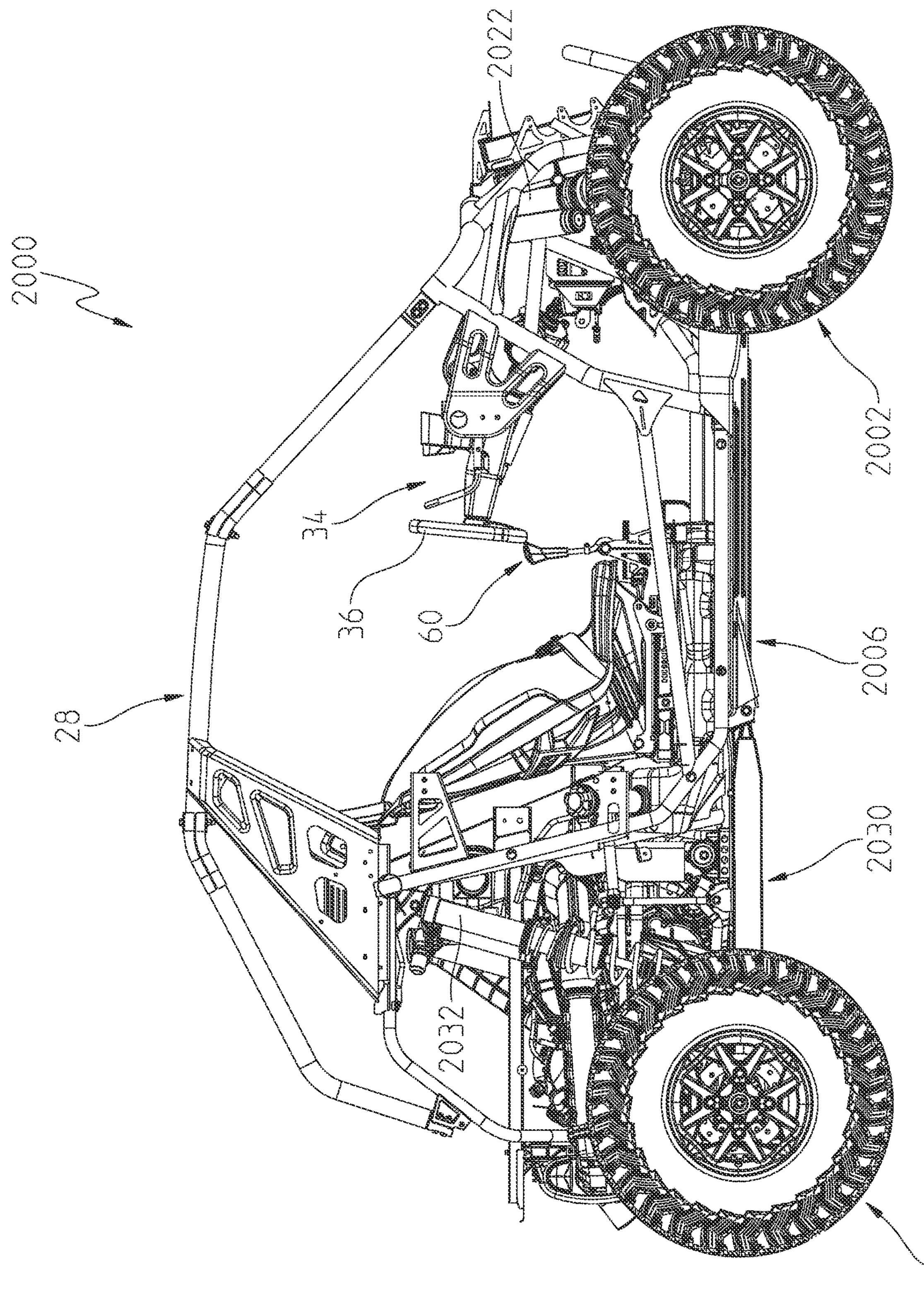

Fig. 28

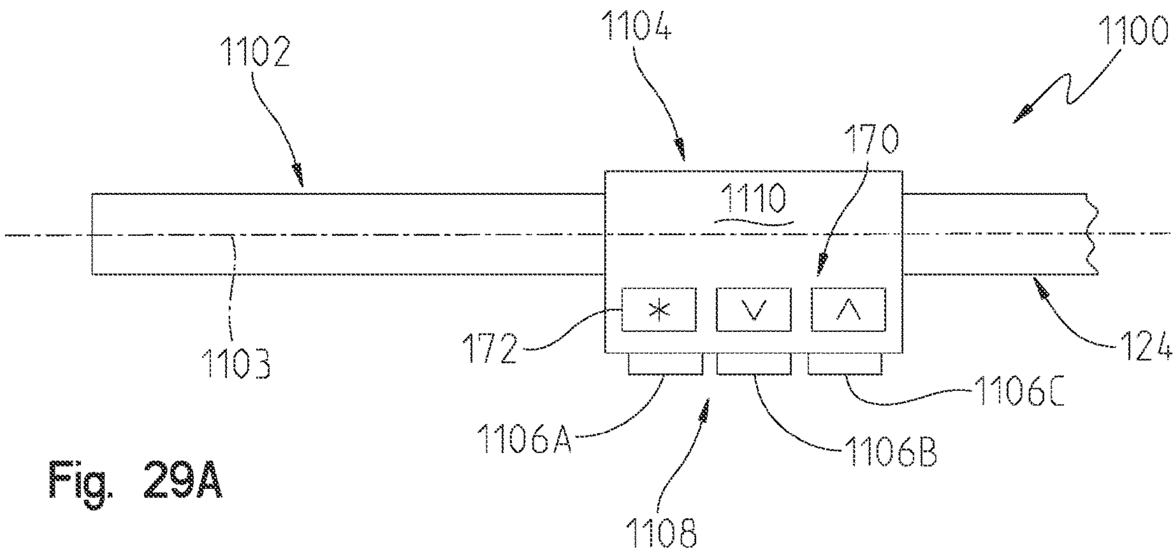

Fig. 29A

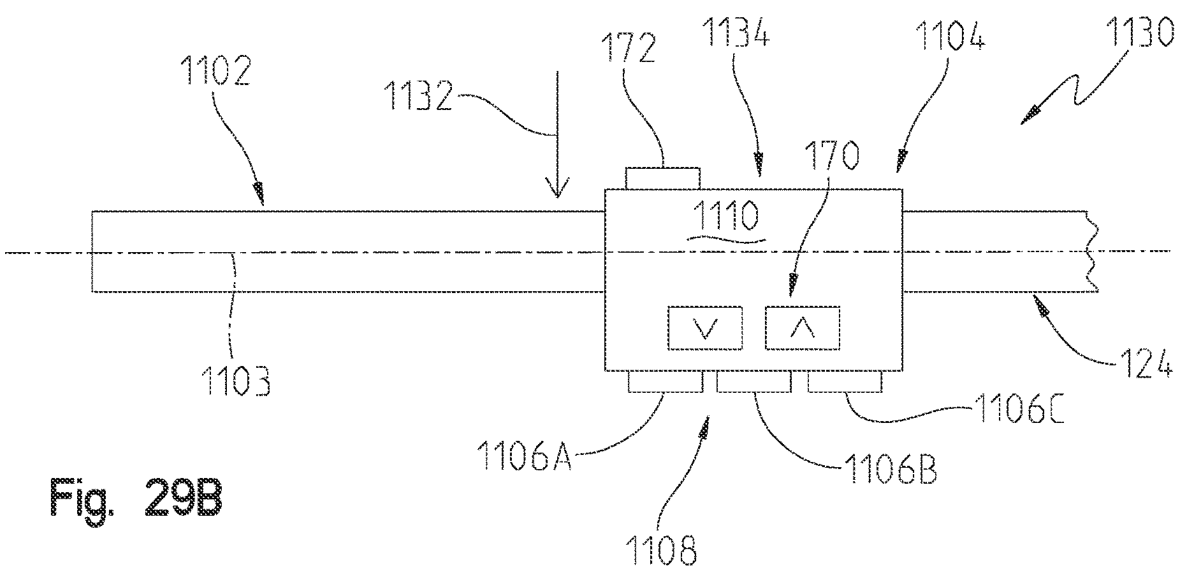

Fig. 29B

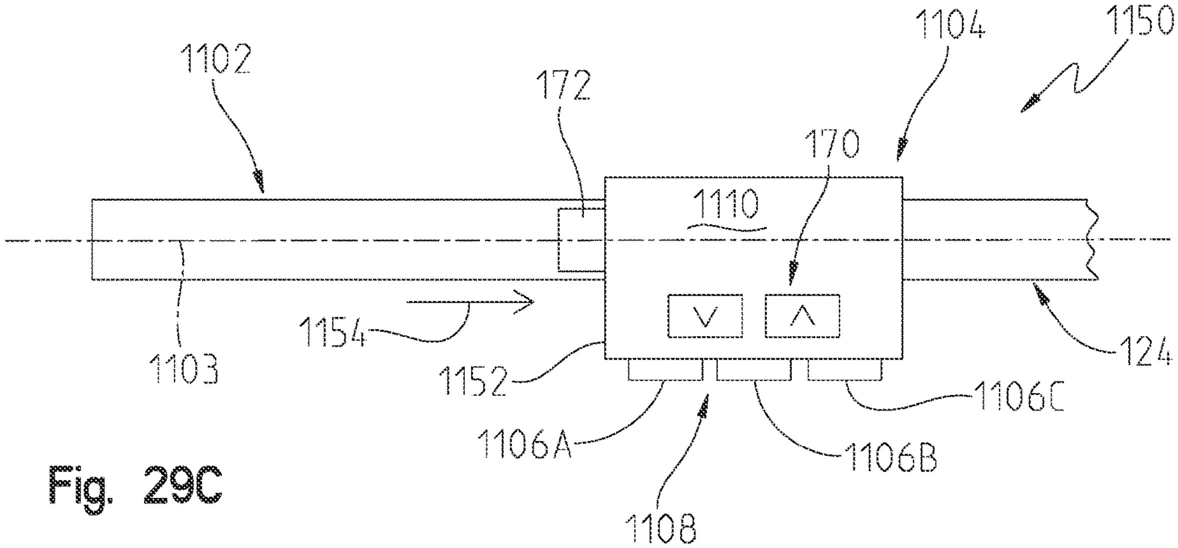

Fig. 29C

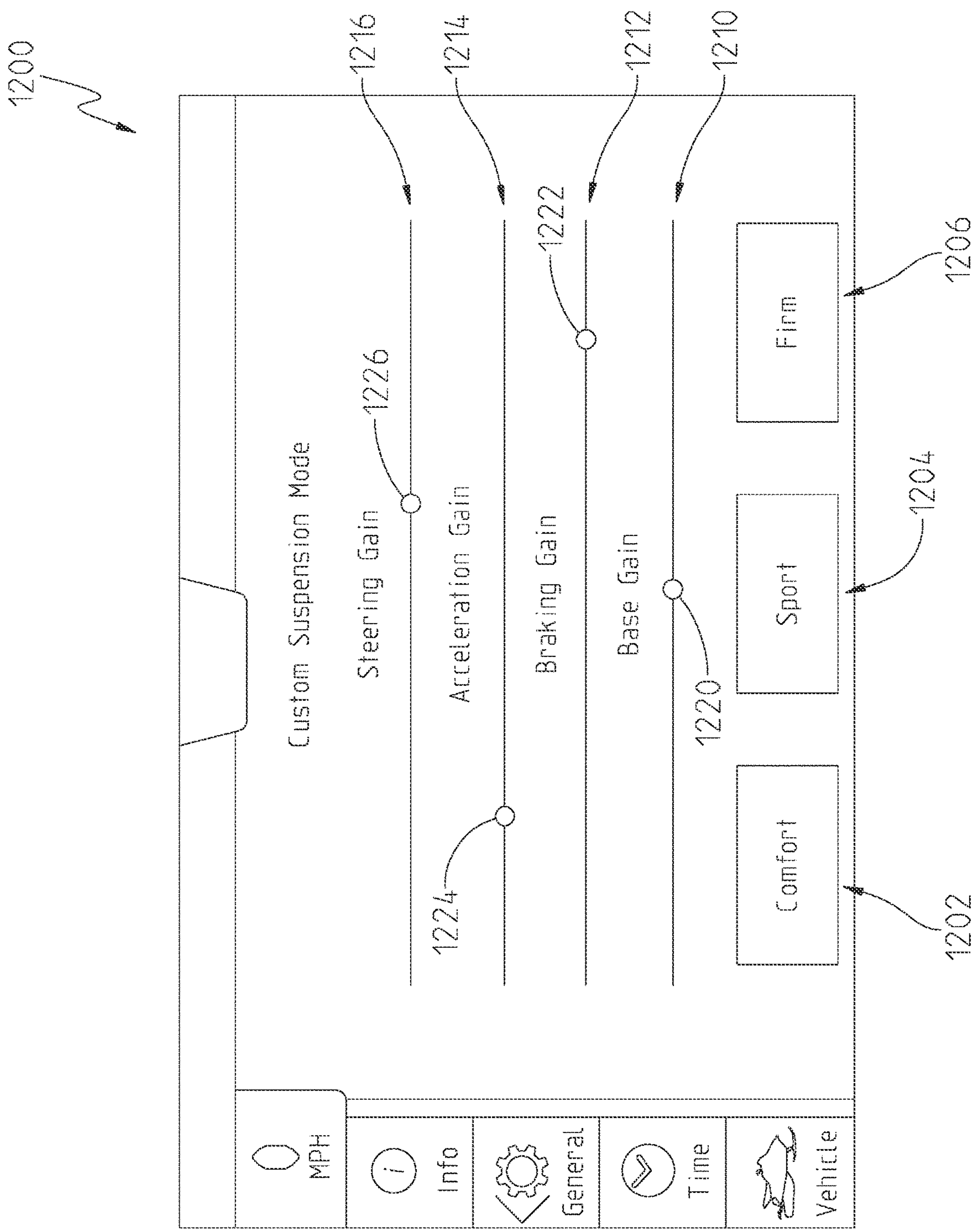
1200
1216
1214
1212
1210
1222
1226
1224
1220
1206
1204
1202
Custom Suspension Mode
Steering Gain
Acceleration Gain
Braking Gain
Base Gain
Firm
Sport
Comfort
MPH
Info
General
Time
Vehicle

Fig. 30

SYSTEMS AND METHODS OF ADJUSTABLE SUSPENSIONS FOR OFF-ROAD RECREATIONAL VEHICLES

RELATED APPLICATION

The present application is related to U.S. Provisional Application No. 63/027,833, filed May 20, 2021, titled SYSTEMS AND METHODS OF ADJUSTABLE SUSPENSIONS FOR OFF-ROAD RECREATIONAL VEHICLES, the entire disclosure of which is expressly incorporated by reference herein.

BACKGROUND AND SUMMARY OF THE DISCLOSURE

The present disclosure relates to improved suspension for off-road recreational vehicles and, in particular, to systems and methods of damping control for shock absorbers of off-road recreational vehicles.

Currently some off-road vehicles include adjustable shock absorbers. These adjustments include spring preload, high and low speed compression damping and/or rebound damping. In order to make these adjustments, the vehicle is stopped and the operator makes an adjustment at each shock absorber location on the vehicle. A tool is often required for the adjustment. Some off-road vehicles also include adjustable electric shocks along with sensors for active ride control systems. Exemplary systems are disclosed in U.S. Pat. No. 9,010,768 and US Published Patent Application No. 2016/0059660, both assigned to the present assignee and the entire disclosures of each expressly incorporated by reference herein.

In an exemplary embodiment of the present disclosure, a snowmobile for propelled movement relative to the ground is provided. The snowmobile comprising a plurality of ground engaging members including an endless track positioned along a centerline vertical longitudinal plane of the snowmobile have having a lateral width, a left front ski positioned on a left side of the centerline vertical longitudinal plane of the snowmobile, and a right front ski positioned on a right side of the centerline vertical longitudinal plane of the snowmobile; a frame supported by the plurality of ground engaging members; a steering system supported by the frame and operatively coupled to the left front ski and the right front ski to control a direction of travel of the snowmobile; a left ski suspension operatively coupling the left front ski to the frame; a right ski suspension operatively coupling the right front ski to the frame; a track suspension operatively coupling the endless track to the frame, the track suspension including a first adjustable shock absorber, the first adjustable shock absorber having at least one adjustable damping characteristic, the first adjustable shock absorber being laterally positioned within the lateral width of the endless track; a plurality of sensors supported by the ground engaging members; and at least one electronic controller operatively coupled to the first adjustable shock absorber, the at least one electronic controller based on inputs from the plurality of sensors alters at least one damping characteristic of the first adjustable shock absorber.

In an example thereof, the first adjustable shock absorber of the first suspension is positioned within an interior bounded by the endless track.

In another example thereof, the track suspension further comprises a second adjustable shock absorber having at least one adjustable damping characteristic, the second adjustable shock absorber being laterally positioned within the lateral width of the endless track, the first adjustable shock absorber is a front track adjustable shock absorber and the second adjustable shock absorber is a rear track adjustable shock absorber. In a variation thereof, the second adjustable shock absorber of the track suspension is positioned within the interior bounded by the endless track. In another variation thereof, the second adjustable shock absorber of the track suspension is positioned outside of the interior bounded by the endless track. In a further variation thereof, the second adjustable shock absorber of the track suspension is positioned above the endless track. In a refinement thereof, the front track adjustable shock absorber is positioned forward of the rear track adjustable shock absorber.

In a further example thereof, the at least one electronic controller alters the at least one damping characteristic of the first adjustable shock absorber while the snowmobile is moving relative to the ground. In a variation thereof, the at least one electronic controller alters a compression damping characteristic of the first adjustable shock absorber. In another variation thereof, the at least one electronic controller alters a rebound damping characteristic of the first adjustable shock absorber. In a further variation thereof, the at least one electronic controller alters both a compression damping characteristic of the first adjustable shock absorber and a rebound damping characteristic of the first adjustable shock absorber.

In still another example thereof, the snowmobile further comprises a third adjustable shock absorber and a fourth adjustable shock absorber, the third adjustable shock absorber is part of the left ski suspension operatively coupling the left front ski to the frame and the fourth adjustable shock absorber is part of the right ski suspension operatively coupling the right front ski to the frame.

In still a further example, a first damping characteristic of the first adjustable shock absorber is adjusted by the at least one electronic controller based on a longitudinal acceleration of the snowmobile. In a variation thereof, a first damping characteristic of the first adjustable shock absorber and a first damping characteristic of the second adjustable shock absorber are adjusted by the at least one electronic controller based on a longitudinal acceleration of the snowmobile. In another variation thereof, a first damping characteristic of the first adjustable shock absorber, a first damping characteristic of the second adjustable shock absorber, a first damping characteristic of the third adjustable shock absorber, and a first damping characteristic of the fourth adjustable shock absorber are adjusted by the at least one electronic controller based on a longitudinal acceleration of the snowmobile. In a further variation thereof, the longitudinal acceleration of the snowmobile is measured by the plurality of sensors. In still a further variation thereof, the longitudinal acceleration of the snowmobile is estimated by the at least one electronic controller. In yet another variation thereof, the longitudinal acceleration of the snowmobile is predicted by the at least one electronic controller.

In yet another example thereof, a first damping characteristic of the first adjustable shock absorber is adjusted by the at least one electronic controller based on a vehicle pitch motion of the snowmobile. In a variation thereof, a first damping characteristic of the first adjustable shock absorber and a first damping characteristic of the second adjustable shock absorber are adjusted by the at least one electronic controller based on a vehicle pitch motion of the snowmobile. In another variation thereof, a first damping characteristic of the first adjustable shock absorber, a first damping characteristic of the second adjustable shock absorber, a first damping characteristic of the third adjustable shock absorber, and a first damping characteristic of the fourth adjustable shock absorber are adjusted by the at least one electronic controller based on a predicted vehicle pitch motion of the snowmobile. In still another variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal acceleration, the at least one electronic controller alters a rebound damping characteristic for the third adjustable shock absorber and alters a rebound damping characteristic for the fourth adjustable shock absorber. In yet another variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal acceleration, the at least one electronic controller increases a rebound damping characteristic for the third adjustable shock absorber and increases a rebound damping characteristic for the fourth adjustable shock absorber. In yet still another variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal acceleration, the at least one electronic controller alters a compression damping characteristic for the third adjustable shock absorber and alters a compression damping for the fourth adjustable shock absorber. In still yet a further variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal acceleration, the at least one electronic controller decreases a compression damping characteristic for the third adjustable shock absorber and decreases a compression damping for the fourth adjustable shock absorber. In still another variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile and alters a compression damping characteristic of the first adjustable shock absorber. In yet another variation, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile and increases a compression damping characteristic of the first adjustable shock absorber. In still yet another variation, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile and alters a compression damping characteristic of the second adjustable shock absorber. In a further still variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile and increases a compression damping characteristic of the second adjustable shock absorber. In a yet further still variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile and alters a rebound damping characteristic of the second adjustable shock absorber. In another still variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile and increases a rebound damping characteristic of the second adjustable shock absorber. In a further variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile, the at least one electronic controller increases a rebound damping characteristic for the third adjustable shock absorber, increases a rebound damping characteristic for the fourth adjustable shock absorber, alters the compression damping characteristic of the first adjustable shock absorber, and increases the compression damping characteristic of the second adjustable shock absorber. In still another variation, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile based on an indication of a throttle increase. In yet still another variation, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile based on an indication of an engine torque increase. In a further still variation, the at least one electronic controller determines the predicted vehicle pitch motion indicates a rearward pitch of the snowmobile based on an indication of an increased vehicle acceleration. In a further yet still variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal deceleration, the at least one electronic controller increases a compression damping characteristic for the third adjustable shock absorber and increases a compression damping characteristic for the fourth adjustable shock absorber.

In yet a further example, thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal deceleration, the at least one electronic controller increases a rebound damping characteristic for the first adjustable shock absorber.

In still another example thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal deceleration, the at least one electronic controller alters a compression damping for the first adjustable shock absorber. In a variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal deceleration, the at least one electronic controller increases a rebound damping characteristic of the second adjustable shock absorber.

In another variation thereof, the at least one electronic controller determines the snowmobile is predicted to travel at a generally constant longitudinal deceleration, the at least one electronic controller alters a compression damping of the second adjustable shock absorber. In still a further variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile and alters a rebound damping characteristic of the first adjustable shock absorber. In yet still a further variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile and increases a damping characteristic of the first adjustable shock absorber. In a further yet variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile and alters a rebound damping characteristic of the second adjustable shock absorber. In a still further variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile and increases a rebound damping characteristic of the second adjustable shock absorber. In a yet still further variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile and alters a compression damping characteristic of the second adjustable shock absorber. In a further still variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile, the at least one electronic controller increases a compression damping characteristic for the third adjustable shock absorber, increases a compression damping characteristic for the fourth adjustable shock absorber, alters a rebound damping characteristic of the first adjustable shock absorber, and alters a rebound damping characteristic of the second adjustable shock absorber. In yet a further still variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile, the at least one electronic controller increases a compression damping characteristic for the third adjustable shock absorber, increases a compression damping characteristic for the fourth adjustable shock absorber, increases a rebound damping characteristic of the first adjustable shock absorber, and increases a rebound damping characteristic of the second adjustable shock absorber. In another variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile based on an indication of a throttle decrease. In a further variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile based on an indication of an application of the brake. In yet another variation thereof, the at least one electronic controller determines the predicted vehicle pitch motion indicates a forward pitch of the snowmobile based on an indication of a decreased vehicle acceleration.

In still another example thereof, at least one of a first damping characteristic of the first adjustable shock absorber, a first damping characteristic of the second adjustable shock absorber, a first damping characteristic of the third adjustable shock absorber, and a first damping characteristic of the fourth adjustable shock absorber are adjusted by the at least one electronic controller based on a turning of the snowmobile. In yet a further variation thereof, the at least one of the first damping characteristic of the first adjustable shock, the first damping characteristic of the second adjustable shock, the first damping characteristic of the third adjustable shock, and the first damping characteristic of the fourth adjustable shock are adjusted by the at least one electronic controller based on the turning of the snowmobile corresponding to a corner entry. In still another variation thereof, the at least one of the first damping characteristic of the first adjustable shock, the first damping characteristic of the second adjustable shock, the first damping characteristic of the third adjustable shock, and the first damping characteristic of the fourth adjustable shock are adjusted by the at least one electronic controller based on the turning of the snowmobile corresponding to a time subsequent to a corner entry. In a further still variation, the at least one electronic controller determines the snowmobile is making a left turn, the at least one electronic controller increases a compression damping characteristic for the fourth adjustable shock absorber. In another variation thereof, the at least one electronic controller determines the snowmobile is making the left turn, the at least one electronic controller decreases a compression damping characteristic of the fourth adjustable shock absorber. In a further variation thereof, the at least one electronic controller determines the snowmobile is making the left turn, the at least one electronic controller increases a rebound damping characteristic of the fourth adjustable shock absorber. In still another variation thereof, the at least one electronic controller determines the snowmobile is making the left turn, the at least one electronic controller decreases a compression damping characteristic of the first adjustable shock absorber. In yet still another variation thereof, the at least one electronic controller determines the snowmobile is making the left turn, the at least one electronic controller increases a rebound damping characteristic of the first adjustable shock absorber. In another variation thereof, the at least one electronic controller determines the snowmobile is making a right turn, the at least one electronic controller increases a compression damping characteristic for the third adjustable shock absorber. In a further variation thereof, the at least one electronic controller determines the snowmobile is making the right turn, the at least one electronic controller decreases a compression damping characteristic of the fourth adjustable shock absorber. In another variation thereof, the at least one electronic controller determines the snowmobile is making the right turn, the at least one electronic controller increases a rebound damping characteristic of the fourth adjustable shock absorber. In a yet still further variation thereof, the at least one electronic controller determines the snowmobile is making the right turn, the at least one electronic controller decreases a compression damping characteristic of the first adjustable shock absorber. In a further variation thereof, the at least one electronic controller determines the snowmobile is making the right turn, the at least one electronic controller increases a rebound damping characteristic of the first adjustable shock absorber.

In still a further example thereof, the at least one electronic controller adjusts a first damping characteristic of the first adjustable shock absorber to promote a lifting of the skis of the snowmobile. In a variation thereof, a compression damping characteristic of the first adjustable shock absorber is altered to promote a lifting of the skies of the snowmobile. In another variation thereof, the compression damping characteristic of the first adjustable shock absorber is decreased to promote a lifting of the skies of the snowmobile. In a further variation thereof, the compression damping characteristic of the first adjustable shock absorber is increased to promote a lifting of the skies of the snowmobile. In still another variation thereof, the at least one electronic controller adjusts a first damping characteristic of the first adjustable shock absorber and a first damping characteristic of the second adjustable shock absorber to promote a lifting of the skis of the snowmobile. In a further variation thereof, a compression damping characteristic of the second adjustable shock absorber is altered to promote the lifting of the skies of the snowmobile and to cause a wheelie. In a further still variation thereof, the compression damping characteristic of the second adjustable shock absorber is decreased to promote the lifting of the skies of the snowmobile and to cause the wheelie. In another variation thereof, a rebound damping characteristic of the second adjustable shock absorber is altered to promote the lifting of the skies of the snowmobile and to prevent a wheelie. In yet another variation thereof, the rebound damping characteristic of the second adjustable shock absorber is increased to promote the lifting of the skies of the snowmobile and to prevent the wheelie. In a further variation thereof, the at least one electronic controller adjusts a first damping characteristic of the first adjustable shock absorber, a first damping characteristic of the second adjustable shock absorber, a first damping characteristic of the third adjustable shock absorber, and a first damping characteristic of the fourth adjustable shock absorber to promote a lifting of the skis of the snowmobile. In a yet a further variation, a compression damping characteristic of the second adjustable shock absorber is altered to promote the lifting of the skies of the snowmobile and to cause a wheelie. In a further yet variation, the compression damping characteristic of the second adjustable shock absorber is decreased to promote the lifting of the skies of the snowmobile and to cause the wheelie. In a further still variation, a rebound damping characteristic of the third adjustable shock absorber is altered and a rebound damping characteristic of the fourth adjustable shock absorber is altered to promote the lifting of the skies of the snowmobile and to cause the wheelie. In another variation, the rebound damping characteristic of the third adjustable shock absorber is decreased and the rebound damping characteristic of the fourth adjustable shock absorber is decreased to promote the lifting of the skies of the snowmobile and to cause the wheelie. In yet a further variation, a rebound damping characteristic of the second adjustable shock absorber is altered to promote the lifting of the skies of the snowmobile and to prevent a wheelie. In still a variation, the rebound damping characteristic of the second adjustable shock absorber is increased to promote the lifting of the skies of the snowmobile and to prevent the wheelie. In yet still a further variation, a rebound damping characteristic of the third adjustable shock absorber is altered and a rebound damping characteristic of the fourth adjustable shock absorber is altered to promote the lifting of the skies of the snowmobile and to prevent the wheelie. In further still variation, the rebound damping characteristic of the third adjustable shock absorber is increased and the rebound damping characteristic of the fourth adjustable shock absorber is increased to promote the lifting of the skies of the snowmobile and to prevent the wheelie. In yet another variation, the damping characteristic of the first adjustable shock absorber is adjusted for a first condition and thereafter allowed to further adjust based on the plurality of sensors. In yet a further variation, the damping characteristic of the first adjustable shock absorber and the damping characteristic of the second adjustable shock absorber are adjusted for a first condition and thereafter allowed to further adjust based on the plurality of sensors. In a further still variation, the damping characteristic of the first adjustable shock absorber, the damping characteristic of the second adjustable shock absorber, the damping characteristic of the third adjustable shock absorber, and the damping characteristic of the fourth adjustable shock absorber are adjusted for a first condition and thereafter allowed to further adjust based on the plurality of sensors. In a still further variation, the damping characteristic of the first adjustable shock absorber and the damping characteristic of the second adjustable shock absorber are adjusted for a first condition and thereafter allowed to further adjust based on the plurality of sensors and the damping characteristic of the third adjustable shock absorber and the damping characteristic of the fourth adjustable shock absorber are adjusted for a second condition and thereafter allowed to further adjust based on the plurality of sensors. In a further variation, the first condition is a duration of a timer. In a still further variation, the second condition is a duration of a second timer.

In yet another example, the at least one electronic controller adjusts a first damping characteristic of the first adjustable shock absorber in response to the snowmobile being airborne. In a variation thereof, the adjustment of the first damping characteristic of the first adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In another variation thereof, the at least one electronic controller increases a compression damping characteristic of the first adjustable shock absorber in response to the snowmobile being airborne. In still a further variation, the at least one electronic controller continues to hold the increased compression damping characteristic of the first adjustable shock absorber for a first period of time after the snowmobile has landed. In a further still variation, the at least one electronic controller alters a rebound damping characteristic of the first adjustable shock absorber in response to the snowmobile being airborne. In yet another variation, the adjustment of the rebound damping characteristic of the first adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In yet a further variation, the at least one electronic controller increases the rebound damping characteristic of the first adjustable shock absorber for a first post landing period of time after the snowmobile has landed. In a further variation, the at least one electronic controller adjusts a first damping characteristic of the first adjustable shock absorber and a first damping characteristic of the second adjustable shock absorber in response to the snowmobile being airborne. In another variation, the adjustment of the first damping characteristic of the first adjustable shock absorber and the first damping characteristic of the second adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In a further variation, the at least one electronic controller increases a compression damping characteristic of the first adjustable shock absorber and increases a compression damping characteristic of the second adjustable shock absorber in response to the snowmobile being airborne. In yet a further variation, the at least one electronic controller continues to hold the increased compression damping characteristic of the first adjustable shock absorber and the increased compression damping characteristic of the second adjustable shock absorber for a first period of time after the snowmobile has landed. In yet a further variation, the at least one electronic controller alters a rebound damping characteristic of the first adjustable shock absorber and alters a rebound damping characteristic of the second adjustable shock absorber in response to the snowmobile being airborne. In another variation, the adjustment of the rebound damping characteristic of the first adjustable shock absorber and the adjustment of the rebound damping characteristic of the second adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In a further variation, the at least one electronic controller increases the rebound damping characteristic of the first adjustable shock absorber and increases the rebound damping characteristic of the second adjustable shock absorber for a first post landing period of time after the snowmobile has landed. In yet another variation, the at least one electronic controller adjusts a first damping characteristic of the third adjustable shock absorber, a first damping characteristic of the fourth adjustable shock absorber, a first damping characteristic of the first adjustable shock absorber, and a first damping characteristic of the second adjustable shock absorber in response to the snowmobile being airborne. In yet another variation, the adjustment of at least one of the first damping characteristic of the third adjustable shock absorber, the first damping characteristic of the fourth adjustable shock absorber, the first damping characteristic of the first adjustable shock absorber, and the first damping characteristic of the second adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In still another variation, the at least one electronic controller increases a compression damping characteristic of the third adjustable shock absorber, increases a compression damping characteristic of the fourth adjustable shock absorber, increases a compression damping characteristic of the first adjustable shock absorber, and increases a compression damping characteristic of the second adjustable shock absorber in response to the snowmobile being airborne. In yet still another embodiment, the at least one electronic controller continues to hold the increased compression damping characteristic of the third adjustable shock absorber, the increased compression damping characteristic of the fourth adjustable shock absorber, the increased compression damping characteristic of the first adjustable shock absorber, and the increased compression damping characteristic of the second adjustable shock absorber for a first period of time after the snowmobile has landed. In a further variation, the at least one electronic controller alters a rebound damping characteristic of the third adjustable shock absorber, alters a rebound damping characteristic of the fourth adjustable shock absorber, alters a rebound damping characteristic of the first adjustable shock absorber, and alters a rebound damping characteristic of the second adjustable shock absorber in response to the snowmobile being airborne. In a further still variation, the adjustment of the rebound damping characteristic of the third adjustable shock absorber and the adjustment of the rebound damping of the fourth adjustable shock absorber is adjusted in a first configuration in response to the snowmobile being airborne for less than a first time duration and in a second configuration in response to the snowmobile being airborne for longer than the first time duration. In a further still variation, the at least one electronic controller increases the rebound damping characteristic of the third adjustable shock absorber and increases the rebound damping characteristic of the fourth adjustable shock for a first post landing period of time after the snowmobile has landed in the second configuration. In yet another variation, the at least one electronic controller one of maintains or alters the rebound damping characteristic of the third adjustable shock absorber and one of maintains or alters the rebound damping characteristic of the fourth adjustable shock absorber for a first post landing period of time after the snowmobile has landed in the first configuration. In another variation, the adjustment of the rebound damping characteristic of the first adjustable shock absorber and the adjustment of the rebound damping characteristic of the second adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In yet another variation, the at least one electronic controller increases the rebound damping characteristic of the third adjustable shock absorber, increases the rebound damping characteristic of the fourth adjustable shock absorber, increases the rebound damping characteristic of the first adjustable shock absorber, and increases the rebound damping characteristic of the second adjustable shock absorber for a first post landing period of time after the snowmobile has landed.

In still another example, the snowmobile further comprises a driver actuatable suspension adjust input, wherein the at least one electronic controller adjusts a damping characteristic of the first adjustable shock absorber in response to a first actuation of the driver actuatable suspension input. In a variation thereof, the at least one electronic controller increases the compression damping of the first adjustable shock absorber in response to the first actuation of the driver actuatable suspension input. In another variation thereof, the driver actuatable suspension adjust input is supported by the steering system. In a further variation thereof, the driver actuatable suspension adjust input is positioned on a left hand portion of a handlebar of the steering system. In a further still variation thereof, the driver actuatable suspension adjust input is moveable along a longitudinal axis of the left hand portion of the handlebar of the steering system. In still another variation thereof, a first actuation characteristic of the driver actuatable suspension adjust input results in a first type of damping characteristic of the first adjustable shock absorber and a second actuation characteristic of the driver actuatable suspension adjust input results in a second type of damping characteristic of the first adjustable shock absorber. In a further still variation thereof, the first actuation characteristic is a single depress of a first time duration. In yet a further variation thereof, the second actuation characteristic is a single depress of a second time duration longer than the first time duration. In still another variation thereof, the second actuation characteristic is a plurality of depresses within a first time period.

In a further example thereof, the snowmobile further comprises at least one mode input supported by the snowmobile, the at least one electronic controller selecting at least one damping characteristic of the first adjustable shock absorber based on a mode selected with the at least one mode input. In a variation thereof, an operator may select a first mode from a plurality of available modes with the at least one mode input.

In still a further example thereof, the plurality of sensors include an internal measurement unit. In a variation thereof, the internal measurement unit is located between a spindle of the left ski and a rear end of a fuel tank supported by the plurality of ground engaging members. In another variation thereof, the internal measurement unit is located laterally within the lateral width of the endless track. In still another variation, the internal measurement unit is located laterally within the lateral width of an engine of the snowmobile. In yet another variation, the internal measurement unit is located between an engine and a rear end of a fuel tank both supported by the plurality of ground engaging members. In a further variation, the internal measurement unit is supported by a tunnel covering the endless track. In yet another variation, the internal measurement unit is located vertically in line with a steering post of the steering system. In yet a further variation thereof, the frame includes an over-structure positioned above an engine of the snowmobile and supporting a steering post of the steering system, wherein the internal measurement unit is located within an interior of the over-structure. In still a further variation, the internal measurement unit is mounted to a first portion of the snowmobile and is vibration isolated from the first portion of the snowmobile to reduce engine vibration. In yet still another variation thereof, the internal measurement unit is integrated into the at least one electronic controller. In a further still variation thereof, the internal measurement unit is spaced apart from the at least one electronic controller.

In another exemplary embodiment of the present disclosure, a method of controlling ride characteristics of a snowmobile is provided. The method comprising the steps of monitoring with at least one electronic controller a plurality of sensors supported by the snowmobile while the snowmobile is moving; and adjusting with the at least one electronic controller at least one damping characteristic of an adjustable shock absorber while the vehicle is moving, the adjustable shock absorber being apart of a suspension of an endless track of the snowmobile.

In an example thereof, the method further comprises the step of maintaining the at least one damping characteristic of the adjustable shock absorber when the vehicle is stationary.

In another example thereof, the method further comprises the step of discontinuing adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber when the vehicle is stationary.

In still another example thereof, the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a longitudinal acceleration of the snowmobile. In a variation thereof, the longitudinal acceleration of the snowmobile is measured by at least one sensor. In another variation thereof, the longitudinal acceleration of the snowmobile is estimated by the at least one electronic controller. In still another variation thereof, the longitudinal acceleration of the snowmobile is predicted by the at least one electronic controller.

In still another example thereof, the method further comprising the steps of determining the longitudinal acceleration of the snowmobile indicates an acceleration of the snowmobile; and changing a rebound damping characteristic of the adjustable shock absorber. In yet still another variation, the method further comprises the steps of determining the longitudinal acceleration of the snowmobile indicates an acceleration of the snowmobile; and changing a compression damping characteristic of the adjustable shock absorber.

In another example, the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a predicted pitch motion of the snowmobile. In a variation thereof, the method further comprising the steps of determining the predicted longitudinal pitch motion of the snowmobile indicates a rearward pitch of the snowmobile; and increasing a compression damping characteristic of the adjustable shock absorber. In another variation thereof, the method further comprising the steps of determining the predicted longitudinal pitch motion of the snowmobile indicates a forward pitch of the snowmobile; and changing a rebound damping characteristic of the adjustable shock absorber. In still another variation thereof, the method further comprising the steps of: determining the predicted pitch motion of the snowmobile indicates a forward pitch of the snowmobile; and changing a compression damping characteristic of the adjustable shock absorber.

In a further example thereof, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a turning of the snowmobile. In a variation thereof, the step of adjusting the at least one damping characteristic based on the turning of the snowmobile includes the step of altering a compression damping characteristic of the adjustable shock absorber. In another variation thereof, the step of adjusting the at least one damping characteristic based on the turning of the snowmobile includes the step of decreasing a compression damping characteristic of the adjustable shock absorber. In a further variation thereof, the step of adjusting the at least one damping characteristic based on the turning of the snowmobile includes the step of altering a rebound damping characteristic of the adjustable shock absorber. In yet another variation thereof, the step of adjusting the at least one damping characteristic based on the turning of the snowmobile includes the step of increasing a rebound damping characteristic of the adjustable shock absorber. In yet a further variation thereof, the step of adjusting the at least one damping characteristic based on the turning of the snowmobile includes the steps of decreasing a compression damping characteristic of the adjustable shock absorber and increasing a rebound damping characteristic of the adjustable shock absorber.

In still another example thereof, the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a braking of the snowmobile. In a variation thereof, the step of adjusting the at least one damping characteristic based on the braking of the snowmobile includes the step of altering a compression damping characteristic of the adjustable shock absorber. In another variation thereof, the step of adjusting the at least one damping characteristic based on the braking of the snowmobile includes the step of decreasing a compression damping characteristic of the adjustable shock absorber. In a further variation thereof, the step of adjusting the at least one damping characteristic based on the braking of the snowmobile includes the step of altering a rebound damping characteristic of the adjustable shock absorber. In still a further variation thereof, the step of adjusting the at least one damping characteristic based on the braking of the snowmobile includes the step of increasing a rebound damping characteristic of the adjustable shock absorber. In yet still a further variation, the step of adjusting the at least one damping characteristic based on the braking of the snowmobile includes the steps of decreasing a compression damping characteristic of the adjustable shock absorber and increasing a rebound damping characteristic of the adjustable shock absorber.

In a still further example, the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic to promote a lifting of the skis of the snowmobile. In a variation thereof, the step of adjusting the at least one damping characteristic to promote the lifting of the skis of the snowmobile includes the steps of decreasing a compression damping characteristic of the adjustable shock absorber. In another variation thereof, the step of adjusting the at least one damping characteristic to promote the lifting of the skis of the snowmobile includes the steps of increasing a rebound damping characteristic of a second adjustable shock absorber associated with the suspension of the endless track.

In a further example thereof, the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on the snowmobile being airborne. In a variation thereof, the step of adjusting the at least one damping characteristic based on the snowmobile being airborne includes the steps of increasing a compression damping characteristic of the adjustable shock absorber. In another variation thereof, a magnitude of the increase of the compression damping characteristic of the adjustable shock absorber is dependent on a length of time that the snowmobile has been airborne. In a further variation thereof, the method further comprises the step of continuing to hold the increased compression damping characteristic of the adjustable shock absorber for a first period of time after the snowmobile has landed. In still a further variation thereof, the step of adjusting the at least one damping characteristic based on the snowmobile being airborne further includes the step of decreasing a rebound damping characteristic of the adjustable shock absorber. In still another variation thereof, the method further comprising the step of increasing the rebound damping characteristic of the adjustable shock absorber for a second period of time after the snowmobile has landed. In a further still variation, the step of adjusting the at least one damping characteristic based on the snowmobile being airborne includes the step of decreasing a rebound damping characteristic of the adjustable shock absorber. In yet a further still variation, the method further comprising the step of increasing the rebound damping characteristic of the adjustable shock absorber for a second period of time after the snowmobile has landed.

In a still further exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising receiving with an electronic controller a plurality of inputs from a plurality of sensors supported by the vehicle; predictively determining a longitudinal acceleration of the vehicle; and adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the predicted longitudinal acceleration of the vehicle.

In an example thereof, the method further comprising the steps of predictively determining a longitudinal pitch motion of the vehicle; and adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the predicted longitudinal pitch motion of the vehicle.

In another example thereof, the predicted longitudinal acceleration is determined by the steps of: determining a predicted power for a prime mover of the snowmobile; determining an output power of the drivetrain based on the determined predicted power, the drivetrain including a CVT; determining a forward moving force of the snowmobile based on the determined output power of the drivetrain; determining a resultant forward moving force by subtracting at least one of a coast down force and an applied braking force from the determined forward moving force; and dividing the resultant forward moving force by a mass of the snowmobile to determine the predicted vehicle longitudinal acceleration.

In a further still exemplary embodiment of the present disclosure, a method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver is provided. The method comprising receiving with an electronic controller a plurality of inputs from a plurality of sensors supported by the vehicle; predictively determining a longitudinal pitch motion of the vehicle; and adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the predicted longitudinal pitch motion of the vehicle.

In yet still a further exemplary embodiment of the present disclosure, a snowmobile for propelled movement relative to the ground is provided. The snowmobile comprising a plurality of ground engaging members including an endless track positioned along a centerline vertical longitudinal plane of the snowmobile having a lateral width, a left front ski positioned on a left side of the centerline vertical longitudinal plane of the snowmobile, and a right front ski positioned on a right side of the centerline vertical longitudinal plane of the snowmobile; a frame supported by the plurality of ground engaging members; a steering system supported by the frame and operatively coupled to the left front ski and the right front ski to control a direction of travel of the snowmobile; a left ski suspension operatively coupling the left front ski to the frame; a right ski suspension operatively coupling the right front ski to the frame; a track suspension operatively coupling the endless track to the frame, the track suspension including a plurality of shock absorbers, the plurality of shock absorbers including a first adjustable shock absorber, the first adjustable shock absorber having at least one adjustable damping characteristic, the first adjustable shock absorber being laterally positioned within the lateral width of the endless track and being the forwardmost of the plurality of shock absorbers of the track suspension; a plurality of sensors supported by the ground engaging members; and at least one electronic controller operatively coupled to the first adjustable shock absorber, the at least one electronic controller based on inputs from the plurality of sensors alters at least one damping characteristic of the first adjustable shock absorber.

In an example thereof, the plurality of shock absorbers of the track suspension includes a second shock absorber positioned within the lateral width of the endless track. In a variation thereof, the second shock absorber is an adjustable shock absorber and the electronic controller is operatively coupled to the second shock absorber.

In another example thereof, the left ski suspension includes a third adjustable shock absorber and the right ski suspension includes a fourth adjustable shock absorber, the electronic controller is operatively coupled to the third adjustable shock absorber and the fourth adjustable shock absorber.

Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of the following detailed description of illustrative embodiments exemplifying the best mode of carrying out the invention as presently perceived.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing aspects and many additional features of the present system and method will become more readily appreciated and become better understood by reference to the following detailed description when taken in conjunction with the accompanying drawings.

FIG. **1** illustrates a left side view of an exemplary snowmobile;

FIG. **1**A illustrates a left front perspective view of the exemplary snowmobile of FIG. **1** illustrating a yaw-axis, a pitch-axis, and a roll-axis for the snowmobile;

FIG. **1**B illustrates a right front perspective view of another exemplary snowmobile having another exemplary shock setup relative to the endless track of the snowmobile;

FIG. **1**C illustrates a left side view of the exemplary snowmobile of FIG. **1**B;

FIG. **2** illustrates a right side view of the exemplary snowmobile of FIG. **1**;

FIG. **3** illustrates the track suspension of the exemplary snowmobile of FIG. **1**;

FIG. **4** illustrates a representative view of components of the exemplary snowmobile of FIG. **1** including a suspension with a plurality of continuous damping control shock absorbers, an operator interface, and a plurality of sensors integrated with a controller of the snowmobile;

FIG. **5** illustrates a partial left rear perspective view of the exemplary snowmobile of FIG. **1**;

FIG. **6** illustrates a representative view of components of the exemplary snowmobile of FIG. **1** including a suspension with a plurality of continuous damping control shock absorbers, an operator interface, and a plurality of sensors integrated with a controller of the snowmobile;

FIG. **7** illustrates a right front perspective view of the exemplary snowmobile of FIG. **1** illustrating a tunnel covering an endless track ground engaging member and a pyramidal frame;

FIG. **8**A illustrates a representative view of an exemplary shock damping logic of the electronic controller of FIG. **6**;

FIG. **8**B illustrates a representative view of another exemplary shock damping logic of the electronic controller of FIG. **6**;

FIG. **8**C illustrates a representative view of yet another exemplary shock damping logic of the electronic controller of FIG. **6**;

FIGS. **9**A-C illustrate an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for a continuous damping control shock absorber provided as part of the suspension of a left front ski of the exemplary snowmobile of FIG. **1**;

FIGS. **10**A-C illustrate an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for a continuous damping control shock absorber provided as part of the suspension of a right front ski of the exemplary snowmobile of FIG. **1**;

FIGS. **11**A and **11**B illustrate an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for a continuous damping control shock absorber provided as part of the suspension for the endless track of the exemplary snowmobile of FIG. **1**;

FIGS. **12**A and **12**B illustrate an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for a continuous damping control shock absorber provided as part of the suspension for the endless track of the exemplary snowmobile of FIG. **1**;

FIG. **13** illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting a g-out event or a chatter event;

FIG. **14** illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary brake event;

FIG. **15** illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary airborne event;

FIGS. **16**A-C illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary anti-squat event;

FIGS. **17**A-D illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary cornering event;

FIG. **18** illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary driver actuatable suspension adjust input event;

FIGS. **19**A and **19**B illustrate an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary launch mode event;

FIG. **20** illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6** for detecting an exemplary one ski event;

FIG. **21** illustrates the exemplary snowmobile of FIG. **1** airborne;

FIG. **22** illustrates a pitch of snowmobile **10** in deep snow without an adjustment of the damping characteristics of the adjustable shock absorbers;

FIG. **23** illustrates a pitch of snowmobile **10** in deep snow with an adjustment of the damping characteristics of the adjustable shock absorbers;

FIG. **24** illustrates a representative view of an exemplary shock damping logic of the electronic controller of FIG. **6**;

FIG. **25** illustrates an exemplary processing sequence of the shock damping logic of the electronic controller of FIG. **6**;

FIG. **26** illustrates a left, front, perspective view of a side-by-side off-road recreational vehicle;

FIG. **27** illustrates a left side view of the side-by-side off-road recreational vehicle of FIG. **26**;

FIG. **28** illustrates a right side view of the side-by-side off-road recreational vehicle of FIG. **26**.

FIGS. **29**A-C illustrate exemplary arrangements for inputs of exemplary operator interfaces; and

FIG. **30** illustrates an exemplary input screen for adjusting a gain for various damping profiles.

Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of various features and components according to the present disclosure, the drawings are not necessarily to scale and certain features may be exaggerated in order to better illustrate and explain the present disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, which are described below. The embodiments disclosed below are not intended to be exhaustive or limited to the precise form disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings.

Referring now to FIGS. **1**, **1**A, **2**-**4**, and **7**, an illustrative embodiment of a snowmobile **10** is shown. Snowmobile **10** includes a chassis or frame **12** having a front frame portion **12***a* and a rear frame portion **12***b*. A body assembly **14** generally surrounds at least front frame portion **12***a* of frame **12**. Front frame portion **12***a* is supported by front ground-engaging members, illustratively skis **16**, and rear frame portion **12***b* is supported by a rear ground-engaging member, illustratively an endless track **18**. The rider uses a steering assembly **20**, which is operably coupled to at least skis **16** through a steering linkage **17** (see FIG. **7**), when operating snowmobile **10**. A seat assembly **22** is provided generally rearward of steering assembly **20** and is configured to support the rider.

Front skis **16** are operably coupled to a front left suspension assembly **24** and a right front suspension assembly **24**, and endless track **18** cooperates with a rear suspension assembly **26**. A powertrain assembly is positioned generally intermediate front suspension assembly **24** and rear suspension assembly **26**, and provides power to endless track **18** to move snowmobile **10**. More particularly, the powertrain assembly **30** includes an engine (see prime mover **110**, see FIG. **1**C), a transmission, and a drive shaft. In one embodiment, the transmission includes a shiftable transmission and a continuously variable transmission ("CVT") **111** (see FIG. **1**C). In embodiments, the shiftable transmission includes a forward low gear, a forward high gear, a reverse gear, a neutral setting, and a park setting.

As shown in FIG. **3**, rear suspension assembly **26** includes a plurality of slide rails **32**, a front track adjustable shock absorber **144** positioned inside of an interior of endless track **18**, a rear track adjustable shock absorber **146** positioned inside of the interior of endless track **18**, a plurality of torque arms **36** operably coupled to a forward, lower end of rear track adjustable shock absorber **146**, and a link assembly **38** operably coupled to a rear, upper end of rear track adjustable shock absorber **146**. In one embodiment, torque arms **36** may be comprised of forged aluminum, which may reduce the overall weight of snowmobile **10**. Additionally, rear suspension assembly **26** may include front track adjustable shock absorber **144** positioned forward of rear track adjustable shock absorber **146** and operably coupled to torque arms **36** and slide rails **32**.

Rear suspension assembly **26** also includes a plurality of rear idler wheels **42** rotatably coupled to the rear end of slide rails **32** and a plurality of carrier wheels **44** laterally adjacent the rear, upper end of rear track adjustable shock absorber **146**. Rear idler wheels **42** and carrier wheels **44** are configured to maintain tension in endless track **18**. Additionally, the position of rear idler wheels **42** on slide rails **32** may be adjusted to adjust the tension in endless track **18**. As shown in FIGS. **1** and **2**, endless track **18** generally surrounds rear suspension assembly **26** and is supported on at least slide rails **32**, rear idler wheels **42**, and carrier wheels **44**. Rear suspension assembly **26** is configured to cooperate with endless track **18** when snowmobile **10** is operating. In particular, rear suspension assembly **26** is configured to move longitudinally and vertically during operation of snowmobile **10**, and the tension in endless track **18** is maintained throughout the movement of rear suspension assembly **26** by at least rear idler wheels **42**.

Referring to FIGS. **1**B and **1**C, a snowmobile **10'** is shown having a different rear suspension **24'** than snowmobile **10**. Rear suspension **24'** positions front track adjustable shock absorber **144** inside the interior of the endless track **18** and rear track adjustable shock absorber **146** outside of the interior of endless track **18** and, in particular, above endless track **18**. As shown in FIG. **1**A, endless track **18** has a lateral width W. For each of snowmobile **10** and snowmobile **10'**, front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** are laterally positioned within the later width W of endless track **18**. Throughout the disclosure, snowmobile **10** should be interpreted to read snowmobile **10**, **10'** in discussing the damping characteristics of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**.

Right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** are adjustable shock absorbers, the damping characteristics of which are continuously controlled by an electronic controller **100**. In embodiments, endless track **18** includes one adjustable shock absorbers and a standard shock absorber, such as a manually adjustable shock absorber. In embodiments, electronic controller **100** updates the damping characteristics of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** during movement of snowmobile **10**, **10'**. Electronic controller **100** continuously controls right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** by updating the desired damping characteristics of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on monitored sensor values, received operator inputs, and/or other inputs at discrete instances of time. An exemplary time interval is about 1 milli-seconds to about 5 milliseconds. For example, electronic controller **100** updates targets for each of right front adjustable shock absorber **140**, f**142**, front track adjustable shock absorber **144**, rear track adjustable shock absorber **146** about every 5 milliseconds and updates the current control loop about every milli-second.

In embodiments, the damping characteristics of front track adjustable shock absorber **144** are controlled by electronic controller **100** while rear track adjustable shock absorber **146** is manually adjustable independent of electronic controller **100**. In embodiments, the damping characteristics of rear track adjustable shock absorber **146** are controlled by electronic controller **100** while front track adjustable shock absorber **144** is manually adjustable independent of electronic controller **100**. In embodiments, the arrangement and control of one or both of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** of rear suspension **26** is applicable to single front ski vehicles, such as snowbikes. A snowbike may have one or more shocks associated with the front ski. These one or more shocks may be adjustable shock absorbers controlled by controller **100** is a similar manner as right front adjustable shock absorber **140** and left front adjustable shock absorber **142**.

In embodiments, right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** include solenoid valves mounted at the base of the shock body or internal to a damper piston of the respective right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. The stiffness of the shock absorber is increased or decreased by introducing additional fluid to the interior of the shock absorber, removing fluid from the interior of the shock absorber, and/or increasing or decreasing the ease with which fluid can pass from a first side of a damping piston of the shock absorber to a second side of the damping piston of the shock absorber. In another embodiments, right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** include a magnetorheological fluid internal to the respective right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. The stiffness of the shock is increased or decreased by altering a magnetic field experienced by the magnetorheological fluid. Additional details on exemplary adjustable shocks are provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein. In one embodiment, right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** each include a first controllable proportional valve to adjust compression damping and a second controllable proportional valve to adjust rebound damping. In another embodiment, right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** each include a combination proportional valve which controls both compression damping and rebound damping.

Additional details of frame **12**, body assembly **14**, endless track **18**, front suspension assembly **24**, rear suspension assembly **26**, and the powertrain assembly for snowmobile **10**, **10'** and other exemplary snowmobiles are disclosed in U.S. Pat. Nos. 7,891,454, 8,590,654, 8,820,458, 8,944,204, 9,428,232, 9,540,072, 9,809,195, and 10,358,187, the complete disclosures of which are expressly incorporated by reference herein

Referring to FIG. **4**, electronic controller **100** includes at least one processor **104** and at least one non-transitory computer readable medium, memory **106**. In embodiments, electronic controller **100** is a single unit that controls the operation of various systems of snowmobile **10**. In embodiments, electronic controller **100** is a distributed system comprised of multiple controllers each of which control one or more systems of snowmobile **10** and may communicate with each other over one or more wired and/or wireless networks. For example, electronic controller **100** may include a suspension controller **200** (see FIG. **6**) which controls the damping characteristics of each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** and an engine controller **204** which controls the operator of a prime mover **110** (see FIG. **4**) of snowmobile **10**.

Electronic controller **100** includes shock damping logic **150** which controls the damping characteristics of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. The term "logic" as used herein includes software and/or firmware executing on one or more programmable processors, application-specific integrated circuits, field-programmable gate arrays, digital signal processors, hardwired logic, or combinations thereof. Therefore, in accordance with the embodiments, various logic may be implemented in any appropriate fashion and would remain in accordance with the embodiments herein disclosed. A non-transitory machine-readable medium comprising logic can additionally be considered to be embodied within any tangible form of a computer-readable carrier, such as solid-state memory, magnetic disk, and optical disk containing an appropriate set of computer instructions and data structures that would cause a processor to carry out the techniques described herein. This disclosure contemplates other embodiments in which electronic controller **100** is not microprocessor-based, but rather is configured to control operation of right front adjustable shock absorber **140**, left Front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on one or more sets of hardwired instructions.

Returning to FIG. **4**, electronic controller **100** provides the electronic control of and/or monitors the various components of snowmobile **10**, illustratively prime mover **110**, a steering system **120**, and a braking system **122**. Exemplary prime movers include two-cycle combustion engines, four-cycle combustion engines, electric motors, and other suitable motive devices. Prime mover **110** is operatively coupled to endless track **18** through a transmission. Exemplary transmissions include shiftable transmissions, continuously variable transmissions, and combinations thereof. An exemplary steering system **120** is shown in FIG. **7** including handlebars **124** attached to a steering post **126** which in turn is coupled to a steering linkage **128**. The steering linkage **128** is coupled to skis **16** to adjust an orientation of skis **16** relative to the ground.

Further, electronic controller **100** is operatively coupled to a plurality of sensors **130** (see FIG. **4**) which monitor various parameters of snowmobile **10** or the environment surrounding snowmobile **10**. Exemplary sensors **130** include a global positioning sensor (GPS) **131**, an inertial measurement unit ("IMU") **132**, an engine speed sensor **133**, a brake switch **134**, a brake pressure sensor **135**, a steering angle sensor **136**, a transmission gear selection sensor **137**, a throttle position sensor **138**, a vehicle speed sensor **139**, an air pressure sensor **141**.

GPS sensor **131** provides a location of snowmobile **10** on the surface of the earth. IMU **132** includes a three-axis accelerometer and a three-axis gyroscope. Referring to FIG. **1**A, the three-axis accelerometer provides acceleration data of snowmobile **10** along axes **160**, **162**, and **164** and the three-axis gyroscope provides angular information regarding rotation about each of axes **160**, **162**, and **164**. IMU **132** is illustratively supported (see FIG. **7**) on the chassis **150** of snowmobile **10** to provide an indication of acceleration forces of the vehicle during operation. In embodiments, IMU **132** is attached to a tunnel **154** of chassis **150** which covers track **18**. In embodiments, IMU **132** is attached to an over structure frame **156** of chassis **150**. In embodiments, IMU **132** is located within an interior of an over structure frame **156** of chassis **150**. In embodiments, IMU **132** is located vertically in line with a steering post **126** (see FIG. **7**) of snowmobile **10** within an interior of an over structure frame **156** of chassis **150**. In embodiments, IMU **132** is mounted proximate to frame **156**, such as on a front side of a fuel tank **114** (see FIG. **1**C) or integrated into a wall of the fuel tank. In embodiments, IMU **132** is located longitudinally between a fuel tank and a prime mover **110** of snowmobile **10**. In embodiments, IMU **132** is located between a spindle **116** (see FIG. **1**C) of the left ski **16** and a rear end **118** of a fuel tank **114** supported by the plurality of ground engaging members. In embodiments, IMU **132** is located along a longitudinal centerline plane snowmobile **10**. In embodiments, IMU **132** is located at a center of gravity of snowmobile **10**. In embodiments, IMU **132** is offset from the center of gravity of snowmobile **10** and the readings of the three-axis gyroscope are used by electronic controller **100** to determine the acceleration values of snowmobile **10** at the center of gravity of snowmobile **10**. In one embodiment, IMU is integrated into electronic controller **100**, such as integrated into suspension controller **200**. In one embodiment, IMU is spaced apart from electronic controller **100**. In embodiments, IMU **132** is isolated from chassis **150** with isolation mounts, such as rubber mounts, to reduce the amount of engine vibration experienced by the IMU **132**.

Engine speed sensor **133** monitors a crankshaft speed of the engine. In embodiments, the crankshaft speed is provided to controller **100** by the engine control module ("ECU").

Brake switch **134** monitors an actuation of an operator brake input, such as a foot actuated input or a hand actuated input. Steering angle sensor **136** monitors a rotation angle of steering post **126** (see FIG. **7**). In embodiments, steering system **120** includes an electronic power steering unit. In embodiments, steering system does not include an electronic power steering unit. In the embodiments without an electronic power steering unit, steering angle sensor **136** is supported either in front of the engine below the exhaust ports of the engine or rearward of the engine and above the throttle bodies of the engine.

Throttle position sensor **138** monitors an actuation of a throttle input, such as a foot actuated input or a hand actuated input. Vehicle speed sensor **139** monitors a ground speed of snowmobile **10**. In one example, a speed of endless track **18** is monitored by vehicle speed sensor **139** and used as an indication of the speed of snowmobile **10** relative to the ground. In embodiments, snowmobile **10** has a single vehicle speed sensor **139** which monitors a movement of endless track **18**.

Brake pressure sensor **135** is a transducer that measures the pressure in applied by the operator in the brake lines of the braking system.

Steering angle sensor **136** monitors a rotational movement of the steering column or monitors other movements of portions of the steering system.

Transmission gear selection sensor **137** monitors an input from the engine control module of the gear selection for a shiftable transmission and/or a rotational direction of the crankshaft for forward and reverse directions.

Vehicle speed sensor **139** measures ground speed by monitoring a rotational speed of the track drive shaft or other measurements of rotational members of the drive line which are indicative of ground speed.

Air pressure sensor **141** monitors one of ambient barometric pressure and/or the manifold air pressure in the engine intake system.

Electronic controller **100** also interacts with an operator interface **180** (see FIGS. **4** and **5**) which includes at least one input device **182** and at least one output device **184**. Exemplary input devices **182** include levers, buttons, switches, soft keys, and other suitable input devices. Exemplary output devices **184** include lights, displays, audio devices, tactile devices, and other suitable output devices. In embodiments, operator interface **180** includes a display **190**, such as a touch screen display, and electronic controller **100** interprets various types of touches to the touch screen display as inputs and controls the content displayed on touch screen display.

Referring to FIG. **5**, operator interface **180** may include one or more switches **192**, one or more inputs **194** supported by handlebars **124**, and one or more parts of an instrument cluster **196** which may include display **190**. In embodiments, referring to FIG. **6**, operator interface **180** includes a mode select input **170**, a driver actuatable suspension adjust input **172**, and a launch mode input **174**. The mode select input permits an operator to select between at least three setups of snowmobile **10**, each having predefined base damping characteristics for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**.

Exemplary modes selectable with mode select input **170** include a comfort mode, a handling mode, and a rough trail mode. In the comfort mode both compression and rebound damping of each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** are lower than the rough trail mode. In one example, the compression and rebound damping are generally constant for vehicle speed and longitudinal acceleration. In the rough trail mode, the compression damping of each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** are generally higher than the comfort mode and the handling mode based on vehicle speed and longitudinal acceleration. The rebound damping in the rough trail mode will be maintained or lowered the respective shock absorbers will be more prone to extend and therefore have more shock length to absorb compression events. In the handling mode, the compression damping of each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** is lower than the comfort mode and the rough handling mode based on vehicle speed and longitudinal acceleration and the rebound damping is higher than the comfort mode based on vehicle speed and longitudinal acceleration. The dynamic ride height for snowmobile **10** is lowest for the handling mode and highest for the rough trail mode. In embodiments, the selected mode provides base damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. These base damping characteristics are updated by controller **200** based on the movement or predicted movement of snowmobile **10**. Exemplary updates for the damping characteristics are provided herein. In embodiments, for the predicted longitudinal acceleration of the snowmobile **10**, controller **200** actively reviews the engine torque and/or throttle position and adjusting the compression and rebound damping to counter predicted motion of snowmobile **10**, such as diving or squatting. In an example, snowmobile **10** is traveling 80 MPH and the operator drops the throttle to 0%. In response, controller **200** increases the compression damping in right front adjustable shock absorber **140** and left front adjustable shock absorber **142** to counter a front end dive of snowmobile **10** and increases the rebound damping in front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** to counter the lifting of the rear end.

In embodiments, a longitudinal acceleration of snowmobile **10** is measured based on one or more inputs, such as IMU **132**, estimated based on one or more inputs, such as a monitored throttle position and/or a monitored engine rpm, or predicted based on one or more inputs as described herein.

In embodiments, an operator may provide a gain coefficient to one or more damping profiles. Referring to FIG. **30**, an exemplary input screen **1200** for display **190** is shown. Input screen **1200** is a custom suspension mode screen. An operator selects which mode to modify, by selecting one of comfort mode input **1202**, sport mode input **1204**, and firm mode input **1206**. Also shown in FIG. **30** are slider inputs, illustratively base gain input **1210**, braking gain input **1212**, acceleration gain input **1214**, and steering or cornering gain input **1216**. Each of base gain input **1210**, braking gain input **1212**, acceleration gain input **1214**, and steering post **126** have a respective slider **1220**, **1222**, **1224**, and **1226**. When a slider is centered, such as base gain slider **1220** of base gain input **1210**, no change is made to the standard base damping profile for the selected mode. Moving a respective slider to the left on input screen **1200** reduces the standard base damping profile for the selected mode while moving a respective slider to the right on steering system **120** increases the standard base damping profile for the selected mode. In embodiments, a first screen is provided for right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and a second screen is provided for front track adjustable shock absorber **144** and rear track adjustable shock absorber **146**.

In embodiments, input screen **1200** further includes inputs for selecting one passenger (driver only) or two passengers and for indicating cargo weight. In embodiments, one or more of the damping profiles are adjusted automatically based on an input of the operator, such as the recognition of the operator and their rider profile settings in a rider profile. The rider profile may be stored in the electronic controller of the snowmobile **10**, in a personal computing device (such as a mobile phone), a key fob, or retrieval from the cloud or remote computing device.

Referring to FIG. **29**A, an exemplary embodiment **1100** of operator interface **180** is illustrated. FIG. **29**A illustrates a portion of the left handlebar **124** including a hand grip portion **1102** to be grasped by the left hand of the operator and an input housing **1104**. Hand grip portion **1102** includes a longitudinal axis **1103**. Input housing **1104** includes a plurality of actuatable inputs **1106**A-C on a front face **1108** of input housing **1104** facing the seat **22**. Exemplary inputs include a light input, heated grips input, and control inputs for display **190**. Inputs **1106**A-C may be any type of actuatable inputs including buttons, membrane switches, rockers, and other suitable inputs. Additional inputs are provided on a top face **1110** of input housing **1104**. In embodiments, mode selection inputs **170** (up and down buttons for cycling through modes) and driver actuatable suspension adjust input **172** are provided on top face **1110** of input housing **1104**. Inputs **170** and **172** may be any type of actuatable inputs including buttons, membrane switches, rockers, levers, triggers, and other suitable inputs. In embodiments, input **172** is positioned within a finger's or thumb's touch distance of a hand grip so that it may be actuated without requiring the operator to remove their hand from the grip. In embodiments, one or both of input **170** and input **172** may be positioned on front face **1108** and actuatable therefrom.

Referring to FIG. **29**B, an exemplary embodiment **1130** of operator interface **180** is illustrated. FIG. **29**B illustrates a portion of the left handlebar **124** including a hand grip portion **1102** to be grasped by the left hand of the operator and an input housing **1104**. Hand grip portion **1102** includes a longitudinal axis **1103**. Input housing **1104** includes a plurality of actuatable inputs **1106**A-C on a front face **1108** of input housing **1104** facing the seat **22**. Exemplary inputs include a light input, heated grips input, and control inputs for display **190**. Inputs **1106**A-C may be any type of actuatable inputs including buttons, membrane switches, rockers, and other suitable inputs. Additional inputs are provided on a top face **1110** of input housing **1104**. In embodiments, mode selection inputs **170** (up and down buttons for cycling through modes) is provided on top face **1110** of input housing **1104**. Additional inputs are provided on a rear face **1134** of input housing **1104**, rear face **1134** being opposite of front face **1108**. In embodiments, driver actuatable suspension adjust input **172** is provided on rear face **1134** of input housing **1104**. Inputs **170** and **172** may be any type of actuatable inputs including buttons, membrane switches, rockers, levers, triggers, and other suitable inputs. Driver actuatable suspension adjust input **172** is depressed in direction **1132** towards front face **1108** to actuate driver actuatable suspension adjust input **172**. In embodiments, input **172** is a lever or trigger which is one of pulled toward the operator in actuation or pushed away from the operator in actuation.

Referring to FIG. **29**C, an exemplary embodiment **1150** of operator interface **180** is illustrated. FIG. **29**C illustrates a portion of the left handlebar **124** including a hand grip portion **1102** to be grasped by the left hand of the operator and an input housing **1104**. Hand grip portion **1102** includes a longitudinal axis **1103**. Input housing **1104** includes a plurality of actuatable inputs **1106**A-C on a front face **1108** of input housing **1104** facing the seat **22**. Exemplary inputs include a light input, heated grips input, and control inputs for display **190**. Inputs **1106**A-C may be any type of actuatable inputs including buttons, membrane switches, rockers, and other suitable inputs. Additional inputs are provided on a top face **1110** of input housing **1104**. In embodiments, mode selection inputs **170** (up and down buttons for cycling through modes) is provided on top face **1110** of input housing **1104**. Additional inputs are provided on a hand grip side face **1152** of input housing **1104**. In embodiments, driver actuatable suspension adjust input **172** is provided on hand grip side face **1152** of input housing **1104**. Inputs **170** and **172** may be any type of actuatable inputs including buttons, membrane switches, rockers, and other suitable inputs. Driver actuatable suspension adjust input **172** is depressed in direction **1154** along **1103** to actuate driver actuatable suspension adjust input **172**. In embodiments, the stroke of input **172** in direction **1154** is about 0.25 inches. In embodiments, input **172** includes tactile feedback to the operator of depression. In one example, input **172** and input housing **1104** cooperate to provide a "click" feel at the end of stroke of input **172** in direction **1154**. In embodiments an operator interface surface of input **172** is a half-moon shape and can be actuated from hand grip face **1152** or front face **1108**.

Although FIGS. **29**A-C, illustrate exemplary placements for driver actuatable suspension adjust input **172**, other inputs may occupy the respective positions. In embodiments, launch mode input **174** replaces driver actuatable suspension adjust input **172**.

In embodiments, input **172** may be provided as one input of a multi-functional input device. For example, the multi-functional input device may provide the functionality of input **172** in response to a first actuation characteristic(s) (such as a sequence) and another functionality, such as mode change, in response to a second actuation characteristic(s) (such as a sequence).

The driver actuatable suspension adjust input **172** permits an operator to request a maximum stiffness for one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. Exemplary operation of a driver actuatable suspension adjust input is discussed in U.S. Pat. No. 10,406,884, assigned to the assignee of the present application, the entire disclosure of which is incorporated by reference herein. Electronic controller **100** controls the operation of output devices **184** and monitors the actuation of input devices **182**. Referring to FIG. **6**, in embodiments, display **190** is operatively coupled to electronic controller **100** (illustratively suspension controller **200**) over a network, illustratively a CAN network.

Referring to FIG. **8**A, shock damping logic **150** includes one or more processing sequences **202** which control the damping characteristics of one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. In embodiments, shock damping logic **150** includes one or more functions which based on one or more inputs output a desired damping characteristic for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**.

Referring to FIG. **8**B, shock damping logic **150** includes one or more processing sequences **202** which control the damping characteristics of one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** and one or more lookup tables **204** which based on one or more inputs provides damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**.

Referring to FIG. **8**C, shock damping logic **150** includes one or more processing sequences **202** which control the damping characteristics of one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** and one or more look-up tables **204** which based on one or more inputs provides damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. Exemplary look-up tables include airborne time event table **210**, driver actuatable suspension adjust input event table **212**, brake event table **214**, anti-dive event table **216**, one ski event table **218**, brake corner event table **220**, chatter corner event table **222**, g-out event table **224**, chatter event table **226**, corner event table **228**, base damping table **230**, launch mode event table **232**, anti-squat event table **234**, and a predictive longitudinal acceleration and predictive vehicle pitch table **236**. The use of look-up tables **204** by processing sequences **202** are illustrated in the exemplary processing sequences of FIGS. **9-20** and **24**. In embodiments, electronic controller **200** executes the exemplary processing sequences generally continuously while monitoring for operator inputs and sensor values. In one embodiment, electronic controller **200** updates the damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** every about 1 to about 5 milliseconds. For example, electronic controller **100** updates targets for each of right front adjustable shock absorber **140**, f**142**, front track adjustable shock absorber **144**, rear track adjustable shock absorber **146** about every 5 milliseconds and updates the current control loop about every 1 millisecond.

Electronic controller **200** when executing processing sequences **202** arbitrates which damping characteristics to use for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on the sensed values from sensors **130** and inputs from operator interface **180**. In embodiments, the arbitration of electronic controller **200** for compression damping may be prioritized with airborne detection compression damping having the highest priority, followed by driver actuatable suspension adjust input **172** compression damping having the next highest priority, followed by cornering detection compression damping, followed by braking detection compression damping, followed by the maximum of the base compression damping based on mode selection and acceleration based detection compression damping. In one example, wherein the predictive longitudinal acceleration and predictive vehicle pitch processing sequence is used, the braking detection damping is taken into account with the predictive acceleration and pitch and thus following cornering detection compression damping is the maximum of the base compression damping based on mode selection and the predictive longitudinal acceleration and predictive vehicle pitch based detection compression damping. In embodiments, the arbitration of electronic controller **200** for rebound damping may be prioritized with airborne detection rebound damping having the highest priority including post landing time, followed by cornering detection rebound damping, followed by braking detection rebound damping, followed by the maximum of the base rebound damping based on mode selection and acceleration based detection rebound damping. In one example, wherein the predictive longitudinal acceleration and predictive vehicle pitch processing sequence is used the braking detection rebound damping is taken into account with the predictive acceleration and pitch and thus following cornering detection rebound damping is the maximum of the base rebound damping based on mode selection and the predictive longitudinal acceleration and predictive vehicle pitch based detection rebound damping. The above are exemplary arbitration priorities and different processing sequences may include different arbitration priorities based on the event detection tables provided and the desired performance of the vehicle.

Referring to FIG. **24**, predictive longitudinal acceleration and predictive pitch table **236** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on one or both of a predicted longitudinal acceleration of snowmobile **10** and a predicted pitch of snowmobile **10**. Shock damping logic **150** based on a predicted vehicle longitudinal acceleration **240** and/or a predicted vehicle pitch **242** alters the damping characteristics of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**.

When both the predicted longitudinal acceleration **240** and the predicted vehicle pitch **242** are taken into account, shock damping logic **150** is able to adjust right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** for acceleration of snowmobile **10** including minimizing vehicle squat during acceleration and minimizing vehicle dive during deceleration. In one example, a predicted increase in the longitudinal acceleration results in shock damping logic **150** altering, such as increasing, the compression damping of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** to reduce the chance of skis **16** being lifted off of the snow and alters, such as increases the rebound damping of right front adjustable shock absorber **140** and left front adjustable shock absorber **142**. Further, the rebound damping may be increased on the front track adjustable shock absorber **144**. In another example, when the predicted vehicle pitch indicates the rear of snowmobile **10** is predicted to drop and the predicted longitudinal acceleration is increasing, shock damping logic **150** increases the compression damping of right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and decreases the rebound damping of right front adjustable shock absorber **140** and left front adjustable shock absorber **142** to promote a lifting of skis **16** off of the snow. In a further embodiment, when the predicted longitudinal acceleration is decreasing and the predicted vehicle pitch indicates the front of snowmobile **10** is predicted to drop, shock damping logic **150** increases the compression damping of right front adjustable shock absorber **140** and left front adjustable shock absorber **142** to reduce diving of the front of snowmobile **10** and, alter, such as increases or decreases the compression damping of front track adjustable shock absorber **144** to promote track **18** pressure on the snow and to absorb bumps.

The predicted longitudinal acceleration **240** and the predicted vehicle pitch **242** are also applicable to other off-road recreational vehicles including side-by-side vehicles, such as vehicle **2000** shown in FIGS. **26-28**. Vehicle **2000** includes a pair of front wheels **2002** and a pair of rear wheels **2004** supporting a frame **2006**. Vehicle **2000** includes a driver seat **2010** and a passenger seat **2012** arranged in a side-by-side arrangement. Each of the front wheels **2002** are coupled to frame **2006** through front suspensions **2020** including shock absorbers **2022**, **2024** which may be adjustable shock absorbers as described herein. Each of the rear wheels **2004** are coupled to frame **2006** through rear suspensions **2030** including shock absorbers **2032**, **2034** which may be adjustable shock absorbers as described herein. Additional details regarding vehicle **2000** are provided in US Published Patent Application US2019/0210668, filed Jan. 10, 2019, titled VEHICLE, the entire disclosure of which is expressly incorporated by reference herein.

Referring to FIG. **25**, an exemplary processing sequence **250** of suspension controller **200** for determining a predicted longitudinal acceleration **240** and a predicted vehicle pitch **242** of snowmobile **10** is illustrated. A predicted power for prime mover **110**, for example an internal combustion engine, is determined, as represented by block **252**. In one example, an engine torque is provided from an engine controller of the vehicle. The engine torque is multiplied by a measured engine speed measured by engine speed sensor **133** to determine a power output of the engine. In another example, a throttle position is measured and with a look-up table a corresponding engine torque is provided. Again, the engine torque is multiplied by an engine speed to obtain the output power of the engine. In embodiments, a value is measured by air pressure sensor **141** and the look-up table used to determine engine torque is multi-dimensional and includes torque values for different air pressures. In one example, air pressure sensor **141** measures an air pressure associated with an airbox of the snowmobile **10**. In another example, air pressure is measured indirectly by GPS sensor **131** which determines a location of snowmobile **10** and based on a look-up table provides an ambient air pressure reading for that elevation either actual from a third party service or typical based on a look-up table.

The determined engine power is then multiplied by an efficiency factor for the transmission of snowmobile **10** to provide an output power for the drivetrain, as represented by block **254**. In one example, the efficiency factor accounts for losses associated with the CVT transmission. The output power of the drivetrain is converted to a forward moving force of the vehicle by dividing the output power of the drivetrain by the vehicle speed measured by vehicle speed sensor **139**, as represented by block **256**.

A resultant or composite forward moving force is determined by subtracting from the determined forward moving force of block **256** a coast down force of the vehicle and a braking force, as represented by block **258**. The coast down force of the vehicle is determined through a look-up table as a function of a measured vehicle speed measured by vehicle speed sensor **139**. The braking force is determined through a look-up table of braking force as a function of a measured brake pressure measured by brake pressure sensor **135**.

A predicted vehicle longitudinal acceleration is determined by dividing the resultant forward moving force by the mass the vehicle, as represented by block **260**. In one example, a standard mass of the vehicle is used.

The predicted vehicle longitudinal acceleration is compared to tractive limits and set equal to the respective tractive limit (a negative tractive limit for a deceleration of snowmobile **10** and a positive tractive limit for an acceleration of snowmobile **10**) if the predicted longitudinal acceleration exceeds the respective tractive limit, as represented by block **262**.

In embodiments, the predicted vehicle acceleration from block **262** is filtered, as represented by block **264**, to provide a smoother response. The filtering is helpful to account for the time difference between a determined engine output power and an acceleration of snowmobile **10** and to account for different sampling rates of the various sensors.

The filtered predicted vehicle longitudinal acceleration is used to determine a predicted pitch motion of snowmobile **10**. A direction of travel of snowmobile **10** is determined, as represented block **266**. Once a direction of travel is known, forward or reverse, the effect of the acceleration or deceleration on the front and rear of the vehicle can be considered. In one example, a gear selection sensor **137** is provided as part of the shiftable transmission of snowmobile **10** and provides an indication of whether the shiftable transmission is in a forward gear or a reverse gear. In another example, the gear selection sensor monitors in an electric reverse input (not shown) has been actuated. Upon actuation of the electric reverse input, the engine of snowmobile **10** is stalled and then fired in a reverse direction to change the rotation of the output shaft of the engine.

The predicted magnitude of the pitch motion is determined by taking the derivative of the filtered predicted vehicle longitudinal acceleration, as represented by block **268**. This predicted vehicle pitch motion value is filtered to provide a smother result over time, as represented by block **270**. The predicted vehicle pitch motion **242** and/or the predicted vehicle longitudinal acceleration **240** are used by shock damping logic **150** to adjust the damping characteristics of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**, as represented by block **272**.

In embodiments, the predicted vehicle longitudinal acceleration and the predicted vehicle pitch motion are used to alter the base damping of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** which may be set by the selected vehicle mode (comfort, handling, and rough trail). The damping characteristic tables for compression of each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** and the damping characteristics tables for rebound of each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** may be two-dimensional (one input, one output damping characteristic), three-dimensional (two inputs, one output damping characteristic), or x dimensional ($x-1$ inputs, one output damping characteristic).

In embodiments, the base damping tables are two-dimensional map for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** and each of compression damping characteristic and rebound characteristic (two inputs, one output). The two inputs are vehicle speed and predicted longitudinal vehicle acceleration and the output depending on the table is one of compression damping and rebound damping. In one example, vehicle speed is measured by vehicle speed sensor **139** and the predicted longitudinal vehicle acceleration is determined by processing sequence **250**.

Referring to FIG. **8**C, predictive longitudinal acceleration and pitch motion table **236** provides damping characteristics for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on a predicted longitudinal vehicle acceleration and a predicted vehicle pitch motion. Assuming snowmobile **10** is traveling forward, for a predicted constant longitudinal acceleration (no or minimal predicted pitch motion) shock damping logic **150** increases the rebound damping characteristic for right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and, optionally, reduce the compression damping characteristic for right front adjustable shock absorber **140** and left front adjustable shock absorber **142**. For a predicted increase in the vehicle pitch motion (a rearward pitch) shock damping logic **150** further increases the rebound damping characteristic of right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and increase the compression damping characteristic for front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** for the duration of the predicted vehicle pitch motion.

Assuming snowmobile **10** is traveling forward, for a predicted constant longitudinal deceleration (no or minimal pitch) shock damping logic **150** increases the rebound damping characteristic and decreases the compression damping characteristic for right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and, optionally, increases the rebound damping characteristic for front track adjustable shock absorber **144** and rear track adjustable shock absorber **146**. For a predicted decrease in the vehicle longitudinal pitch motion (a forward pitch) shock damping logic **150** further increases the compression damping characteristic of right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and alters, such as increases, the rebound damping characteristic for front track adjustable shock absorber **144** and alters, such as increases, a rebound damping of the rear track adjustable shock absorber **146** for the duration of the predicted vehicle pitch motion. For a predicted increase in the vehicle longitudinal pitch motion (a rearward pitch) shock damping logic **150** alters, such as increases, the compression damping characteristic front track adjustable shock absorber **144** and alters, such as increases, a compression damping of the rear track adjustable shock absorber **146** for the duration of the predicted vehicle pitch motion. Further, for a predicted increase in the vehicle longitudinal pitch motion (a rearward pitch) shock damping logic **150** alters, such as increases, the rebound damping characteristic of the rear track adjustable shock absorber **146** for the duration of the predicted vehicle pitch motion. Additionally, for a predicted increase in the vehicle longitudinal pitch motion (a rearward pitch) shock damping logic **150** alters, such as increases, the rebound damping characteristic of the right front adjustable shock absorber **140** and the left front adjustable shock absorber **142** for the duration of the predicted vehicle pitch motion

Returning to FIG. **8**C, airborne time table **210** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on the elapsed time that snowmobile **10** has been airborne. An example of snowmobile **10** being airborne is illustrated in FIG. **21**. In embodiments, airborne time table **210** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping for one or more time period ranges of airborne detection. For example, a first percentage for a first time range of airborne elapsed time detection and a second percentage for a second time range of airborne elapsed time detection.

In embodiments, upon detection of an airborne state, the compression damping characteristics for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** provide an increased damping percentage over the base damping tables and continue to increase as the detected airborne time continues to increase. The increased compression damping is held after airborne event has concluded to make sure the added compression damping is used through the entire compression stroke of the respective shocks. In one example, the increased compression damping is held for about 300 milliseconds. The increased compression damping is stepped back down to the base damping level for normal operation of snowmobile **10** or is altered to another level if a different event is detected, such as acceleration, braking or cornering.

In embodiments, upon detection of an airborne state, the rebound damping characteristics for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** go to a minimal value, such as zero to promote shock extension. Once snowmobile **10** has landed, the rebound damping characteristic increases to stabilize the landing and reduce the occurrence of vehicle hops. The amount of rebound increase is based on airborne time (higher for longer airborne times). This provides different responses between small airborne events, like whoops, and larger airborne events so it can activate differently in whoops (medium size bumps on trail) versus big jumps which have longer air times. Rebound damping is held at the minimal level while snowmobile **10** is airborne and upon initial landing increased and held for a time period generally equal to the first extension stroke of the respective shock after compression. An advantage among others is that this keeps the track and skis in contact with the snow more for increased traction and stability.

In embodiments, IMU **132** is used to detect when snowmobile **10** is airborne by monitoring acceleration along axis **164** (see FIG. **1**A). Exemplary detection of an airborne event is described in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein. Additional exemplary detection methodologies for an airborne event are described in U.S. Pat. No. 9,381,810, filed Jun. 3, 2011, titled ELECTRONIC THROTTLE CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

Driver actuatable suspension adjust input event table **212** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** in response to an actuation of driver actuatable suspension adjust input event table **212** of snowmobile **10**. In embodiments, driver actuatable suspension adjust input event table **212** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping for an actuation of driver actuatable suspension adjust input event table **212**. In embodiments, actuation of driver actuatable suspension adjust input event table **212** results in the compression damping for each of right front adjustable shock absorber

140, left front adjustable shock absorber 142, front track adjustable shock absorber 144, and rear track adjustable shock absorber 146 is set at an increased level. In one example, the compression damping for each of right front adjustable shock absorber 140, left front adjustable shock absorber 142, front track adjustable shock absorber 144, and rear track adjustable shock absorber 146 are set at 100 percent. The driver actuatable suspension adjust input 172 is actuatable by an operator to increase the compression damping to absorb rough terrain that the operator views as upcoming.

In embodiments, a second actuation of driver actuatable suspension adjust input 172 within a given time period is interpreted by electronic controller 100 to reduce compression damping for right front adjustable shock absorber 140, left front adjustable shock absorber 142, front track adjustable shock absorber 144, and rear track adjustable shock absorber 146 to absorb chatter bumps (small bumps on trail).

In embodiments, a second actuation of driver actuatable suspension adjust input 172 within a given time period is interpreted by electronic controller 100 to hold compression damping for right front adjustable shock absorber 140, left front adjustable shock absorber 142, front track adjustable shock absorber 144, and rear track adjustable shock absorber 146 until a third actuation of driver actuatable suspension adjust input 172 is received or a driver mode (comfort, handling, rough trail) is changed.

In embodiments, a continued actuation of driver actuatable suspension adjust input 172, such as holding a button down, for a given time period is interpreted by electronic controller 100 to hold compression damping for right front adjustable shock absorber 140, left front adjustable shock absorber 142, front track adjustable shock absorber 144, and rear track adjustable shock absorber 146 until a subsequent actuation of driver actuatable suspension adjust input 172 is received or a driver mode (comfort, handling, rough trail) is changed.

In embodiments, driver actuatable suspension adjust input 172 is not a separate input, but is recognized as a characteristic of another driver input. In one example, a quick partial actuation of a brake lever for a short duration of time that is detected by brake switch 134 is interpreted by controller 100 as an input from driver actuatable suspension adjust input 172 while a longer actuation and/or a more complete actuation of the brake lever is interpreted by controller 100 as an actuation of the brakes of snowmobile 10.

Brake event table 214 includes damping characteristics for each of right front adjustable shock absorber 140, left front adjustable shock absorber 142, front track adjustable shock absorber 144, and rear track adjustable shock absorber 146 based on an actuation of a brake input of snowmobile 10 as monitored by brake switch 134 on the brake lever or based on a pressure value from brake pressure sensor 135 (actual pressure reading or pressure switch).

In embodiments, upon detection of braking while snowmobile 10 is traveling forward, the compression damping characteristics for right front adjustable shock absorber 140 and left front adjustable shock absorber 142 are increased to minimize forward pitch of snowmobile 10 and bottoming out of right front adjustable shock absorber 140 and left front adjustable shock absorber 142. For front track adjustable shock absorber 144, the compression damping characteristic is decreased to increase the ability of snowmobile 10 to absorb trail bumps and to collect snow in front of endless track 18 to assist slowing snowmobile 10. For rear track adjustable shock absorber 146, the compression damping either remains at a current level or is decreased to keep more weight of snowmobile 10 on endless track 18.

In embodiments, upon detection of braking while snowmobile 10 is traveling forward, the rebound damping characteristics for right front adjustable shock absorber 140 and left front adjustable shock absorber 142 either remains the same or is decreased to keep right front adjustable shock absorber 140 and left front adjustable shock absorber 142 extended as much as possible and thereby keep the front of snowmobile 10 higher. For front track adjustable shock absorber 144, the rebound damping characteristic is increased to collect snow in front of endless track 18 to assist slowing snowmobile 10 and keep track weight. For rear track adjustable shock absorber 146, the rebound damping is increased to reduce pitch motion of snowmobile 10 and keep more weight transferred to the rear of snowmobile 10.

In embodiments, a separate brake event table 214 is not provided due to the braking response being accounted for in predictive vehicle pitch table 236 when looking at a predicted deceleration of snowmobile 10.

Anti-dive event table 216 includes damping characteristics for each of right front adjustable shock absorber 140 and left front adjustable shock absorber 142 based on a deceleration rate of the vehicle from the accelerometer input or the vehicle speed input. In embodiments, anti-dive event table 216 specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping based on an actuation of a brake input of snowmobile 10 as monitored by brake switch 134 or based on a pressure value from brake pressure sensor 135. In embodiments, the compression damping characteristics for right front adjustable shock absorber 140 and left front adjustable shock absorber 142 are increased to minimize forward pitch of snowmobile 10 and bottoming out of right front adjustable shock absorber 140 and left front adjustable shock absorber 142. For front track adjustable shock absorber 144, the compression damping characteristic is decreased to increase the ability of snowmobile 10 to absorb trail bumps and to collect snow in front of endless track 18 to assist slowing snowmobile 10. For rear track adjustable shock absorber 146, the compression damping either remains at a current level or is decreased to keep more weight of snowmobile 10 on endless track 18.

In embodiments, upon detection of braking while snowmobile 10 is traveling forward, the rebound damping characteristics for right front adjustable shock absorber 140 and left front adjustable shock absorber 142 either remains the same or is decreased to keep right front adjustable shock absorber 140 and left front adjustable shock absorber 142 extended as much as possible and thereby keep the front of snowmobile 10 higher. For front track adjustable shock absorber 144, the rebound damping characteristic is increased to collect snow in front of endless track 18 to assist slowing snowmobile 10 and keep track weight. For rear track adjustable shock absorber 146, the rebound damping is increased to reduce pitch motion of snowmobile 10 and keep more weight transferred to the rear of snowmobile 10.

In embodiments, the compression damping characteristics for right front adjustable shock absorber 140 and left front adjustable shock absorber 142 are increased over level for a braking event 214. Based on a level of the brake pressure applied, shock damping logic 150 selects between brake event table 214 and anti-dive event table 216 with anti-dive event table 216 being for higher levels of brake pressure. In one example, anti-dive event table 216 is used for an initial timeframe and then brake event table **214** thereafter because the vehicle speed has slowed.

In embodiments, a separate anti-dive event table **216** is not provided. In embodiments, the braking and anti-dive response are accounted for in the predictive vehicle pitch table **236** when looking at a predicted deceleration of snowmobile **10** and at a predicted pitch forward of snowmobile **10**. In embodiments, the braking and anti-dive response is accounted for based on the degree of actuation of the brake level or on the level or monitored brake pressure.

One ski event table **218** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on a steering angle of the steering system and one or more outputs of IMU **132**, such as a lateral acceleration, a roll axis angle, and a roll rate. In embodiments, one ski event table **218** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping based on a roll angle detected by IMU **132**.

In embodiments, if snowmobile **10** is rolling towards the right side the compression damping for left front adjustable shock absorber **142** is increased to counteract the roll and the rebound damping for right front adjustable shock absorber **140** is decreased to extend right front adjustable shock absorber **140** to increase contact with the snow. If snowmobile **10** is rolling towards the left side the compression damping for right front adjustable shock absorber **140** is increased to counteract the roll and the rebound damping for left front adjustable shock absorber **142** is decreased to extend left front adjustable shock absorber **142** to increase contact with the snow. In embodiments, if snowmobile **10** is rolling towards the right then increase compression damping for the right shock and decrease rebound damping for the right shock. In embodiments, if snowmobile **10** is rolling towards left then increase compression damping for left shock and decrease compression damping for left shock.

G-out event table **224** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on a pitch motion of snowmobile **10**. In embodiments, G-out event table **224** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping. In embodiments, the compression damping characteristic for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, and rear track adjustable shock absorber **146** is increased.

Chatter event table **226** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on based on a pitch motion of snowmobile **10**. In embodiments, chatter event table **226** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping. In embodiments, both the compression and rebound damping are decreased for all shock absorbers. In embodiments, chatter events are distinguished from whoops events based on a frequency analysis of the IMU output.

Cornering event table **228** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, and front track adjustable shock absorber **144** based longitudinal acceleration of snowmobile **10** and at least one of steering angle sensor **136** or acceleration along axis **162** (lateral acceleration) for cornering. In embodiments, cornering event table **228** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping.

In embodiments, upon detection of cornering while snowmobile **10** is traveling forward, the compression damping characteristics for the front outside shock (fp**140** if turning to left or left front adjustable shock absorber **142** if turning to the right) is increased minimize roll of snowmobile **10** and for the front inside shock (fp**142** if turning to left or right front adjustable shock absorber **140** if turning to the right) is decreased to absorb trail bumps more smoothly. The level of increase and decrease is dependent on the longitudinal acceleration of snowmobile **10** (larger increases and decreases for higher speeds) to control pitch motion for snowmobile **10**. Regardless of the direction of the turn, the compression damping characteristic for front track adjustable shock absorber **144** is decreased to add ski traction for improved cornering and for rear track adjustable shock absorber **146** the compression damping characteristic is updated to control longitudinal pitch motion and weight transfer throughout the cornering event. For example, at corner entry, the compression damping characteristic of rear track adjustable shock absorber **146** may remain unchanged or be decreased based on longitudinal acceleration, but at mid corner or at corner exit when the throttle is applied the compression damping characteristic of rear track adjustable shock absorber **146** may be increased to prevent lifting of the skis and losing traction on the front of snowmobile **10**.

In embodiments, upon detection of cornering while snowmobile **10** is traveling forward, the rebound damping characteristics for the front inside shock (fp**142** if turning to left or right front adjustable shock absorber **140** if turning to the right) is increased to reduce roll of snowmobile **10**. The level of increase and decrease is dependent on the longitudinal acceleration of snowmobile **10**. In one example, the level of increase and decrease is also dependent on a lateral acceleration of snowmobile **10**. Regardless of the direction of the turn, the rebound damping characteristic for front track adjustable shock absorber **144** is increased to add ski traction for improved cornering and for rear track adjustable shock absorber **146** the rebound damping characteristic remains unchanged or is increased to control longitudinal pitch motion, hold the rear end down, and control vehicle body motion throughout the cornering event.

Brake corner event table **220** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on snowmobile **10** both cornering and braking. In embodiments, shock damping logic **150** follows whichever of brake event table **214** and corner event table **228** has a higher priority. In one example, the cornering event has a higher priority than the braking event.

Chatter corner event table **222** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based on snowmobile **10** both cornering and airborne detection. In embodiments, airborne events override cornering events. Further, a steering angle could be used to modify the airborne damping value increasing the damping of the outside front and/or rear shock. In embodiments, a separate chatter corner event table **222** is not provided, but rather shock damping logic **150** follows whichever of airborne time event table **210** and corner event table **228** has a higher priority.

Base damping table **230** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** in the absence of detecting one or more of airborne time event table **210**, driver actuatable suspension adjust input event table **212**, brake event table **214**, anti-dive event table **216**, one ski event table **218**, brake corner event table **220**, chatter corner event table **222**, g-out event table **224**, chatter event table **226**, corner event table **228**, launch mode event table **232**, anti-squat event table **234**, and predictive vehicle pitch table **236**. In embodiments, base damping table **230** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping for detected vehicle speeds and/or throttle positions. As mentioned herein, each of the selectable modes (comfort, handling, and rough trail) have their own base damping table.

Launch mode event table **232** includes damping characteristics for each of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** based on an actuation of the launch mode input **174** or based on one or more sensor readings (such as vehicle speed is zero and throttle actuation above a first level). In embodiments, launch mode event table **232** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping. In response to an actuation of the launch mode input, different actions are taken based on the selected mode (Comfort, Handling, Rough Trail) of snowmobile **10** for the expected high acceleration of snowmobile **10**. For the Comfort and Handling modes, it is desired to keep skis **16** on the snow. To achieve this outcome, the compression damping characteristics for front track adjustable shock absorber **144** is maintained or decreased, the rebound damping characteristic for front track adjustable shock absorber **144** is increased, the compression damping characteristic for rear track adjustable shock absorber **146** is increased, and the rebound damping characteristic for rear track adjustable shock absorber **146** is maintained or decreased. These changes both promotes keeping the skis **16** on the snow and minimizing a rearward pitch motion. For the Rough Trail mode, it may be desirable to raise skis **16** off of the snow. To achieve this outcome, the compression damping characteristics for front track adjustable shock absorber **144** is increased, the rebound damping characteristic for front track adjustable shock absorber **144** is decreased, the compression damping characteristic for rear track adjustable shock absorber **146** is maintained or decreased, and the rebound damping characteristic for rear track adjustable shock absorber **146** is maintained or increased. These changes both promotes lifting the skis **16** and promoting a rearward pitch motion. Further, when starting from a stop or low speed, the changes to front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** may be further modified to promote snowmobile **10** raising the skis further off the ground to perform a wheelie.

Anti-squat event table **234** includes damping characteristics for each of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based longitudinal acceleration of snowmobile **10** measured by IMU **132**, a pitch rate, and/or a pitch angle measured by IMU **132**. In embodiments, anti-squat event table **234** specifies a percentage of available damping range for compression damping only, for rebound damping only, or for both compression damping and rebound damping for one or more time period ranges of detection. For a measured pitch angle, shock damping logic **150** increases the rebound damping characteristic of right front adjustable shock absorber **140** and left front adjustable shock absorber **142** and increases the compression damping characteristic for front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** for the duration of the predicted vehicle pitch motion.

Having right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** being adjustable is helpful for orienting snowmobile **10** in various environmental situations. Referring to FIG. **22**, snowmobile **10** is shown in deep snow. When the shock absorbers **144** and **146** are not adjustable, the rear of snowmobile **10** sinks in the snow up to the running boards of snowmobile **10**. By sensing the rearward pitch of snowmobile **10** with IMU **132**, the rebound damping characteristic of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** can be automatically increased to promote extension of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** and the compression damping characteristic of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** can be decreased. This results in the rear end of the rear suspension to drop and keeps snowmobile **10** more level in the snow.

Referring to FIGS. **9**A-C, an exemplary processing sequence **300** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for left front adjustable shock absorber **142** provided as part of the suspension **24** of a left front ski **16** of the exemplary snowmobile **10** of FIG. **1**. Referring to FIG. **9**A, processing sequence **300** determines if an airborne event is detected, as represented by block **302**. An exemplary process **700** for detecting an airborne event is illustrated in FIG. **15**. If an airborne event is detected, the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in airborne time table **210** of shock damping logic **150**, as represented by block **304** and processing sequence **300** is completed, as represented by block **306**.

If an airborne event is not detected, shock damping logic **150** determines if a driver actuatable suspension adjust input event is detected, as represented by block **310**. If a driver actuatable suspension adjust input event is detected, the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in driver actuatable suspension adjust input table **212** of shock damping logic **150**, as represented by block **312** and processing sequence **300** is completed, as represented by block **306**.

If a driver actuatable suspension adjust input event is not detected, shock damping logic **150** determines if a braking event is detected, as represented by block **314**. An exemplary process **650** for detecting a braking event is illustrated in FIG. **14**. If a braking event is detected, shock damping logic **150** determines if an anti-dive time threshold has elapsed, as represented by block **316**. If the anti-dive threshold has not elapsed, the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in anti-dive table **216** of shock damping logic **150**, as represented by block **318** and processing sequence **300** is completed, as represented by block **306**.

If the anti-dive threshold has elapsed, the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in brake event table **214** of shock damping logic **150** based on a value of vehicle deceleration from IMU **132**. Shock damping logic **150** next determines if a cornering event is detected, as represented by block **322**. If a cornering event is not detected processing sequence **300** is completed, as represented by block **306**. If a cornering event is detected, shock damping logic **150** determines if the vehicle acceleration is to the left side of snowmobile **10**, as represented by block **324**. If the vehicle acceleration is to the left, left front adjustable shock absorber **142** is designated the outside shock, as represented by block **326**, and the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in brake corner event table **220** for the outside shock, as represented by block **328**, and processing sequence **300** is completed, as represented by block **306**. If the vehicle acceleration is not to the left, left front adjustable shock absorber **142** is designated the inside shock, as represented by block **330**, and the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in brake corner event table for the inside shock, as represented by block **328**, and processing sequence **300** is completed, as represented by block **306**.

If a braking event is not detected, shock damping logic **150** determines if a one ski event is detected, as represented by block **340**. An exemplary process **1000** for detecting a one ski event is illustrated in FIG. **20**. If a one ski event is detected, shock damping logic **150** determines the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in one ski event table **218** of shock damping logic **150**, as represented by block **342** and processing sequence **300** is completed, as represented by block **306**.

If a one ski event is not detected, shock damping logic **150** determines if a G-out event is detected, as represented by block **350**. An exemplary process **600** for detecting a G-out event is illustrated in FIG. **13**. If a G-out event is detected, shock damping logic **150** determines the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in G-out event table **224** of shock damping logic **150**, as represented by block **352** and processing sequence **300** is completed, as represented by block **306**.

If a G-out event is not detected, shock damping logic **150** determines if a chatter event is detected, as represented by block **354**. An exemplary process **600** for detecting a chatter event is illustrated in FIG. **13**. If a chatter event is detected, shock damping logic **150** determines the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in chatter event table **226** of shock damping logic **150**, as represented by block **356**. Shock damping logic **150** then determines if a cornering event is detected, as represented by block **358**. If a cornering event is not detected, processing sequence **300** is completed, as represented by block **306**. If a cornering event is detected, shock damping logic **150** determines if the vehicle acceleration is to the left side of snowmobile **10**, as represented by block **364**. If the vehicle acceleration is to the left, left front adjustable shock absorber **142** is designated the outside shock, as represented by block **366**, and the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in chatter corner event table **222** for the outside shock, as represented by block **368**, and processing sequence **300** is completed, as represented by block **306**. If the vehicle acceleration is not to the left, left front adjustable shock absorber **142** is designated the inside shock, as represented by block **370**, and the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in chatter corner event table **222** for the inside shock, as represented by block **368**, and processing sequence **300** is completed, as represented by block **306**.

If a chatter event is not detected, shock damping logic **150** determines if a cornering event is detected, as represented by block **376**. An exemplary process **800** for detecting a cornering event is illustrated in FIGS. **17**A-D. If a cornering event is detected, shock damping logic **150** determines if the vehicle acceleration is to the left side of snowmobile **10**, as represented by block **378**. If the vehicle acceleration is to the left, left front adjustable shock absorber **142** is designated the outside shock, as represented by block **380**, and the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in the corner event table **228** for the outside shock, as represented by block **382**, and processing sequence **300** is completed, as represented by block **306**. If the vehicle acceleration is not to the left, left front adjustable shock absorber **142** is designated the inside shock, as represented by block **384**, and the damping characteristics for left front adjustable shock absorber **142** is set based on the damping value provided in the corner event table **228** for the inside shock, as represented by block **382**, and processing sequence **300** is completed, as represented by block **306**. In embodiments, shock damping logic **150** also monitors vehicle acceleration/deceleration along axis **160** and this acceleration/deceleration value is a further input to corner event table. By taking into account the vehicle acceleration/deceleration along axis **160**, shock damping logic **150** is able to adjust the damping characteristics for left front adjustable shock absorber **142** differently for when snowmobile **10** is entering a corner (deceleration) and when snowmobile **10** is exiting a corner (acceleration).

If a cornering event is not detected, shock damping logic **150** determines a damping characteristic for left front adjustable shock absorber **142** based on the damping value provided in the base damping table **230**, as represented by block **390** and processing sequence **300** is completed, as represented by block **306**.

Referring to FIGS. **10**A-C, an exemplary processing sequence **400** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for right front continuous damping control shock absorber **142** provided as part of the suspension **24** of a left front ski **16** of the exemplary snowmobile **10** of FIG. **1**. Referring to FIG. **9**A, processing sequence **400** determines if an airborne event is detected, as represented by block **402**. An exemplary process **700** for detecting an airborne event is illustrated in FIG. **15**. If an airborne event is detected, the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in airborne time table **210** of shock damping logic **150**, as represented by block **404** and processing sequence **400** is completed, as represented by block **406**.

If an airborne event is not detected, shock damping logic **150** determines if a driver actuatable suspension adjust input event is detected, as represented by block **410**. If a driver actuatable suspension adjust input event is detected, the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in driver actuatable suspension adjust input table **212** of shock damping logic **150**, as represented by block **412** and processing sequence **400** is completed, as represented by block **406**.

If a driver actuatable suspension adjust input event is not detected, shock damping logic **150** determines if a braking event is detected, as represented by block **414**. An exemplary process **650** for detecting a braking event is illustrated in FIG. **14**. If a braking event is detected, shock damping logic **150** determines if an anti-dive time threshold has elapsed, as represented by block **416**. If the anti-time threshold has not elapsed, the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in anti-dive table **216** of shock damping logic **150**, as represented by block **418** and processing sequence **400** is completed, as represented by block **406**.

If the anti-dive time threshold has elapsed, the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in brake event table **214** of shock damping logic **150** based on a value of vehicle deceleration from IMU **132**. Shock damping logic **150** next determines if a cornering event is detected, as represented by block **422**. If a cornering event is not detected processing sequence **400** is completed, as represented by block **406**. If a cornering event is detected, shock damping logic **150** determines if the vehicle acceleration is to the right side of snowmobile **10**, as represented by block **424**. If the vehicle acceleration is to the right, right front adjustable shock absorber **140** is designated the outside shock, as represented by block **426**, and the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in brake corner event table **220** for the outside shock, as represented by block **428**, and processing sequence **400** is completed, as represented by block **406**. If the vehicle acceleration is not to the left, right front adjustable shock absorber **140** is designated the inside shock, as represented by block **430**, and the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in brake corner event table for the inside shock, as represented by block **428**, and processing sequence **400** is completed, as represented by block **406**.

If a braking event is not detected, shock damping logic **150** determines if a one ski event is detected, as represented by block **440**. An exemplary process **1000** for detecting a one ski event is illustrated in FIG. **20**. If a one ski event is detected, shock damping logic **150** determines the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in one ski event table **218** of shock damping logic **150**, as represented by block **442** and processing sequence **400** is completed, as represented by block **406**.

If a one ski event is not detected, shock damping logic **150** determines if a G-out event is detected, as represented by block **450**. An exemplary process **600** for detecting a G-out event is illustrated in FIG. **13**. If a G-out event is detected, shock damping logic **150** determines the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in G-out event table **224** of shock damping logic **150**, as represented by block **452** and processing sequence **400** is completed, as represented by block **406**.

If a G-out event is not detected, shock damping logic **150** determines if a chatter event is detected, as represented by block **454**. An exemplary process **600** for detecting a chatter event is illustrated in FIG. **13**. If a chatter event is detected, shock damping logic **150** determines the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in chatter event table **226** of shock damping logic **150**, as represented by block **456**. shock damping logic **150** then determines if a cornering event is detected, as represented by block **458**. If a cornering event is not detected, processing sequence **400** is completed, as represented by block **406**. If a cornering event is detected, shock damping logic **150** determines if the vehicle acceleration is to the right side of snowmobile **10**, as represented by block **464**. If the vehicle acceleration is to the right, right front adjustable shock absorber **140** is designated the outside shock, as represented by block **466**, and the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in chatter corner event table **222** for the outside shock, as represented by block **468**, and processing sequence **400** is completed, as represented by block **406**. If the vehicle acceleration is not to the left, right front adjustable shock absorber **140** is designated the inside shock, as represented by block **470**, and the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in chatter corner event table **222** for the inside shock, as represented by block **468**, and processing sequence **400** is completed, as represented by block **406**.

If a chatter event is not detected, shock damping logic **150** determines if a cornering event is detected, as represented by block **476**. An exemplary process **800** for detecting a cornering event is illustrated in FIGS. **17**A-D. If a cornering event is detected, shock damping logic **150** determines if the vehicle acceleration is to the right side of snowmobile **10**, as represented by block **478**. If the vehicle acceleration is to the right, right front adjustable shock absorber **140** is designated the outside shock, as represented by block **480**, and the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in the corner event table **228** for the outside shock, as represented by block **482**, and processing sequence **400** is completed, as represented by block **406**. If the vehicle acceleration is not to the left, right front adjustable shock absorber **140** is designated the inside shock, as represented by block **484**, and the damping characteristics for right front adjustable shock absorber **140** is set based on the damping value provided in the corner event table **228** for the inside shock, as represented by block **482**, and processing sequence **400** is completed, as represented by block **406**. In embodiments, shock damping logic **150** also monitors vehicle acceleration/deceleration along axis **160** and this acceleration/deceleration value is a further input to corner event table. By taking into account the vehicle acceleration/deceleration along axis **160**, shock damping logic **150** is able to adjust the damping characteristics for right front adjustable shock absorber **140** differently for when snowmobile **10** is entering a corner (deceleration) and when snowmobile **10** is exiting a corner (acceleration).

If a cornering event is not detected, shock damping logic **150** determines a damping characteristic for right front adjustable shock absorber **140** based on the damping value provided in the base damping table **230**, as represented by block **490** and processing sequence **400** is completed, as represented by block **406**.

Referring to FIGS. **11**A and **11**B an exemplary processing sequence **500** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for first track continuous damping control shock absorber **144** provided as part of the suspension **26** of the exemplary snowmobile **10** of FIG. **1**.

Shock damping logic **150** determines if a launch mode has been enabled, as represented by block **502**. An exemplary processing sequence **950** for determining if a launch mode has been enabled is provided in FIGS. **19**A and **19**B. If a launch mode has been enabled, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the launch mode event table **232**, as represented by block **504** and processing sequence **500** is completed, as represented by block **506**.

If a launch mode is not detected, shock damping logic **150** determines if an airborne event is detected, as represented by block **510**. If an airborne event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the airborne event table **210**, as represented by block **512** and processing sequence **500** is completed, as represented by block **506**.

If an airborne event is not detected, shock damping logic **150** determines if a driver actuatable suspension adjust input event is detected, as represented by block **514**. If a driver actuatable suspension adjust input event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the driver actuatable suspension adjust input event table **212**, as represented by block **516** and processing sequence **500** is completed, as represented by block **506**.

If a driver actuatable suspension adjust input event is not detected, shock damping logic **150** determines if a brake event is detected, as represented by block **518**. If a brake event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the brake event table **214**, as represented by block **520** and processing sequence **500** is completed, as represented by block **506**.

If a brake event is not detected, shock damping logic **150** determines if a one ski event is detected, as represented by block **522**. If a one ski event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the one ski event table **218**, as represented by block **524** and processing sequence **500** is completed, as represented by block **506**.

If a one ski event is not detected, shock damping logic **150** determines if a g-out event is detected, as represented by block **526**. If a g-out event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the g-out event table **224**, as represented by block **528** and processing sequence **500** is completed, as represented by block **506**.

If a G-out event is not detected, shock damping logic **150** determines if a chatter event is detected, as represented by block **530**. If a chatter event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the chatter event table **226**, as represented by block **532** and processing sequence **500** is completed, as represented by block **506**.

If a chatter event is not detected, shock damping logic **150** determines if an anti-squat event is detected, as represented by block **534**. If an anti-squat event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the anti-squat event table **234**, as represented by block **536** and processing sequence **500** is completed, as represented by block **506**.

If an anti-squat event is not detected, shock damping logic **150** determines if a cornering event is detected, as represented by block **538**. If a cornering event is detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the cornering event table **228**, as represented by block **540** and processing sequence **500** is completed, as represented by block **506**. In embodiments, shock damping logic **150** also monitors vehicle acceleration/deceleration along axis **160** and this acceleration/deceleration value is a further input to corner event table. By taking into account the vehicle acceleration/deceleration along axis **160**, shock damping logic **150** is able to adjust the damping characteristics for front track adjustable shock absorber **144** differently for when snowmobile **10** is entering a corner (deceleration) and when snowmobile **10** is exiting a corner (acceleration).

If a cornering event is not detected, shock damping logic **150** determines a damping characteristic for first track continuous damping control shock absorber **144** based on the damping value provided in the base damping table **230**, as represented by block **542** and processing sequence **500** is completed, as represented by block **506**.

Referring to FIGS. **12**A and **12**B an exemplary processing sequence **550** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for second track continuous damping control shock absorber **146** provided as part of the suspension **26** of the exemplary snowmobile **10** of FIG. **1**.

Shock damping logic **150** determines if a launch mode has been enabled, as represented by block **552**. An exemplary processing sequence for determining if a launch mode has been enabled is provided in FIGS. **19**A and **19**B. If a launch mode has been enabled, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the launch mode event table **232**, as represented by block **554** and processing sequence **550** is completed, as represented by block **556**.

If a launch mode is not detected, shock damping logic **150** determines if an airborne event is detected, as represented by block **558**. If an airborne event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the airborne event table **210**, as represented by block **560** and processing sequence **550** is completed, as represented by block **556**.

If an airborne event is not detected, shock damping logic **150** determines if a driver actuatable suspension adjust input event is detected, as represented by block **562**. If a driver actuatable suspension adjust input event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the driver actuatable suspension adjust input event table **212**, as represented by block **564** and processing sequence **550** is completed, as represented by block **556**.

If a driver actuatable suspension adjust input event is not detected, shock damping logic **150** determines if a brake event is detected, as represented by block **566**. If a brake event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the brake event table **214**, as represented by block **568** and processing sequence **550** is completed, as represented by block **556**.

If a brake event is not detected, shock damping logic **150** determines if a one ski event is detected, as represented by block **570**. If a one ski event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the one ski event table **218**, as represented by block **572** and processing sequence **550** is completed, as represented by block **556**.

If a one ski event is not detected, shock damping logic **150** determines if a g-out event is detected, as represented by block **574**. If a g-out event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the g-out event table **224**, as represented by block **576** and processing sequence **550** is completed, as represented by block **556**.

If a g-out event is not detected, shock damping logic **150** determines if a chatter event is detected, as represented by block **578**. If a chatter event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the chatter event table **226**, as represented by block **580** and processing sequence **550** is completed, as represented by block **556**.

If a chatter event is not detected, shock damping logic **150** determines if an anti-squat event is detected, as represented by block **582**. If an anti-squat event is detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the anti-squat event table **234**, as represented by block **584** and processing sequence **550** is completed, as represented by block **556**.

If an anti-squat event is not detected, shock damping logic **150** determines a damping characteristic for second track continuous damping control shock absorber **146** based on the damping value provided in the base damping table **230**, as represented by block **586** and processing sequence **550** is completed, as represented by block **556**.

Referring to FIG. **13**, an exemplary processing sequence **600** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting a g-out event or a chatter event is illustrated. suspension controller **200** samples from IMU **132** a pitch of snowmobile **10** about axis **162** over a window of time, as represented by block **602**. The sampled values are input to a discrete Fourier transform ("DFT"), as represented by block **604**, to provide a frequency map of the changes in pitch of snowmobile **10**.

The magnitude of the low frequency values, such as from 0 Hz to 3 Hz, are compared a low frequency threshold, as represented by block **606**. If the magnitude of the low frequency values do not satisfy the low frequency threshold, then no event is detected and the low frequency portion of processing sequence **600** ends, as represented by blocks **614** and **612**. If the magnitude of the low frequency values satisfy the low frequency threshold, then processing sequence **600** compares a pitch up rate from a gyroscope input from the IMU **132** to another threshold, as represented by block **608**. If the pitch-up rate does not satisfy the threshold, then no event is detected and the low frequency portion of processing sequence **600** ends, as represented by blocks **614** and **612**. If the pitch-up rate satisfies the threshold, then a g-out event is detected, as represented by block **610**. The detection of a g-out event as explained herein results in the damping characteristics for one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** being selected based on g-out event table **224**. In general, the damping characteristics are increased stiffness of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, and rear track adjustable shock absorber **146**.

The magnitude of the high frequency values, such as from 3 Hz to 10 Hz, are compared a high frequency threshold, as represented by block **616**. If the magnitude of the high frequency values does not satisfy the high frequency threshold, then no event is detected and the high frequency portion of processing sequence **600** ends, as represented by blocks **614** and **612**. If the magnitude of the high frequency values satisfies the high frequency threshold, then a chatter event is detected, as represented by block **618**. The detection of a chatter event as explained herein results in the damping characteristics for one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** being selected based on chatter event table **226**.

Referring to FIG. **14**, an exemplary processing sequence **650** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary brake event is illustrated. shock damping logic **150** determines if a brake switch is activated, as represented by block **652**. In embodiments, a switch is closed when a brake input is actuated. If the brake switch is not activated, no brake event is detected, as represented by block **654**, and processing sequence **650** is completed. If the brake switch is activated, a brake event is detected, as represented by block **656**, and processing sequence **650** is completed.

FIG. **15** illustrates an exemplary processing sequence **700** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary airborne event. shock damping logic **150** if an acceleration of snowmobile **10** along axis **164** detected by IMU **132** is below a threshold value, as represented by block **702**. If the detected acceleration is below the threshold value, an airborne event is detected, as represented by block **704** and processing sequence **700** is completed. If the detected acceleration is greater than the threshold, then shock damping logic **150** checks to see if the current status of snowmobile **10** is that an airborne event is active for snowmobile **10** meaning that snowmobile **10** was recently determined to be in and is still in an airborne event, as represented by block **708**. If an airborne event is not active, no change is made meaning snowmobile **10** is not found to now be in an airborne event, as represented by block **710**, and processing sequence **700** is completed. If an airborne event is active, shock damping logic **150** determines if the detected vertical acceleration exceeds the threshold value plus a hysteresis value of IMU **132**, as represented by block **712**. If not, no change is made meaning snowmobile **10** remains in an airborne event, as represented by block **710**, and processing sequence **700** is completed, as represented by block **706**. If the detected vertical acceleration does exceed the threshold value plus the hysteresis value of IMU **132**, shock damping logic **150** determines if the elapsed time from the determination that snowmobile **10** is in an airborne event exceeds a timeout value, as represented by block **714**. If the elapsed time does not exceed the timeout value, no change is made meaning snowmobile **10** remains in an airborne event, as represented by block **710**, and processing sequence **700** is completed, as represented by block **706**. If the elapsed time does exceed the timeout value, the status of snowmobile **10** is changed to airborne event not detected, as represented by block **716**, and processing sequence **700** is completed, as represented by block **706**.

FIGS. **16**A-C illustrates an exemplary processing sequence **750** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary anti-squat event. shock damping logic **150** determines if a rate of change of the throttle input detected by throttle position sensor **138** exceeds an enable limit, as represented by block **752**. If so, then shock damping logic **150** determines that a throttle rate of change event is detected, as represented by block **754**. If not, shock damping logic **150** determines if a rate of change of the throttle input is less than a disable limit, as represented by block **756**. If the rate of change of the throttle input is less than the disable limit, shock damping logic **150** determines that a throttle rate of change event is not detected, as represented by block **758**. If the rate of change of the throttle input is not less than the disable limit, shock damping logic **150** determines if the detected throttle position is less than a throttle limit meaning that the operator has backed off on the throttle, as represented by block **760**. If the detected throttle position is less than the throttle limit, shock damping logic **150** determines that a throttle rate of change event is not detected, as represented by block **758**. If the detected throttle position is greater than the throttle limit, shock damping logic **150** determines if the elapsed time since a throttle rate of change event was detected exceeds a hold time threshold, as represented by block **762**. If the elapsed time since the throttle rate of change event was detected does not exceed the hold time threshold, no change is made to whether a throttle rate of change event is active or not, as represented by block **764**. If the elapsed time since the throttle rate of change event was detected does exceed the hold time threshold, shock damping logic **150** determines that a throttle rate of change event is not detected, as represented by block **758**.

Turning to FIG. **16**B, shock damping logic **150** next determines if a linear acceleration of snowmobile **10** along axis **160** exceeds a threshold, as represented by block **766**. If the linear acceleration of snowmobile **10** along axis **160** exceeds the threshold, then an acceleration event is detected, as represented by block **768**. If the linear acceleration of snowmobile **10** along axis **160** does not exceed the threshold, shock damping logic **150** determines if the linear acceleration of snowmobile **10** is less than the threshold minus a hysteresis of IMU **132**, as represented by block **770**. If so, an acceleration event is not detected, as represented by block **772**. If not, no change is made to the current status of whether an acceleration event is active or not for snowmobile **10**, as represented by block **774**.

Turning to FIG. **16**C, shock damping logic **150** determines if one of a throttle rate of change event or an acceleration event is active for snowmobile **10**, as represented by block **780**. If either is active, an anti-squat event is determined, as represented by block **782**, and processing sequence **750** is completed, as represented by block **790**. If not, then shock damping logic **150** checks to verify both the throttle rate of change event and the acceleration event are not active, as represented by block **784**. If both are inactive, shock damping logic **150** disables an anti-squat event, as represented by block **786**, and processing sequence **750** is completed, as represented by block **790**. If one of the throttle rate of change event and the acceleration event are active, no change is made to the anti-squat status of snowmobile **10**, as represented by block **788**, and processing sequence **750** is completed, as represented by block **790**.

FIGS. **17**A-D illustrates an exemplary processing sequence **800** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary cornering event. shock damping logic **150** determines if an absolute value of a detected steering angle detected by steering angle sensor **136** exceeds a threshold (a zero steering angle corresponds with the skis pointing directly forward along axis **160**), as represented by block **802**. If the absolute value of the detected steering angle detected by steering angle sensor **136** exceeds the threshold, a steering event is detected, as represented by block **804**. If not, the absolute value of the detected steering angle detected by steering angle sensor **136** is compared to the threshold minus a hysteresis value of the steering angle sensor **136**, as represented by block **806**. If the absolute value of the detected steering angle detected by steering angle sensor **136** is less than the threshold minus the hysteresis value of the steering angle sensor **136**, shock damping logic **150** determines that a steering event is not detected, as represented by block **808**. If the absolute value of the detected steering angle detected by steering angle sensor **136** is not less than the threshold minus the hysteresis value of the steering angle sensor **136**, shock damping logic **150** makes no change to the steering event status of snowmobile **10**, as represented by block **810**.

Turning to FIG. **17**B, shock damping logic **150** determines if a lateral acceleration of snowmobile **10** along axis **162** exceeds a threshold value, as represented by block **812**. If the lateral acceleration of snowmobile **10** along axis **162** exceeds the threshold value, then a corner event is detected, as represented by block **814**. If not, the absolute value of the detected lateral acceleration of snowmobile **10** is compared to the threshold minus a hysteresis value of IMU **132**, as represented by block **816**. If the absolute value of the detected lateral acceleration of snowmobile **10** is less than the threshold minus the hysteresis value of IMU **132**, shock damping logic **150** determines that a cornering event is not detected, as represented by block **818**. If the absolute value of the detected lateral acceleration of snowmobile **10** is not less than the threshold minus the hysteresis value of IMU **132**, shock damping logic **150** makes no change to the corner event status of snowmobile **10**, as represented by block **820**.

Turning to FIG. **17**C, shock damping logic **150** determines if an absolute value of yaw rate and/or angle about axis **164** exceeds a threshold value, as represented by block **830**. If the absolute value of yaw rate and/or angle about axis **164** exceeds the threshold value, then a yaw event is detected, as represented by block **832**. If not, the absolute value of yaw rate and/or angle about axis **164** is compared to the threshold minus a hysteresis value of IMU **132**, as represented by block **834**. If the absolute value of the absolute value of yaw rate and/or angle about axis **164** is not less than the threshold minus the hysteresis value of IMU **132**, shock damping logic **150** makes no change to the yaw status of snowmobile **10**, as represented by block **836**. If the absolute value of absolute value of yaw rate and/or angle about axis **164** is less than the threshold minus the hysteresis value of IMU **132**, shock damping logic **150** determines if a cornering event timeout time has elapsed, as represented by block **838**. If the cornering timeout time has not elapsed, shock damping logic **150** makes no change to the yaw status of snowmobile **10**, as represented by block **836**. If the cornering timeout time has elapsed, then a yaw event is detected, as represented by block **840**.

Turning to FIG. **17**D, shock damping logic **150** determines if a yaw event or a steering event was detected in processing sequence **800**, as represented by block **850**. If one of the yaw event and steering event was detected then a cornering event is detected and the cornering timeout timer is set to zero, as represented by block **852**, and processing sequence **800** is completed, as represented by block **854**. If one of the yaw event and steering event was not detected then shock damping logic **150** determines if both the yaw event and steering event not detected, as represented by block **856**. If not, then there is no change to the cornering event status as represented by block **858**, and processing sequence **800** exits, as represented by block **854**. If so, then shock damping logic **150** determines a cornering event is not detected as represented by block **860**, and processing sequence **800** exits, as represented by block **854**.

FIG. **18** illustrates an exemplary processing sequence **900** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary driver actuatable suspension adjust input event. shock damping logic **150** monitors whether driver actuatable suspension adjust input **172** has been actuated, as represented by block **902**. If driver actuatable suspension adjust input **172** has been actuated, a driver actuatable suspension adjust input event is detected, as represented by block **904** and processing sequence **900** is completed. If driver actuatable suspension adjust input **172** has not been actuated, shock damping logic **150** checks to see of a driver actuatable suspension adjust input timeout counter is active and, if so, has it elapsed, as represented by block **908**. If the driver actuatable suspension adjust input timeout has elapsed, shock damping logic **150** determines that a driver actuatable suspension adjust input event has not been detected, as represented by block **910** and processing sequence **900** is complete. If the driver actuatable suspension adjust input timeout has not elapsed, shock damping logic **150** keeps the current state of the driver actuatable suspension adjust input event status, as represented by block **912**, and processing sequence **900** is complete.

FIGS. **19**A and **19**B illustrate an exemplary processing sequence **950** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary launch mode event. Referring to FIG. **19**A, shock damping logic **150** monitors to determine if a launch mode input **174** has been actuated, as represented by block **952**. If launch mode input **174** has not been actuated then no change is made to the launch mode event status, as represented by block **954**. If launch mode input **174** has been actuated, shock damping logic **150** determines if a launch mode is currently active, as represented by block **956**. If a launch mode is currently active, then the actuation of launch mode input **174** causes the launch mode to be disabled, as represented by block **958**. If a launch mode is not currently enabled, shock damping logic **150** based on vehicle speed sensor **139** determines if the speed of snowmobile **10** is zero, as represented by block **960**. If the vehicle speed is not zero, no change is made to the launch mode status, as represented by block **954**. If the vehicle speed is zero, shock damping logic **150** determines if the launch button **174** has been actuated for timeout, as represented by block **962**. If not, no change is made to the launch mode status, as represented by block **954**. If so, shock damping logic **150** enables the launch mode, as represented by block **964**.

Referring to FIG. **19**B, shock damping logic **150** determines if the launch mode is enabled, as represented by block **966**. If not, no change is made to the launch mode status, as represented by block **968**. If the launch mode is enabled, shock damping logic **150** determines based on vehicle speed sensor **139** if the vehicle speed is above a threshold, as represented by block **970**. If not, no change is made to the launch mode status, as represented by block **968**. If the vehicle speed is above the threshold then shock damping logic **150** determines if the linear acceleration of snowmobile **10** along axis **160** is less than a threshold, as represented by block **972**. If not, no change is made to the launch mode status, as represented by block **968**. If the linear acceleration of snowmobile **10** along axis **160** is less than the threshold, shock damping logic **150** disables the launch mode because snowmobile **10** is traveling closer to a constant speed than quickly accelerating, as represented by block **974**.

FIG. **20** illustrates an exemplary processing sequence **1000** of the shock damping logic **150** of the electronic controller **200** of FIG. **6** for detecting an exemplary one ski event. shock damping logic **150** based on IMU **132** monitors if a roll rate and/or angle of snowmobile **10** about axis **160** is greater than a low threshold, as represented by block **1002**. If so, shock damping logic **150** then checks to see if the monitored roll rate and/or angle is less than a high roll rate and/or angle threshold, as represented by block **1004**. If the vehicle roll rate and/or angle is in the range between the low threshold and the high threshold, a one ski event is detected, as represented by block **1006**, and processing sequence **1000** is completed, as represented by block **1008**. Returning back to block **1002**, if the vehicle roll rate and/or angle is not greater than the low threshold, shock damping logic **150** determines if the vehicle roll rate and/or angle is less than the low threshold minus the hysteresis of the IMU **132**, as represented by block **1010**. If not, no change is made by shock damping logic **150** to the one ski event status, as represented by block **1014**. If so, shock damping logic **150** determines that a one ski event is not detected, as represented by block **1012** and processing sequence **1000** is completed, as represented by block **1008**. Returning to block **1004**, if the vehicle roll rate and/or angle is not less than the high threshold, shock damping logic **150** determines that a one ski event is not detected, as represented by block **1012** and processing sequence **1000** is completed, as represented by block **1008**.

In embodiments, power is provided to front track adjustable shock absorber **144** and/or rear track adjustable shock absorber **146** through a wired connection routed through the suspension components or entering an interior of the track from a lateral side. In embodiments, communication signals are provided to and/or sent by front track adjustable shock absorber **144** and/or rear track adjustable shock absorber **146** through a wired connection routed through the suspension components or entering an interior of the track from a lateral side. In embodiments, both communication signals and power are provided to/from front track adjustable shock absorber **144** and/or rear track adjustable shock absorber **146** through a wired connection routed through the suspension components or entering an interior of the track from a lateral side. In embodiments, at least one of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** include a power source and/or receive/transmit communication signals to controller **100** wirelessly.

In embodiments, snowmobile **10** includes a demo mode to illustrate the functionality of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. In one embodiment, snowmobile **10** includes a battery and an operator input is provided to activate accessory power on snowmobile **10**. In this example, an operator is able to select different operating modes (comfort, handling, rough trail) and/or input adjustments to right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146**. In one example, an operator can select a left turn option and experience the change in the damping characteristics for right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** for a left turn. In another embodiment, snowmobile **10** does not include a battery. In this example, a power connection is provided on snowmobile **10** which allows for an external power source to be plugged into snowmobile **10** to power snowmobile **10** to operate in the demo mode.

In embodiments, during operation of vehicle **10** or **10'**, the damping characteristics of at least one of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146** may be further adjusted to alter handling. In examples, the base damping profile for a given mode (such as comfort or sport), predicted acceleration damping profile (longitudinal or lateral), braking damping profile, and cornering damping profile include adjustments to damping characteristics of at least one of front track adjustable shock absorber **144** and rear track adjustable shock absorber **146**.

During normal straight line driving (no appreciable cornering or change in acceleration), the base damping characteristics of front track adjustable shock absorber **144** may be different from rear track adjustable shock absorber **146** or the same as rear track adjustable shock absorber **146**. Changes in driving, such as cornering or changes in acceleration may result in adjustments to the base damping profile.

For example, in the comfort mode, during normal straight line driving, front track adjustable shock absorber **144** has a first setup relative to rear track adjustable shock absorber **146** in compression damping (stiffer or the same). This setup biases snowmobile **10** balance to have less ski pressure on the ground for skis **16** and generally less positive tracking of skis **16**. The compression damping of front track adjustable shock absorber **144** is still soft enough to not sacrifice ride comfort. When snowmobile **10** brakes, decelerates or corners or is predicted to do so; the compression damping of front track adjustable shock absorber **144** is increased relative to its previous state. This adjustment prevents weight transfer to skis **16** and makes the turning effort of snowmobile **10** less with sacrificed traction on skis **16**.

In another example, in sport mode, during normal driving, front track adjustable shock absorber **144** has a first setup relative to rear track adjustable shock absorber **146** in compression damping (softer or the same). This setup biases snowmobile **10** balance to have more ski pressure on the ground for skis **16** and generally more positive tracking of skis **16**. When snowmobile **10** brakes, decelerates or corners or is predicted to do so; the compression damping of front track adjustable shock absorber **144** is decreased relative to its previous state. This adjustment adds weight transfer to skis **16** and creates increased traction on skis **16** at the sacrifice of increased steering effort.

In embodiments, electronic controller **100** monitors the outputs of IMU **132** (a three-axis accelerometer and a three-axis gyroscope) to evaluate terrain and/or driver aggressiveness. Driver aggressiveness may be monitored by the longitudinal acceleration and lateral acceleration experienced by vehicle **10**. Further, throttle position, brake pressure, and steering angle, and steering velocity may provide indicators. Electronic controller **100** may monitor these factors and adjust the damping characteristics of one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** based thereon. Terrain type may be monitored by longitudinal acceleration, lateral acceleration, vertical acceleration, and all three angular rates of IMU **132**. In embodiments, the outputs are analyzed to determine the frequency response of each. The frequency responses may be determined through one or more bandpass filters, fast Fourier transform, or other methods. For example, the roll angular frequency response may be monitored with a bandpass filter for frequencies in a first range, such as 8-15 Hertz, to provide an indication of chatter. The monitored frequency response for one or more of the outputs are compared to stored ranges for different terrain types and the damping characteristics of one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** are adjusted based thereon.

In embodiments, electronic controller **100** also monitors GPS sensor **131**. A given location might indicate a first terrain type based on the frequency responses of the IMU **132** on a first day, such as fresh snow, and a second terrain type based on the frequency responses of the IMU **132** on a second day, such as hard snow. Based on those different terrain types and the historical data of the location, electronic controller may adjust the damping characteristics of one or more of right front adjustable shock absorber **140**, left front adjustable shock absorber **142**, front track adjustable shock absorber **144**, and rear track adjustable shock absorber **146** to make the hard snow of the second day feel more like the fresh snow of the first day. This provides the ability for snowmobile **10** to have the same feel for the same location on different days even if the terrain characteristics have changed.

In embodiments, sensors **130** includes a sensor which monitors if rear suspension assembly **26** is in a coupled state or uncoupled state. An exemplary sensor is a position sensor, an angular sensor, a pressure sensor, or a contact sensor which monitors when the suspension arm contacts a coupling block of rear suspension assembly **26**. In embodiments, the damping characteristics (compression and/or rebound) of front track adjustable shock absorber **144** and/or rear track adjustable shock absorber **146** are adjusted based on whether rear suspension assembly **26** is in a coupled state or uncoupled state.

While embodiments of the present disclosure have been described as having exemplary designs, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

The invention claimed is:

1. A snowmobile for propelled movement relative to the ground, comprising

a plurality of ground engaging members including an endless track positioned along a centerline vertical longitudinal plane of the snowmobile having a lateral width, a left front ski positioned on a left side of the centerline vertical longitudinal plane of the snowmobile, and a right front ski positioned on a right side of the centerline vertical longitudinal plane of the snowmobile;

a frame supported by the plurality of ground engaging members;

a steering system supported by the frame and operatively coupled to the left front ski and the right front ski to control a direction of travel of the snowmobile;

a left ski suspension operatively coupling the left front ski to the frame;

a right ski suspension operatively coupling the right front ski to the frame;

a track suspension operatively coupling the endless track to the frame, the track suspension including a first adjustable shock absorber, the first adjustable shock absorber having at least one adjustable damping characteristic, the first adjustable shock absorber being laterally positioned within the lateral width of the endless track;

a plurality of sensors supported by the ground engaging members; and

at least one electronic controller operatively coupled to the first adjustable shock absorber, the at least one electronic controller based on a sensor measurement from a sensor spaced apart from each of the left ski suspension, right ski suspension, and track suspension operable to determine an airborne condition, and in response to the airborne condition the electronic controller alters at least one damping characteristic of the first adjustable shock absorber, wherein the at least one damping characteristic includes a rebound damping characteristic and the controller is operable to alter the rebound damping characteristic based upon a length of time that the snowmobile has been airborne.

2. The snowmobile of claim **1**, the first adjustable shock absorber of the first suspension is positioned within an interior bounded by the endless track.

3. The snowmobile of claim **1**, the track suspension further comprises a second adjustable shock absorber having at least one adjustable damping characteristic, the second adjustable shock absorber being laterally positioned within the lateral width of the endless track, the first adjustable shock absorber is a front track adjustable shock absorber and the second adjustable shock absorber is a rear track adjustable shock absorber.

4. The snowmobile of claim **3**, the second adjustable shock absorber of the track suspension is positioned within the interior bounded by the endless track.

5. The snowmobile of claim **3**, the second adjustable shock absorber of the track suspension is positioned outside of the interior bounded by the endless track.

6. The snowmobile of claim **1**, wherein the at least one electronic controller alters the at least one damping characteristic of the first adjustable shock absorber while the snowmobile is moving relative to the ground.

7. The snowmobile of claim **6**, wherein the at least one electronic controller alters a compression damping characteristic of the first adjustable shock absorber.

8. The snowmobile of claim **1**, further comprising a third adjustable shock absorber and a fourth adjustable shock absorber, the third adjustable shock absorber is part of the left ski suspension operatively coupling the left front ski to the frame and the fourth adjustable shock absorber is part of the right ski suspension operatively coupling the right front ski to the frame.

9. The snowmobile of claim **1**, wherein a first damping characteristic of the first adjustable shock absorber is adjusted by the at least one electronic controller based on a longitudinal acceleration of the snowmobile.

10. The snowmobile of claim **1**, wherein a first damping characteristic of the first adjustable shock absorber is adjusted by the at least one electronic controller based on a vehicle pitch motion of the snowmobile.

11. The snowmobile of claim **1**, wherein the at least one electronic controller adjusts a compression damping characteristic of the first adjustable shock absorber in response to the snowmobile being airborne.

12. The snowmobile of claim **11**, wherein the adjustment of the compression damping characteristic of the first adjustable shock absorber is dependent on the length of time that the snowmobile has been airborne.

13. A method of controlling ride characteristics of a snowmobile, the method comprising the steps of:

monitoring with at least one electronic controller a plurality of sensors supported by the snowmobile while the snowmobile is moving;

predictively determining a longitudinal acceleration of the snowmobile based on a current sensor value of at least one sensor of the plurality of sensors; and

adjusting with the at least one electronic controller at least one damping characteristic of an adjustable shock absorber while the vehicle is moving, based on the predicted longitudinal acceleration of the vehicle, the adjustable shock absorber being apart of a suspension of an endless track of the snowmobile.

14. The method of claim **13**, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a longitudinal acceleration of the snowmobile.

15. The method of claim **14**, wherein the longitudinal acceleration of the snowmobile is measured by at least one sensor.

16. The method of claim **15**, further comprising the steps of: determining the longitudinal acceleration of the snowmobile indicates an acceleration of the snowmobile; and changing a rebound damping characteristic of the adjustable shock absorber.

17. The method of claim **15**, further comprising the steps of: determining the longitudinal acceleration of the snowmobile indicates an acceleration of the snowmobile; and changing a compression damping characteristic of the adjustable shock absorber.

18. The method of claim **13**, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a predicted pitch motion of the snowmobile.

19. The method of claim **18**, wherein the predicted pitch motion of the snowmobile is based on the predicted vehicle longitudinal acceleration.

20. The method of claim **18**, wherein step of adjusting the at least one damping characteristic based on the predicted pitch motion of the snowmobile includes determining the predicted vehicle longitudinal acceleration and determining a direction of travel of the snowmobile.

21. The method of claim **13**, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a turning of the snowmobile.

22. The method of claim **13**, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on a braking of the snowmobile.

23. The method of claim **13**, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic based on the snowmobile being airborne.

24. The method of claim **13**, wherein the step of adjusting with the at least one electronic controller the at least one damping characteristic of the adjustable shock absorber which is apart of the suspension of the endless track of the snowmobile while the vehicle is moving, includes the step of adjusting the at least one damping characteristic to promote a lifting of the skis of the snowmobile.

25. The method of claim **24**, wherein the step of adjusting the at least one damping characteristic to promote the lifting of the skis of the snowmobile includes the steps of decreasing a compression damping characteristic of the adjustable shock absorber.

26. The method of claim **25**, wherein the step of adjusting the at least one damping characteristic to promote the lifting of the skis of the snowmobile includes the steps of increasing a rebound damping characteristic of a second adjustable shock absorber associated with the suspension of the endless track.

27. A method of controlling a damping characteristic of at least one adjustable shock absorber of a vehicle being operated by a driver, the method comprising:

receiving with an electronic controller a plurality of inputs from a plurality of sensors supported by the vehicle;

predictively determining a longitudinal acceleration of the vehicle based on a current sensor value of at least one sensor of the plurality of sensors; and

adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the predicted longitudinal acceleration of the vehicle.

28. The method of claim **27**, wherein the step of adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the predicted longitudinal acceleration of the vehicle, includes the step of adjusting the at least one damping characteristic based on a predicted pitch motion of the snowmobile.

29. The method of claim **28**, wherein the at least one adjustable shock absorber includes a left adjustable shock absorber and a right adjustable shock absorber, and in response to a predicted increase in vehicle pitch motion, altering a damping characteristic of the left adjustable shock absorber and a damping characteristic of the right adjustable shock absorber.

30. The method of claim **29**, wherein the at least one adjustable shock absorber includes a rear adjustable shock absorber, and in response to a predicted increase in vehicle pitch motion, altering the damping characteristic of the rear adjustable shock absorber.

31. The method of claim **30**, wherein each of the left adjustable shock absorber, right adjustable shock absorber, and rear adjustable shock absorber have a first damping characteristic and a second damping characteristic, the second damping characteristic being different than the first damping characteristic, and in response to a predicted increase in vehicle pitch motion, increasing the first damping characteristic of the left adjustable shock absorber and the right adjustable shock absorber and increasing the second damping characteristic of the rear adjustable shock absorber.

32. The method of claim **27**, further comprising the steps of: predictively determining a longitudinal pitch motion of the vehicle; and adjusting the damping characteristic of the at least one adjustable shock absorber of the vehicle based on the predicted longitudinal pitch motion of the vehicle.

33. The method of claim **32**, wherein the predicted longitudinal acceleration is determined by the steps of: determining a predicted power for a prime mover of the snowmobile; determining an output power of the drivetrain based on the determined predicted power, the drivetrain including a CVT; determining a forward moving force of the snowmobile based on the determined output power of the drivetrain; determining a resultant forward moving force by subtracting at least one of a coast down force and an applied braking force from the determined forward moving force; and dividing the resultant forward moving force by a mass of the snowmobile to determine the predicted vehicle longitudinal acceleration.

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